19. Troubleshooting Guide

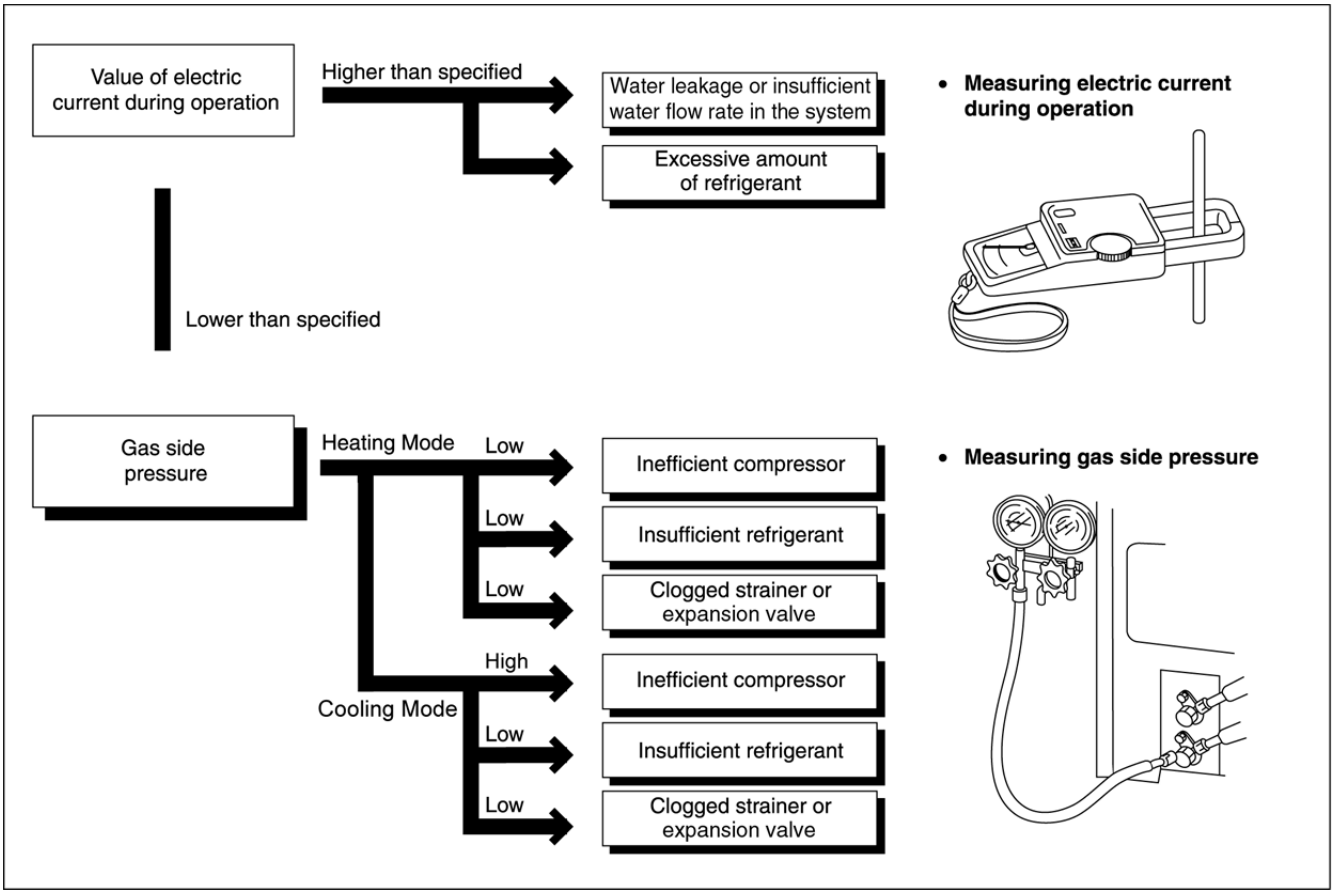
19.1 Refrigeration Cycle System

In order to diagnose malfunctions, make sure that there are no electrical problems before inspecting the refrigeration cycle.
Such problems include insufficient insulation, problem with the power source, malfunction of a compressor and a fan.
The normal pressure of the refrigeration cycle depends on various conditions, the standard values for them are shown in the table on the right.

Normal Pressure (Standard)

	Gas pressure MPa (kg/cm²G)
Heating Mode	2.3 ~ 2.9 (23 ~ 29)
Cooling Mode	0.9 ~ 1.2 (9 ~ 12)

★Condition: ● Outdoor temperature 7°C at heating mode and 35°C at cooling mode.
● Compressor operates at rated frequency.



19.2 Relationship between the Condition of the Air-to-Water Heatpump Indoor and Outdoor Units and Pressure and Electric Current

Condition of the Air-to-Water Heatpump indoor and outdoor units	Heating Mode			Cooling Mode		
	Low Pressure	High Pressure	Electric current during operation	Low Pressure	High Pressure	Electric current during operation
Water leakage or insufficient water flow rate in the system						
Excessive amount of refrigerant						
Inefficient compression						
Insufficient refrigerant (gas leakage)						
Outdoor heat exchange deficiency						
Clogged expansion valve or Strainer						

- Carry out the measurements of pressure, electric current, and temperature fifteen minutes after an operation is started.

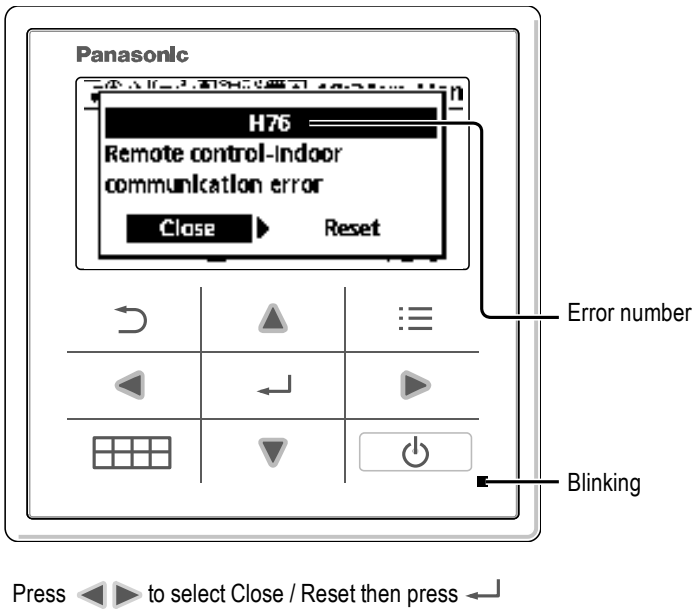
19.3 Breakdown Self Diagnosis Function

19.3.1 Self Diagnosis Function (Three Digits Alphanumeric Code)

- When abnormality occur during operation, the system will stop operation, and OFF/ON control panel LED will blink and error code will display on the control panel.
- Even error code is reset by turning OFF power supply or by selecting ERROR RESET, if the system abnormality is still unrepaired, system will again stop operation, and OFF/ON control panel LED will again blinks and error code will be display.
- The error code will store in IC memory.

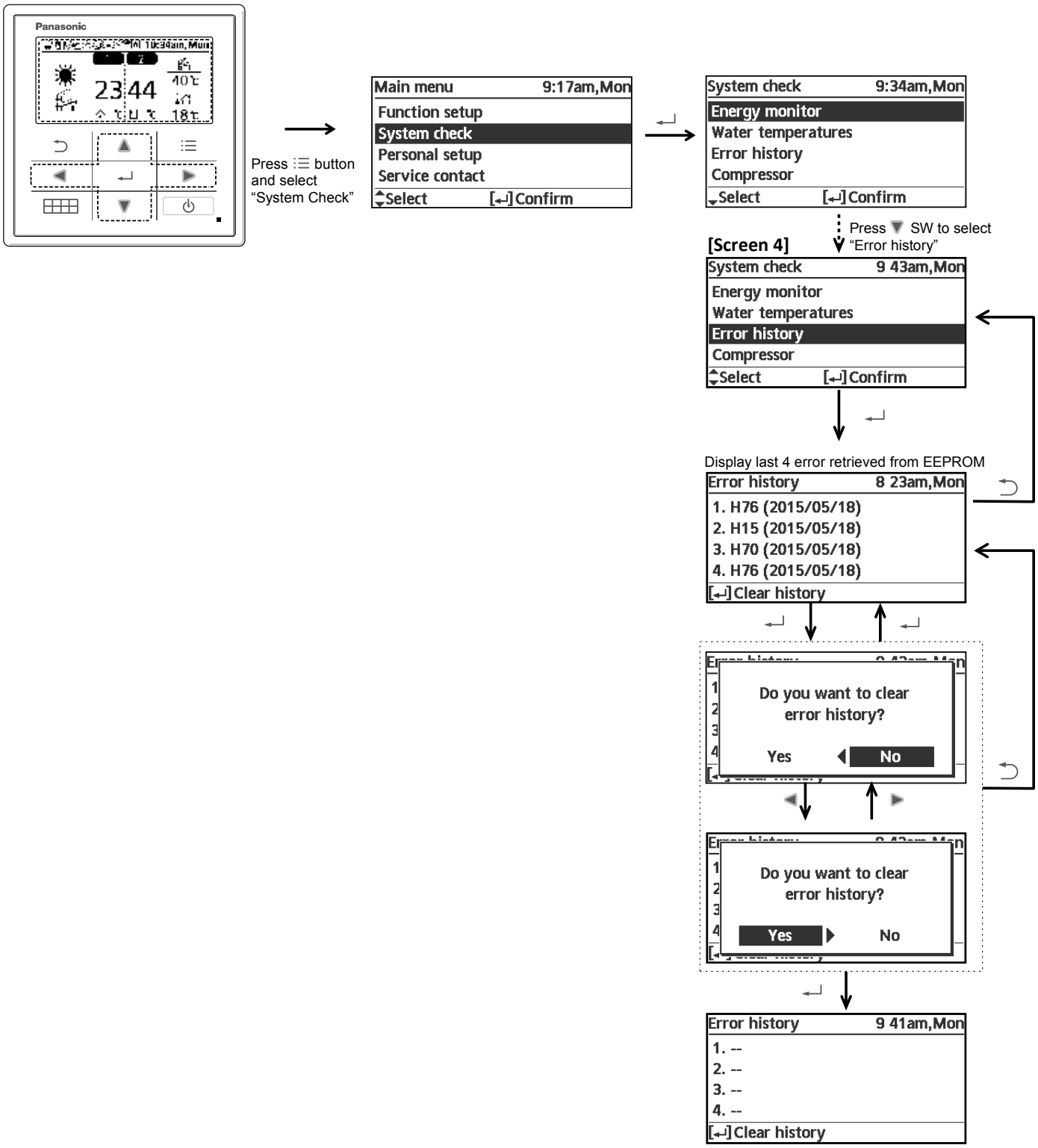
- **To check the error code**
  - When an abnormality occurs, system will stop operation and OFF/ON control panel LED will blink.
  - Error code of the abnormality will be display on the control panel.
  - To determine the abnormality description, the error code table needs to be referred.

eg:



- **To display past/last error code**
  - Turn ON power supply.
  - Refer below procedure to retrieve the error code history.

- **To permanently delete error code from IC memory**
  - Turn ON power supply.
  - Refer below procedure to clear error history.


Panasonic
Press ☰ button and select "System Check"
Main menu 9:17am,Mon
Function setup
System check
Personal setup
Service contact
Select [↵]Confirm
System check 9:34am,Mon
Energy monitor
Water temperatures
Error history
Compressor
Select [↵]Confirm
Press ▼ SW to select "Error history"
[Screen 4]
System check 9 43am,Mon
Energy monitor
Water temperatures
Error history
Compressor
Select [↵]Confirm
Display last 4 error retrieved from EEPROM
Error history 8 23am,Mon
1. H76 (2015/05/18)
2. H15 (2015/05/18)
3. H70 (2015/05/18)
4. H76 (2015/05/18)
[↵]Clear history
Do you want to clear error history?
Yes No
Do you want to clear error history?
Yes No
Error history 9 41am,Mon
1. --
2. --
3. --
4. --
[↵]Clear history

19.4 Error Codes Table

Diagnosis display	Abnormality/Protection control	Abnormality judgement	Primary location to verify
H00	No abnormality detected	—	—
H12	Indoor/Outdoor capacity unmatched	90s after power supply	• Indoor/outdoor connection wire • Indoor/outdoor PCB • Specification and combination table in catalogue
H15	Outdoor compressor temperature sensor abnormality	Continue for 5 sec.	• Compressor temperature sensor (defective or disconnected)
H20	Water pump abnormality	Continue for 10 sec.	• Indoor PCB • Water pump (malfunction)
H23	Indoor refrigerant liquid temperature sensor abnormality	Continue for 5 sec.	• Refrigerant liquid temperature sensor (defective or disconnected)
H27	Service valve error	Continue for 5 minutes	• High pressure sensor (defective or disconnected)
H28	Abnormal solar sensor	Continue for 5 sec.	• Solar temperature sensor (defective or disconnected)
H31	Abnormal swimming pool sensor	Continue for 5 sec.	• Pool temperature sensor (defective or disconnected)
H36	Abnormal buffer tank sensor	Continue for 5 sec.	• Buffer tank sensor (defective or disconnected)
H38	Brand code not match	When indoor and outdoor brand code not same	—
H42	Compressor low pressure abnormality	—	• Outdoor pipe temperature sensor • Clogged expansion valve or strainer • Insufficient refrigerant • Outdoor PCB • Compressor
H43	Abnormal Zone 1 sensor	Continue for 5 sec.	• Water temperature Zone 1 sensor
H44	Abnormal Zone 2 sensor	Continue for 5 sec.	• Water temperature Zone 2 sensor
H62	Water flow switch abnormality	Continue for 1 min.	• Water flow switch
H63	Abnormal low pressure sensor	4 times in 20 minutes	• Low pressure sensor (defective or disconnect)
H64	Refrigerant high pressure abnormality	Continue for 5 sec.	• Outdoor high pressure sensor (defective or disconnected)
H65	Abnormal deice water circulation	water flow > 7 l/min continously for 20 second during anti freeze deice	• Water pump
H67	Abnormal External Thermistor 1	Continue for 5 sec.	• Room temperature Zone 1 sensor
H68	Abnormal External Thermistor 2	Continue for 5 sec.	• Room temperature Zone 2 sensor
H70	Back-up heater OLP abnormality	Continue for 60 sec.	• Back-up heater OLP (Disconnection or activated)
H72	Tank sensor abnormal	Continue for 5 sec.	• Tank sensor
H74	PCB communication error	Communication or transfer error	• Indoor main PCB and Sub PCB
H75	Low water temperature control	Room heater disable and deice request to operate under low water temperature	• Heater operation must enable to increase water temperature
H76	Indoor - control panel communication abnormality	—	• Indoor - control panel (defective or disconnected)
H90	Indoor/outdoor abnormal communication	> 1 min after starting operation	• Internal/external cable connections • Indoor/Outdoor PCB
H91	Tank heater OLP abnormality	Continue for 60 sec.	• Tank heater OLP (Disconnection or activated)
H95	Indoor/Outdoor wrong connection	—	• Indoor/Outdoor supply voltage
H98	Outdoor high pressure overload protection	—	• Outdoor high pressure sensor • Water pump or water leakage • Clogged expansion valve or strainer • Excess refrigerant • Outdoor PCB
H99	Indoor heat exchanger freeze prevention	—	• Indoor heat exchanger • Refrigerant shortage
F12	Pressure switch activate	4 times occurrence within 20 minutes	• Pressure switch
F14	Outdoor compressor abnormal revolution	4 times occurrence within 20 minutes	• Outdoor compressor
F15	Outdoor fan motor lock abnormality	2 times occurrence within 30 minutes	• Outdoor PCB • Outdoor fan motor
F16	Total running current protection	3 times occurrence within 20 minutes	• Excess refrigerant • Outdoor PCB

Diagnosis display	Abnormality/Protection control	Abnormality judgement	Primary location to verify
F20	Outdoor compressor overheating protection	4 times occurrence within 30 minutes	• Compressor tank temperature sensor • Clogged expansion valve or strainer • Insufficient refrigerant • Outdoor PCB • Compressor
F22	IPM (power transistor) overheating protection	3 times occurrence within 30 minutes	• Improper heat exchange • IPM (Power transistor)
F23	Outdoor Direct Current (DC) peak detection	7 times occurrence continuously	• Outdoor PCB • Compressor
F24	Refrigeration cycle abnormality	2 times occurrence within 20 minutes	• Insufficient refrigerant • Outdoor PCB • Compressor low compression
F25	Cooling/Heating cycle changeover abnormality	4 times occurrence within 30 minutes	• 4-way valve • V-coil
F27	Pressure switch abnormality	Continue for 1 min.	• Pressure switch
F29	Low Discharge Superheat	1 times occurrence within 2550 minutes	• Discharge Temperature sensor • Discharge Pressure Sensor • Pressure Switch • Outdoor PCB
F30	Water outlet sensor 2 abnormality	Continue for 5 sec.	• Water outlet sensor 2 (defective or disconnected)
F32	Abnormal Internal Thermostat	Continue for 5 sec.	• Control panel PCB thermostat
F36	Outdoor air temperature sensor abnormality	Continue for 5 sec.	• Outdoor air temperature sensor (defective or disconnected)
F37	Indoor water inlet temperature sensor abnormality	Continue for 5 sec.	• Water inlet temperature sensor (defective or disconnected)
F40	Outdoor discharge pipe temperature sensor abnormality	Continue for 5 sec.	• Outdoor discharge pipe temperature sensor (defective or disconnected)
F41	PFC control	4 times occurrence within 10 minutes	• Voltage at PFC
F42	Outdoor heat exchanger temperature sensor abnormality	Continue for 5 sec.	• Outdoor heat exchanger temperature sensor (defective or disconnected)
F43	Outdoor defrost sensor abnormality	Continue for 5 sec.	• Outdoor defrost sensor (defective or disconnected)
F45	Indoor water outlet temperature sensor abnormality	Continue for 5 sec.	• Water outlet temperature sensor (defective or disconnected)
F46	Outdoor Current Transformer open circuit	—	• Insufficient refrigerant • Outdoor PCB • Compressor low
F48	Outdoor EVA outlet temperature sensor abnormality	Continue for 5 sec.	• Outdoor EVA outlet temperature sensor (defective or disconnected)
F49	Outdoor bypass outlet temperature sensor abnormality	Continue for 5 sec.	• Outdoor bypass outlet temperature sensor (defective or disconnected)
F95	Cooling high pressure overload protection	—	• Outdoor high pressure sensor • Water pump or water leakage • Clogged expansion valve or strainer • Excess refrigerant • Outdoor PCB

19.5 Self-diagnosis Method

19.5.1 Connection Capability Rank Abnormality (H12)

Malfunction Decision Conditions:

During startup operation of cooling and heating, the capability rank of indoor checked by the outdoor is used to determine connection capability rank abnormality.

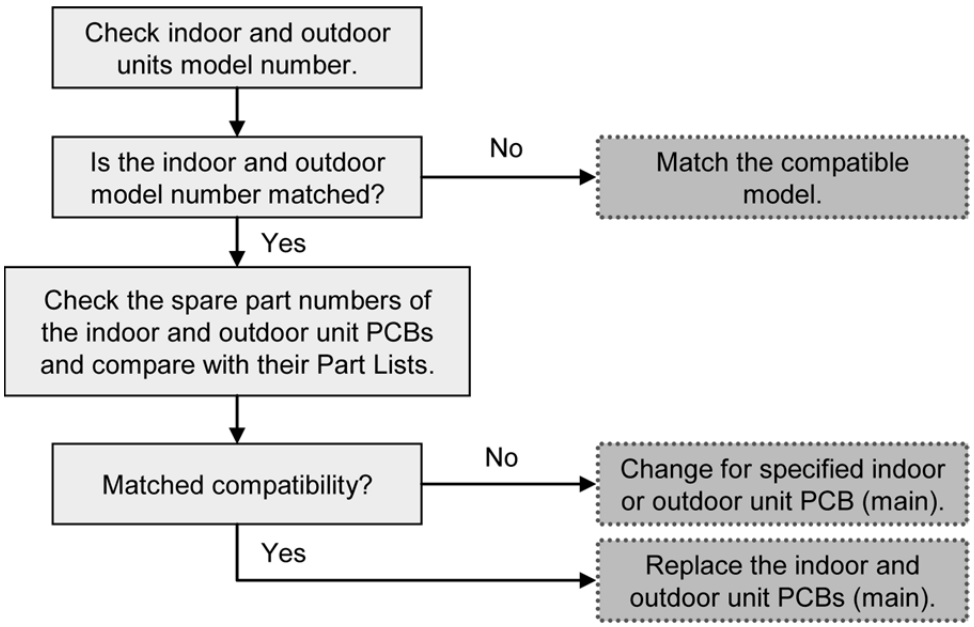
Malfunction Caused:

1 Wrong model interconnected.
2 Wrong indoor unit or outdoor unit PCB (main) used.
3 Faulty indoor unit or outdoor unit PCB (main).

Abnormality Judgment:

Continue for 90 seconds.

Troubleshooting:



19.5.2 Compressor Tank Temperature Sensor Abnormality (H15)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the compressor tank temperature sensor are used to determine sensor error.

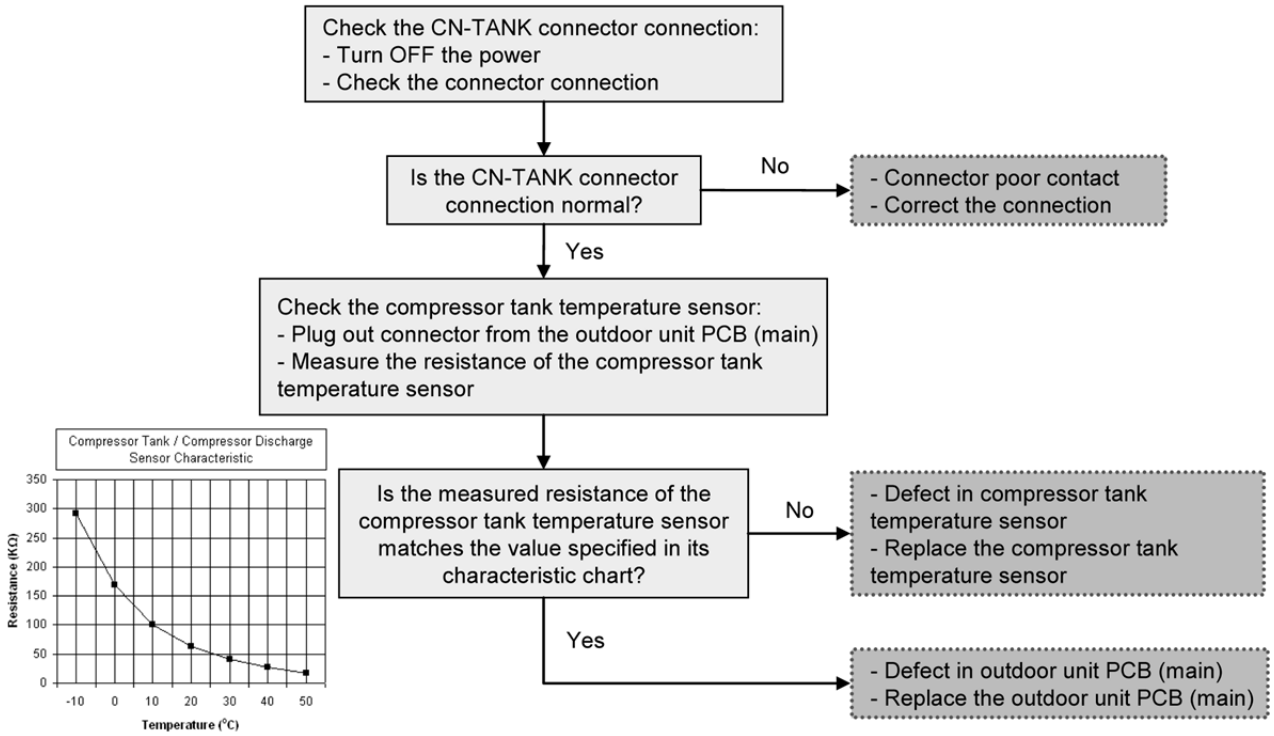
Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:



19.5.3 Water Pump Abnormality (H20)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the rotation speed detected by the IPM of water pump motor during water pump operation is used to determine abnormal water pump (feedback of rotation > 6,000rpm or < 1,000rpm).

Malfunction Caused:

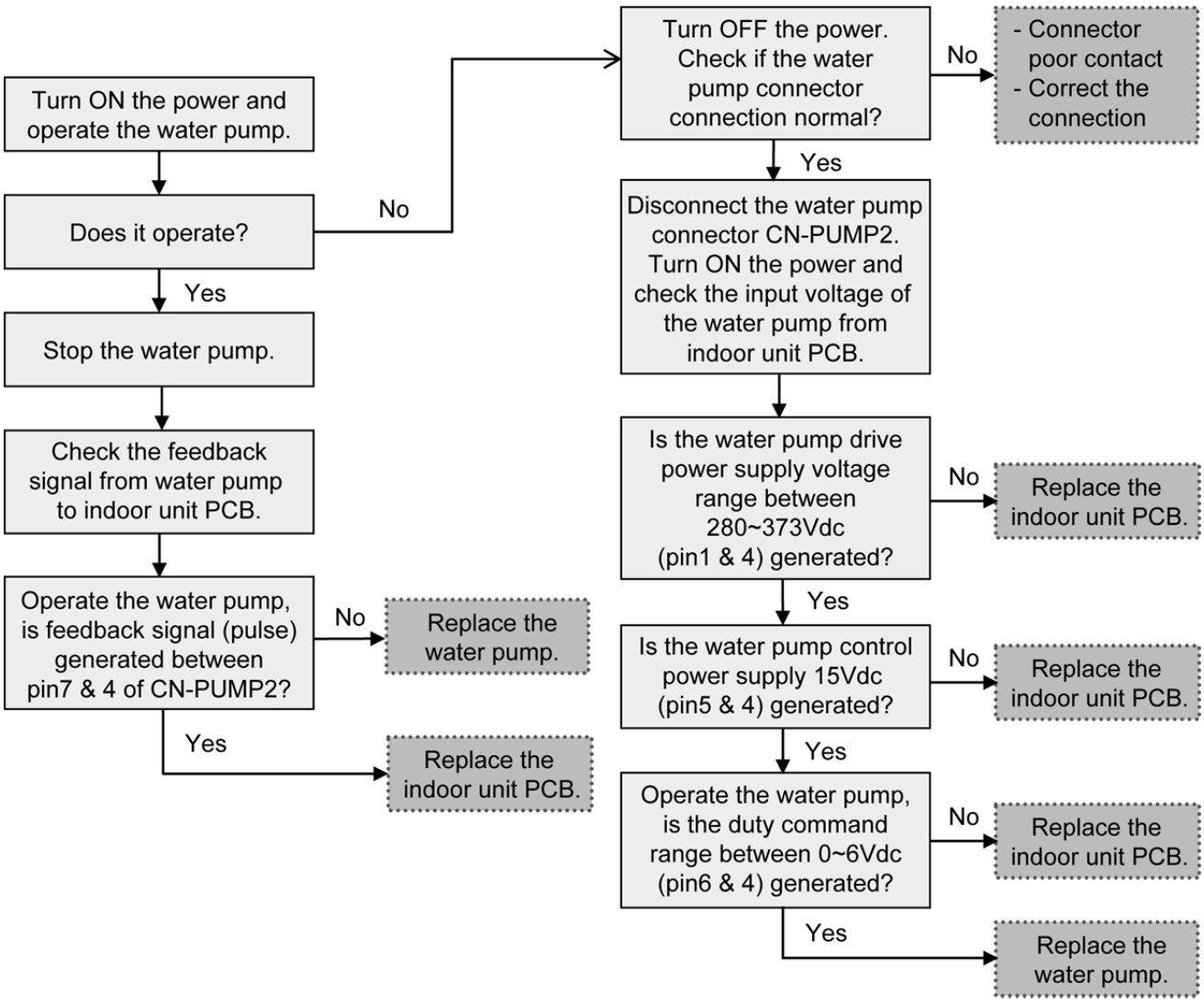
1 Operation stop due to short circuit inside the water pump motor winding.
2 Operation stop due to breaking of wire inside the water pump motor.
3 Operation stop due to breaking of water pump lead wires.
4 Operation stop due to water pump motor IPM malfunction.
5 Operation error due to faulty indoor unit PCB.

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.4 Indoor Refrigerant Liquid Temperature Sensor Abnormality (H23)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the indoor refrigerant liquid temperature sensor are used to determine sensor error.

Malfunction Caused:

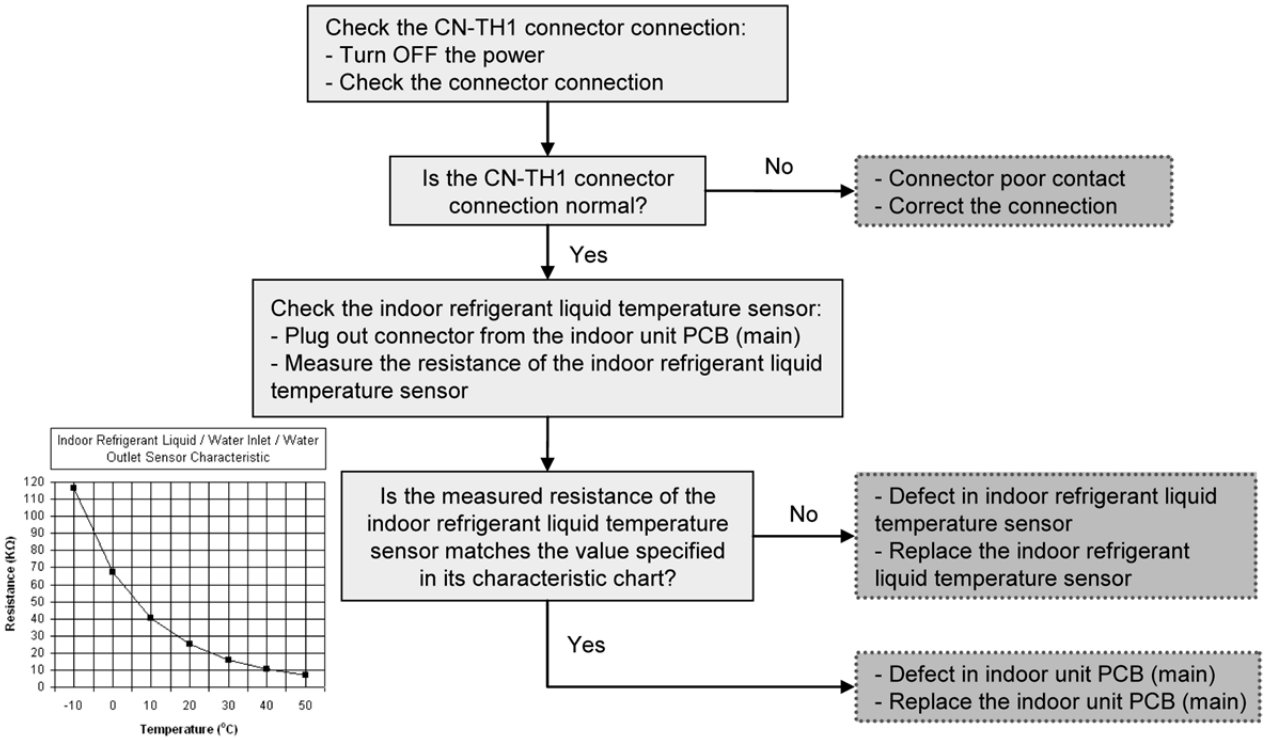
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty indoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.5 Service Valve Error (H27)

Malfunction Decision Conditions:

During cooling operation, when:-
[a] Indoor refrigerant pipe temperature at compressor startup - present indoor refrigerant pipe temperature < 2°C
[b] Present high pressure - high pressure at compressor startup < 5kg/cm^2
**Judgment only for first time cooling operation and not during pump down operation.

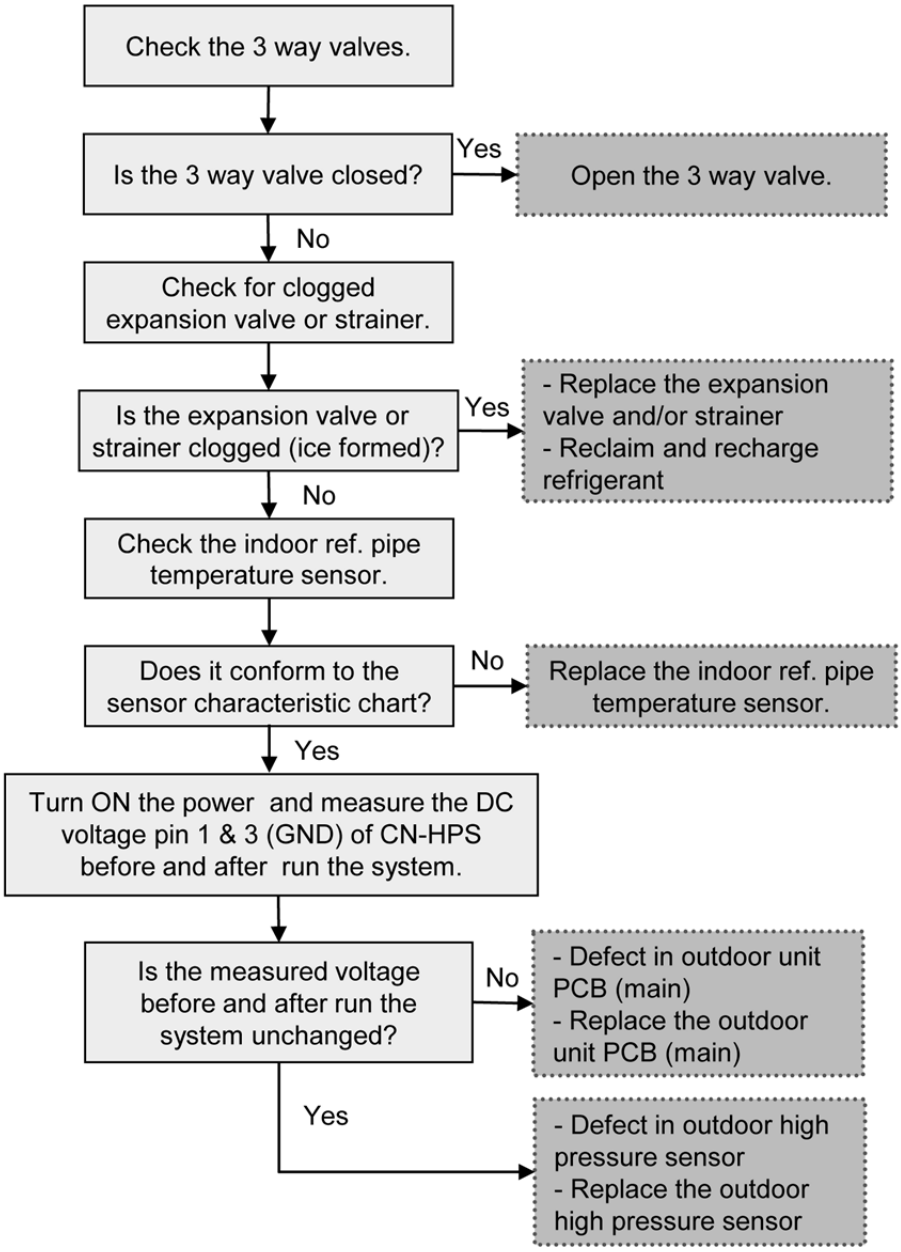
Malfunction Caused:

1 3 way valves closed.
2 Faulty high pressure sensor.
3 Faulty indoor refrigerant pipe temperature sensor
4 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue for 5 minutes.

Troubleshooting: **Caution** For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



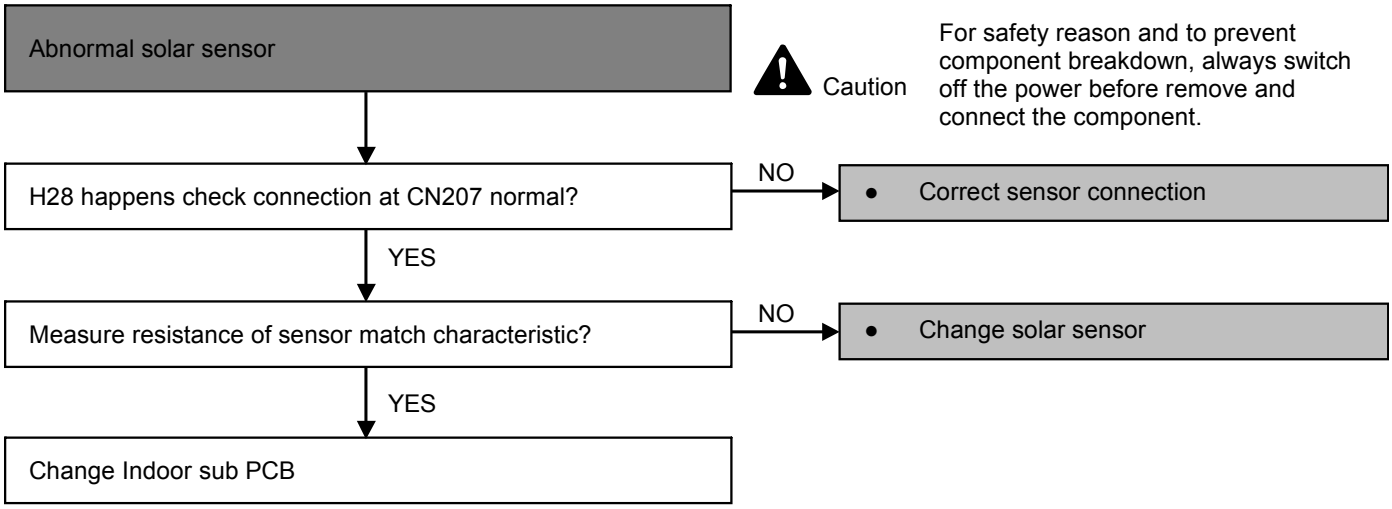
19.5.6 Abnormal Solar Sensor (H28)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty solar sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



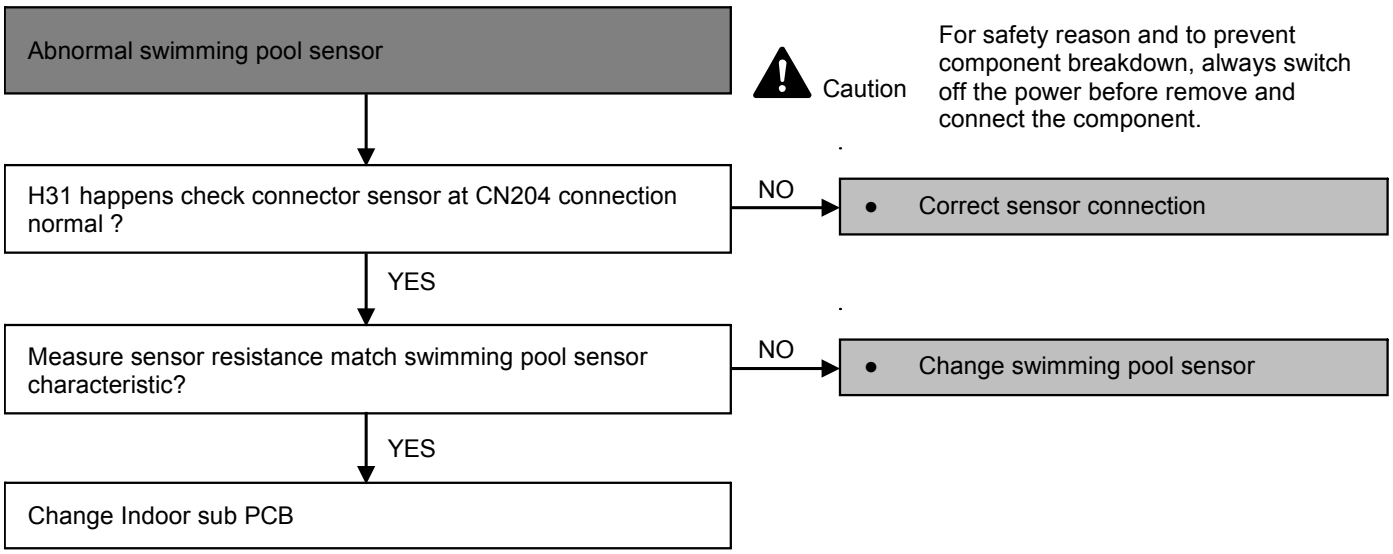
19.5.7 Abnormal Swimming Pool Sensor (H31)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty swimming pool sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



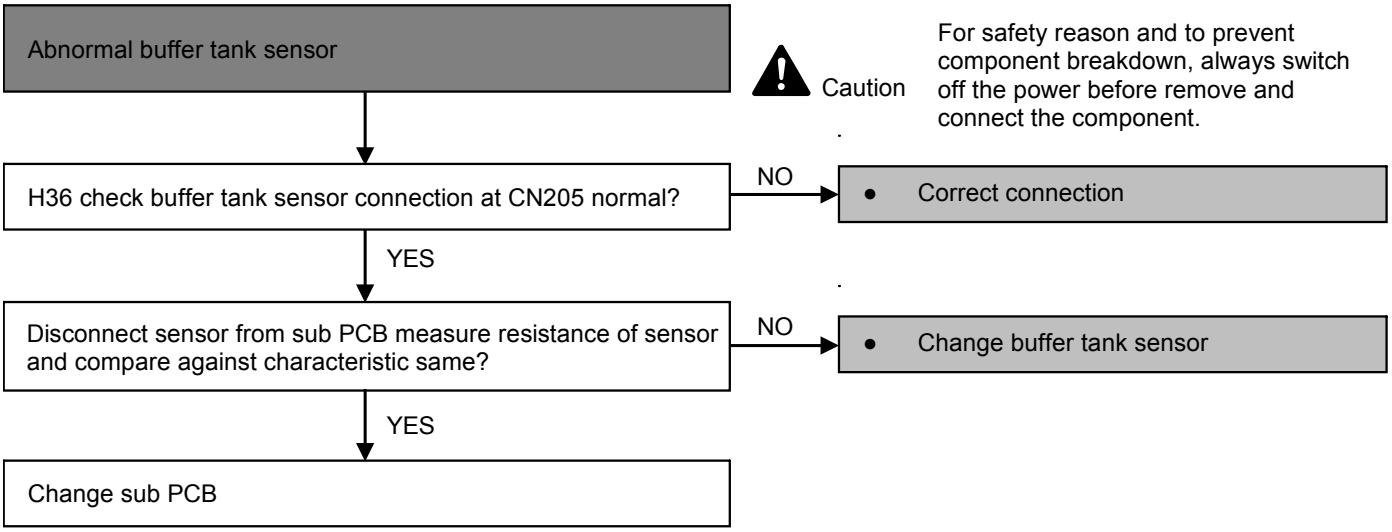
19.5.8 Abnormal Buffer Tank Sensor (H36)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty buffer tank sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

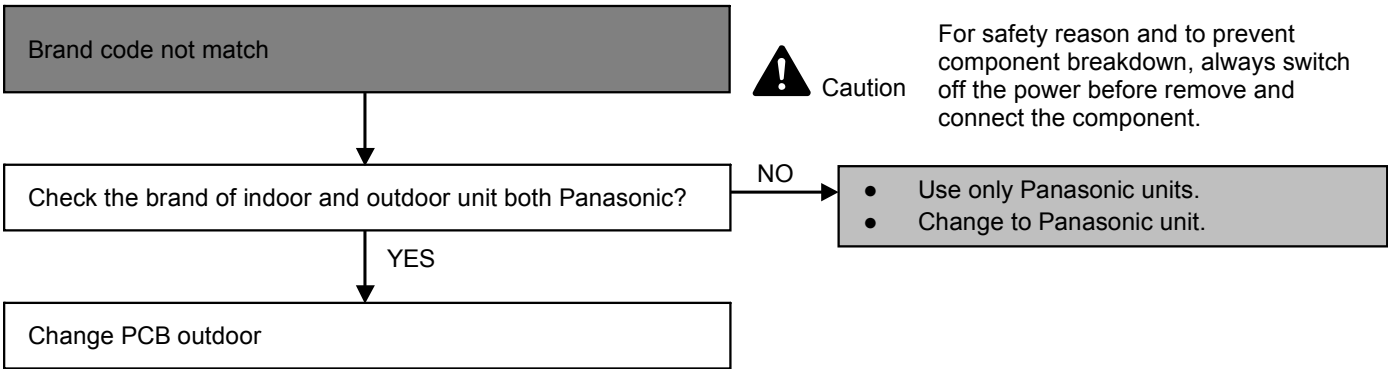
Continue for 5 seconds.



19.5.9 Brand Code Not Matching (H38)

Malfunction Caused:

1 Indoor and outdoor brand code not match.



19.5.10 Compressor Low Pressure Protection (H42)

Malfunction Decision Conditions:

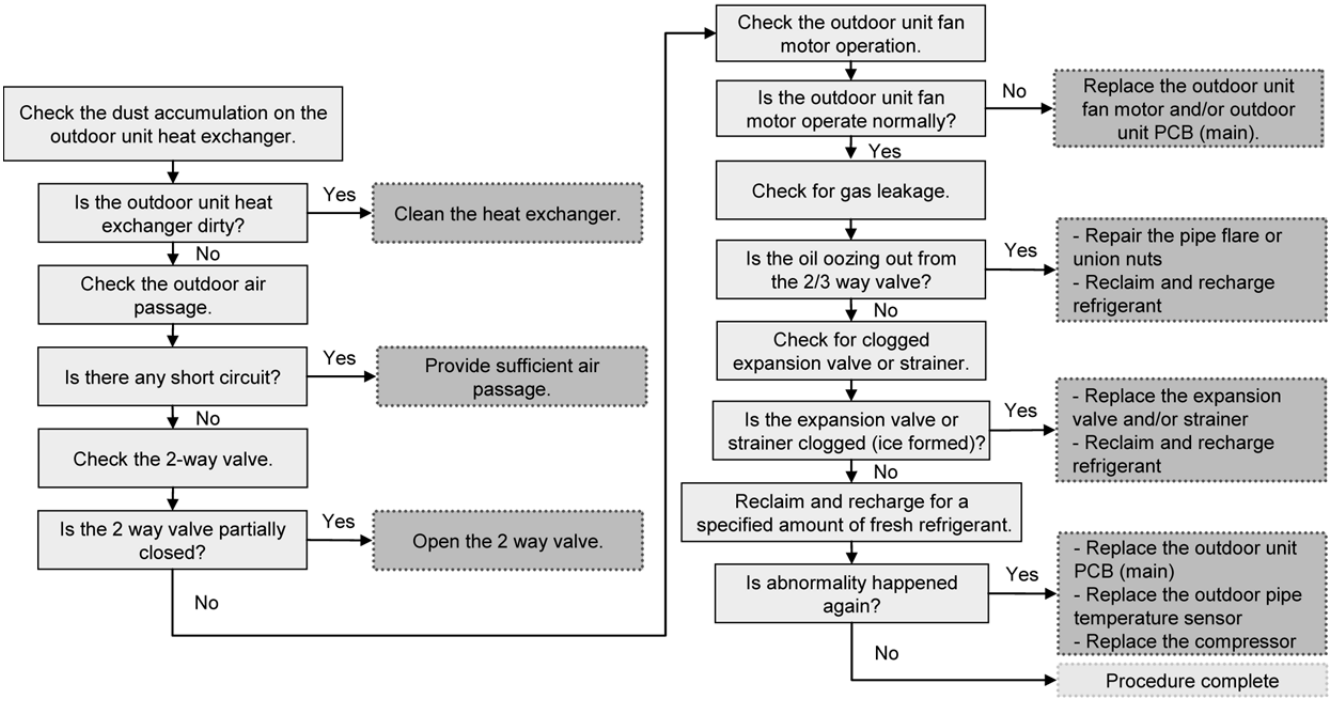
During operation of heating and after 5 minutes compressor ON, when outdoor pipe temperature below -29°C or above 26°C is detected by the outdoor pipe temperature sensor.

Malfunction Caused:

1 Dust accumulation on the outdoor unit heat exchanger.
2 Air short circuit at outdoor unit.
3 2 way valve partially closed.
4 Faulty outdoor unit fan motor.
5 Refrigerant shortage (refrigerant leakage).
6 Clogged expansion valve or strainer.
7 Faulty outdoor pipe temperature sensor.
8 Faulty outdoor unit main PCB (main).

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



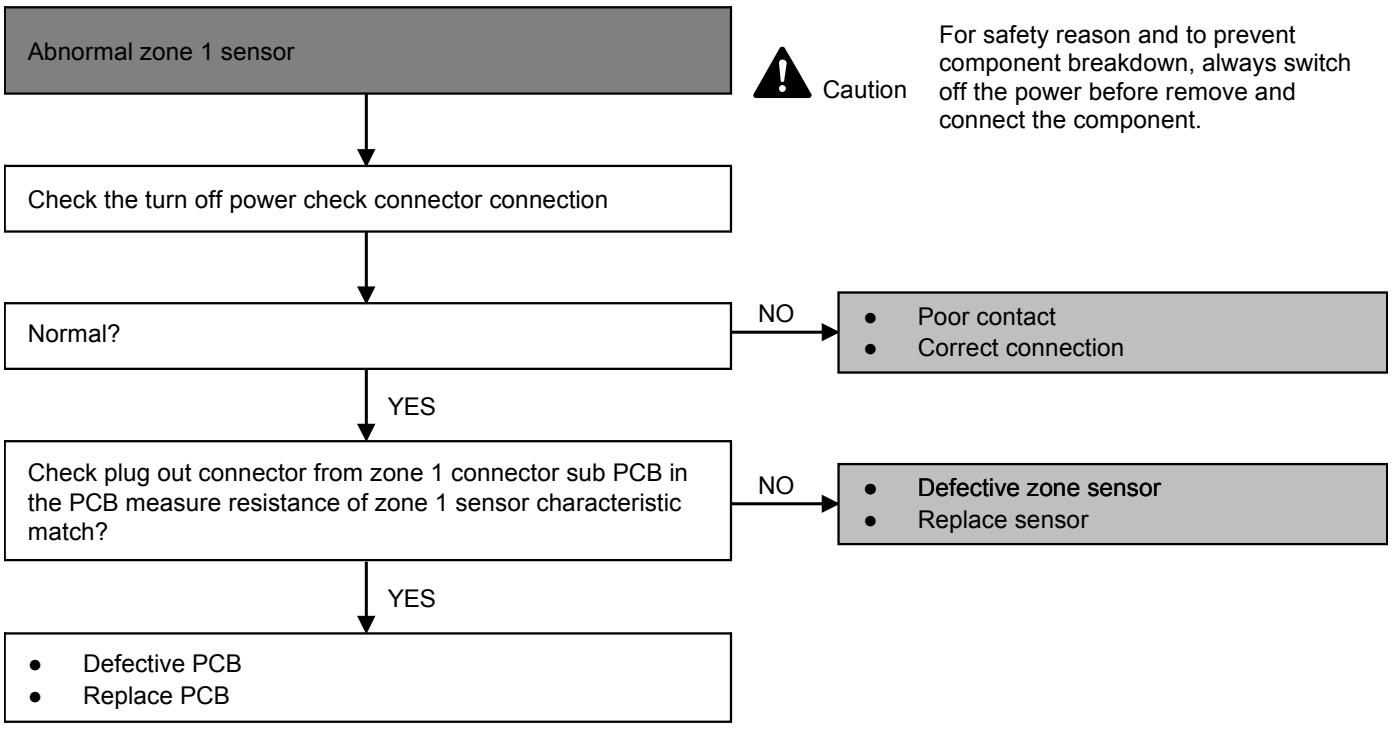
19.5.11 Abnormal Zone 1 Sensor (H43)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty buffer tank sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



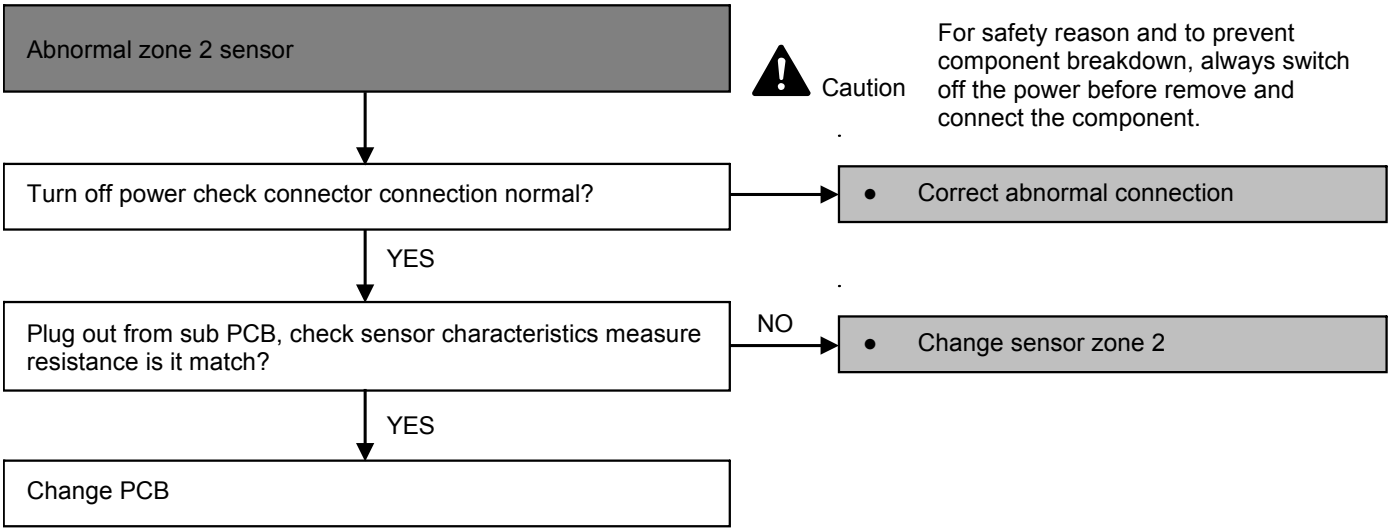
19.5.12 Abnormal Zone 2 Sensor (H44)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty buffer tank sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



19.5.13 Water Flow Switch Abnormality (H62)

Malfunction Decision Conditions:

During operation of cooling and heating, the water flow detected by the indoor water flow switch is used to determine water flow error.

Malfunction Caused:

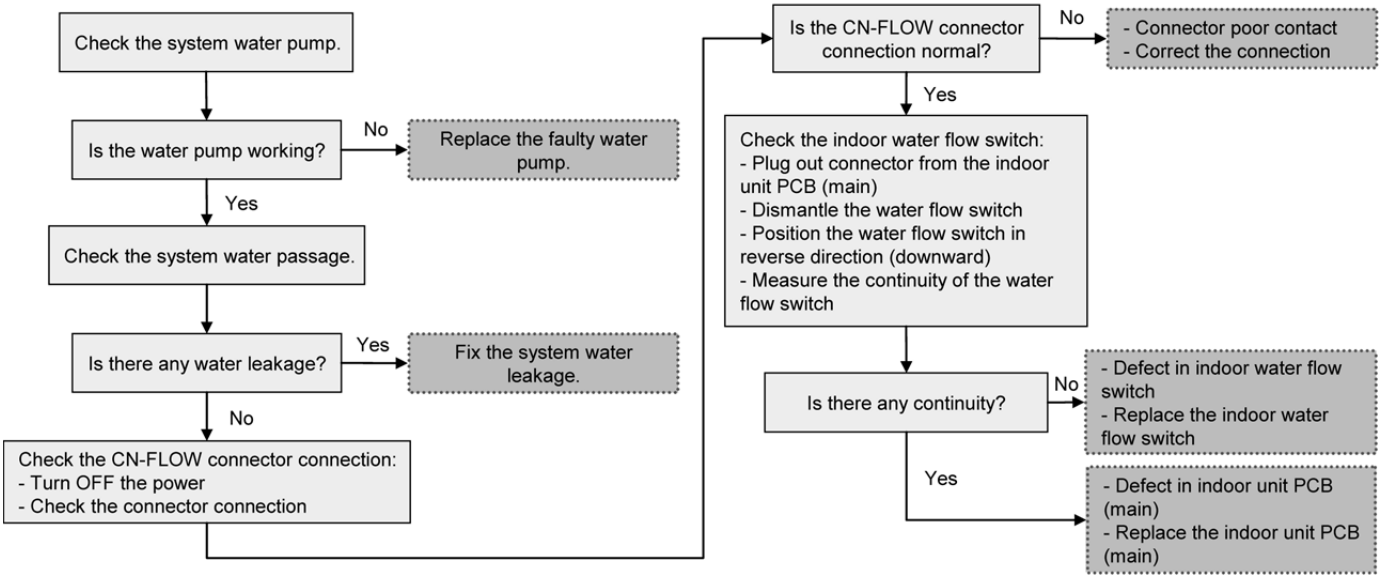
1 Faulty water pump.
2 Water leak in system.
3 Faulty connector connection.
4 Faulty water flow switch.
5 Faulty indoor unit PCB (main).

Abnormality Judgment:

Continue for 10 seconds (but no judgment for 9 minutes after compressor startup/restart).

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.14 Outdoor High Pressure Abnormality (H64)

Malfunction Decision Conditions:

During operation of cooling and heating, when the outdoor high pressure sensor output signal is 0 Vdc or 5 Vdc.

Malfunction Caused:

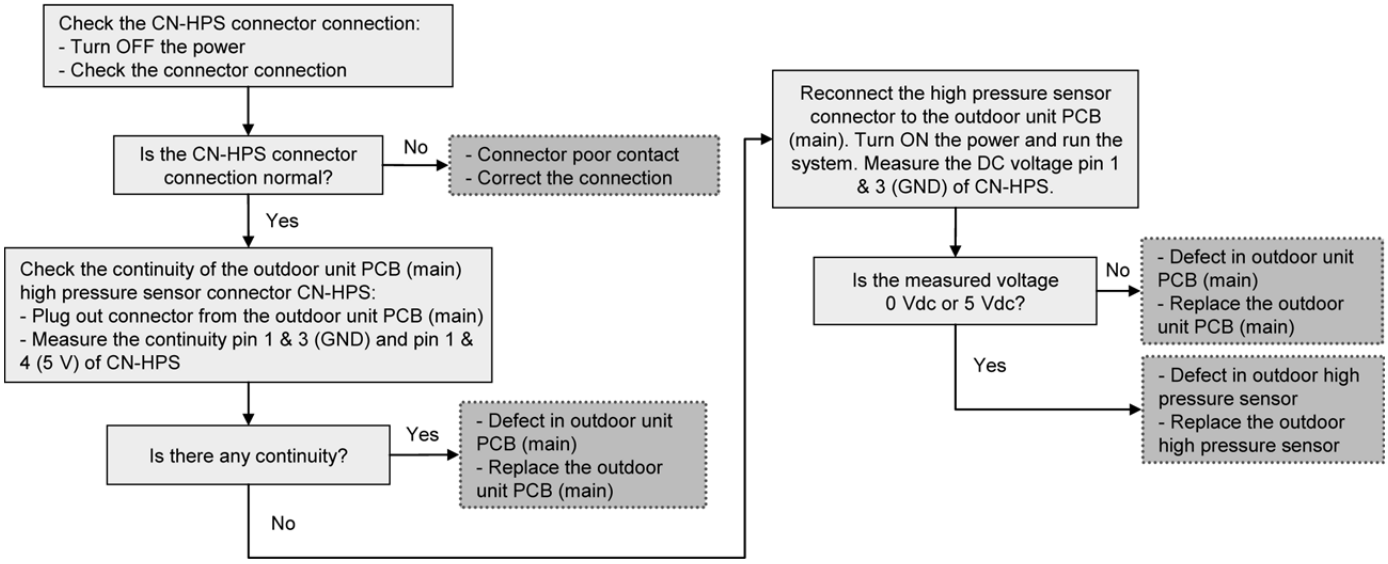
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue 4 times in 20 minutes.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.15 Deice Circulation Error (H65)

Malfunction Decision Conditions:

During startup and operation of deice (mode 2), the water flow (> 10l/min) detected by the water flow switch is used to determine deice circulation error.

Malfunction Caused:

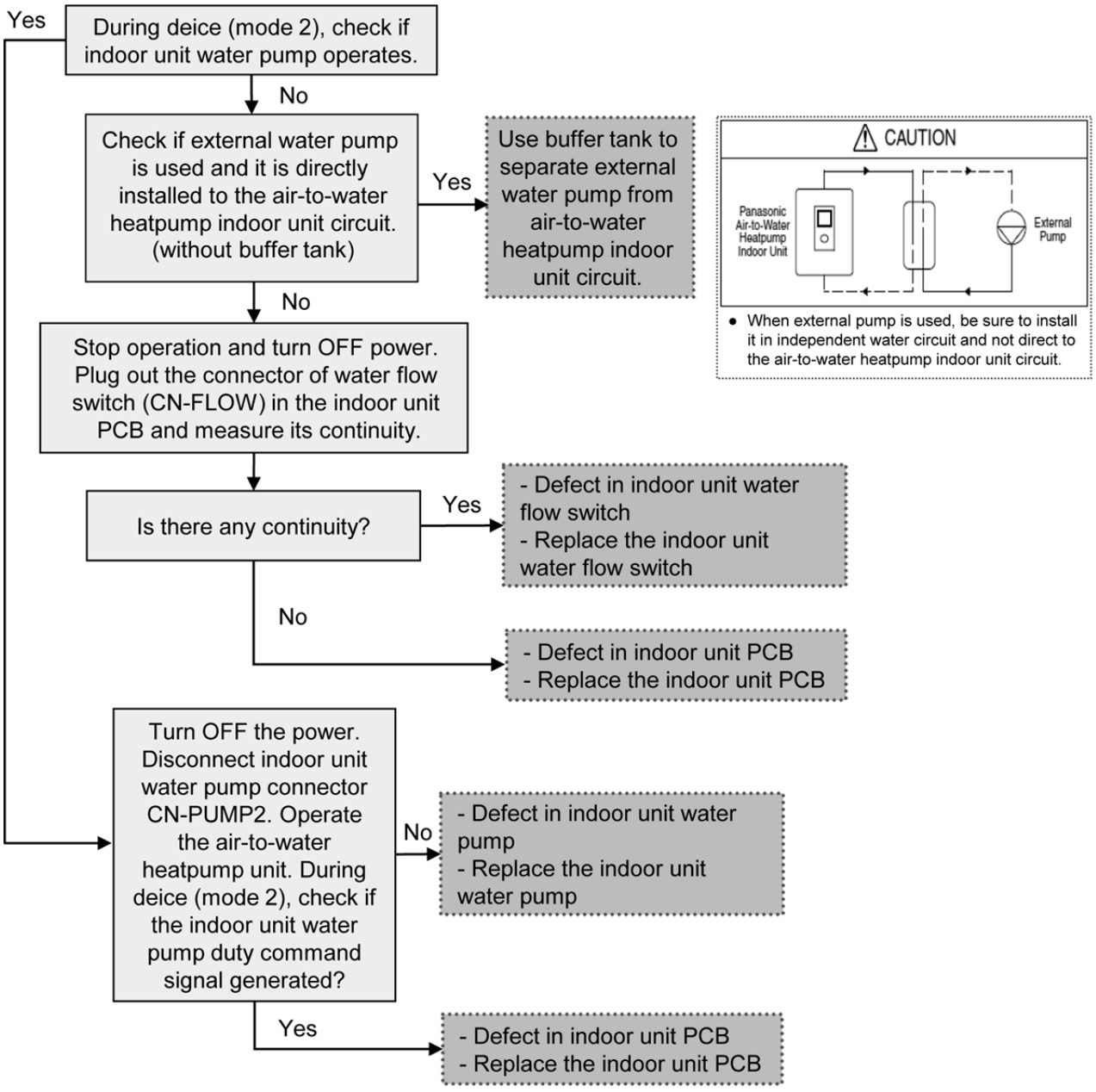
1 Water flow in air-to-water heatpump indoor unit circuitry.
2 Faulty indoor unit water flow switch.
3 Faulty indoor unit water pump.
4 Faulty indoor unit PCB.

Abnormality Judgment:

Continue for 10 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



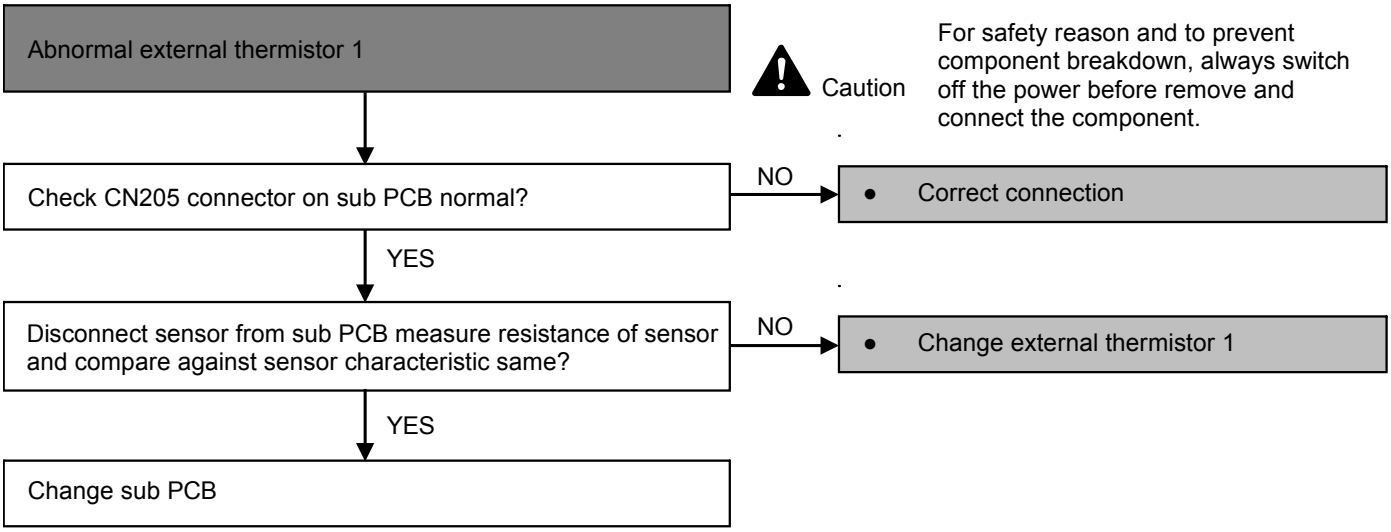
19.5.16 Abnoraml External Thermistor 1 (H67)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty room temperature zone 1 sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



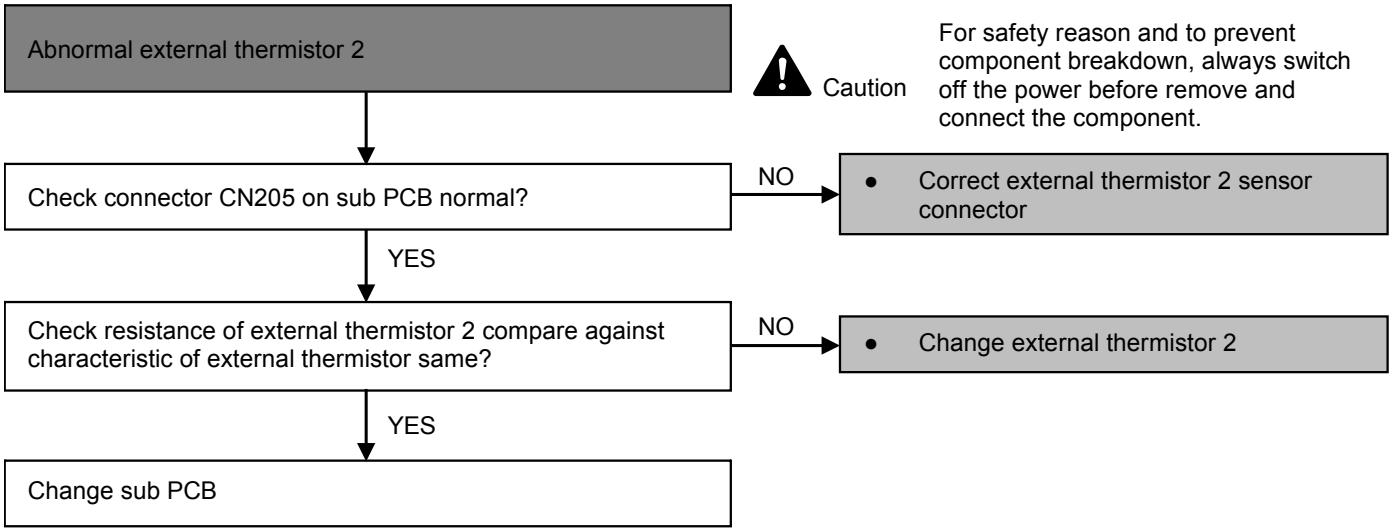
19.5.17 Abnoraml External Thermistor 2 (H68)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty room temperature zone 2 sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



19.5.18 Indoor Backup Heater OLP Abnormality (H70)

Malfunction Decision Conditions:

During operation of indoor backup heater, when no power supplies to indoor backup heater or OLP open circuit.

Malfunction Caused:

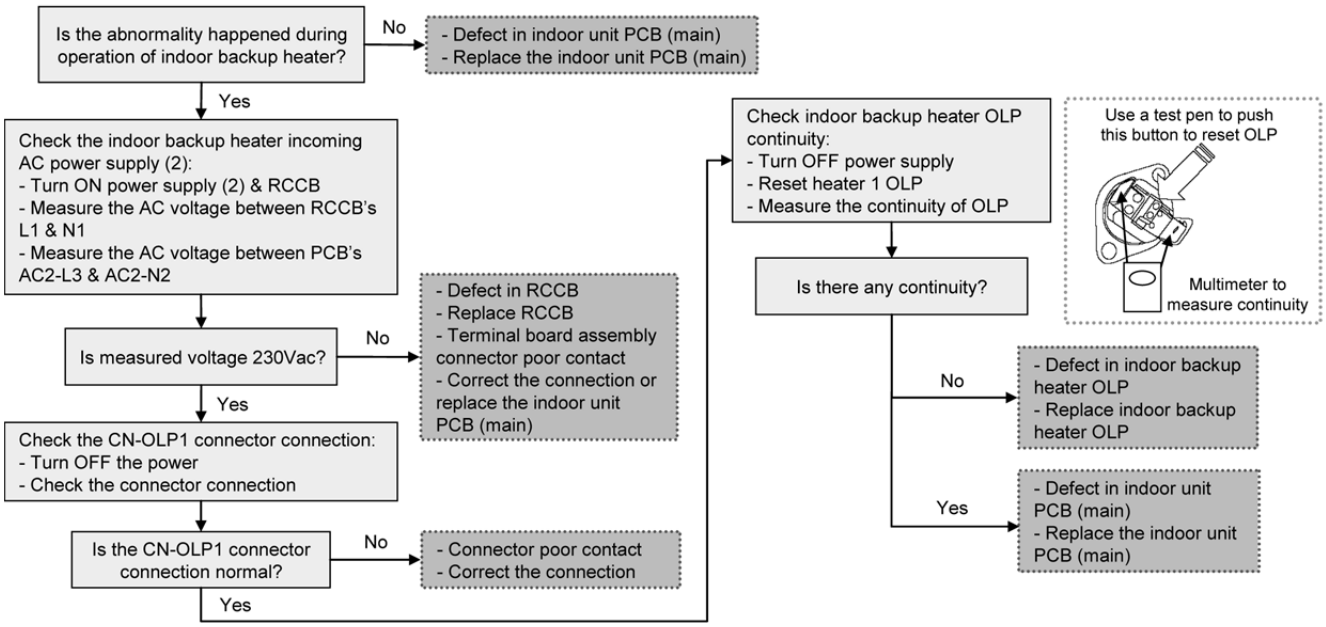
1 Faulty power supply connector connection.
2 Faulty connector connection.
3 Faulty indoor backup heater overload protector (OLP).
4 Faulty indoor unit PCB (main).

Abnormality Judgment:

Continue for 60 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.19 Tank Temperature Sensor Abnormality (H72)

Malfunction Decision Conditions:

When tank connection is set to ON, the temperatures detected by the tank temperature sensor are used to determine sensor error.

Malfunction Caused:

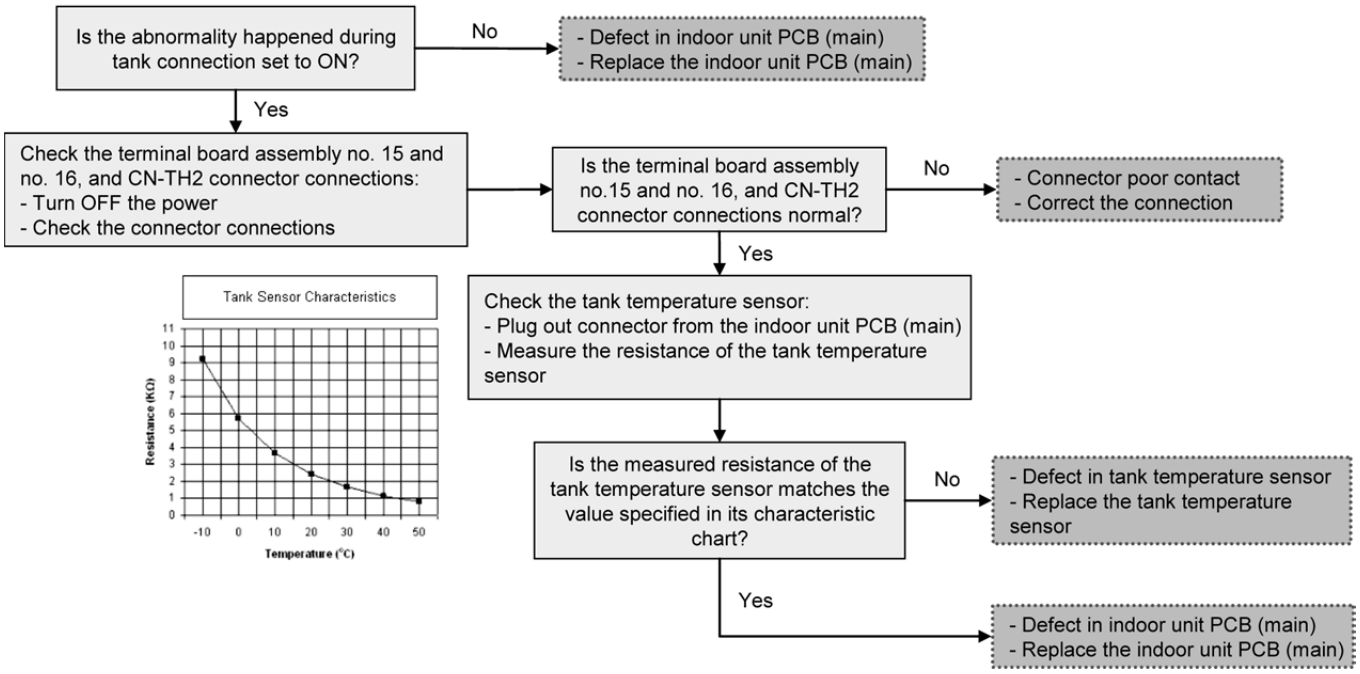
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty indoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.20 PCB Communication Error (H74)

Malfunction Decision Conditions:

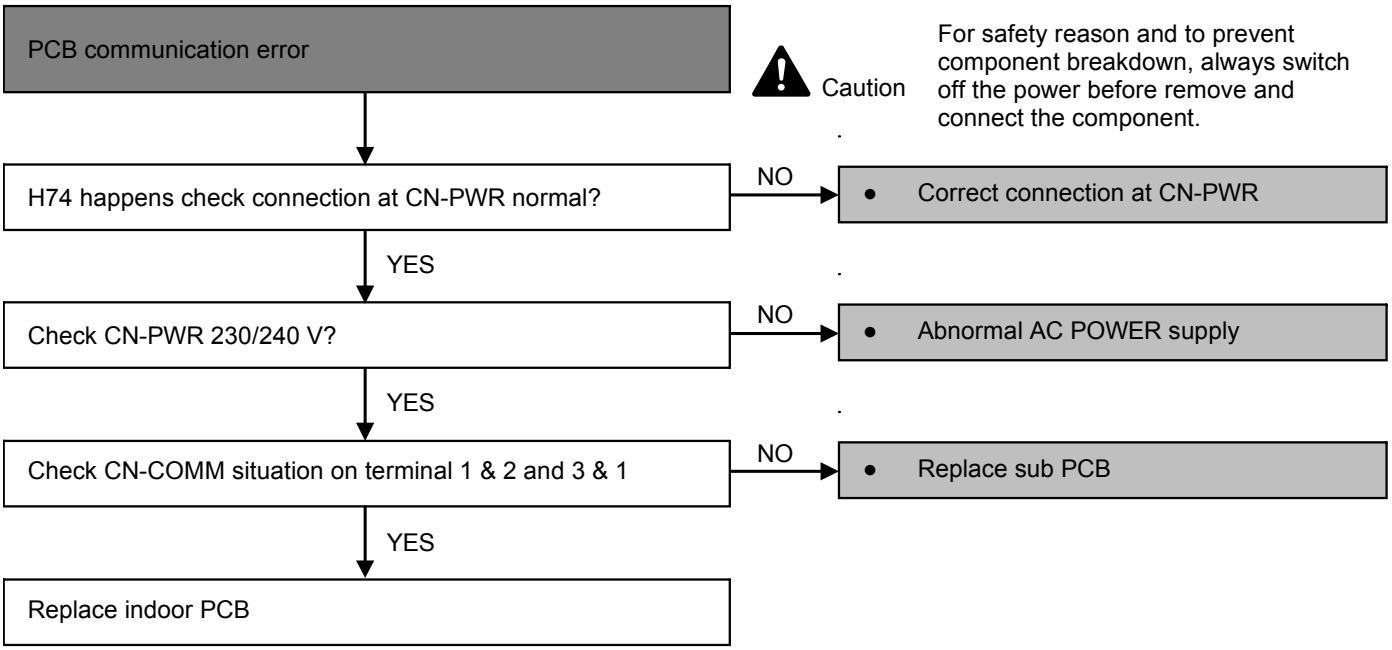
When External PCB connection is select “YES” and no communication with External PCB micon for 10 seconds and above.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty indoor PCB.
3 Faulty indoor sub PCB.

Abnormality Judgment:

After 1 minute operation started.



19.5.21 Indoor-Control Panel Communication Abnormality (H76)

Malfunction Decision Conditions:

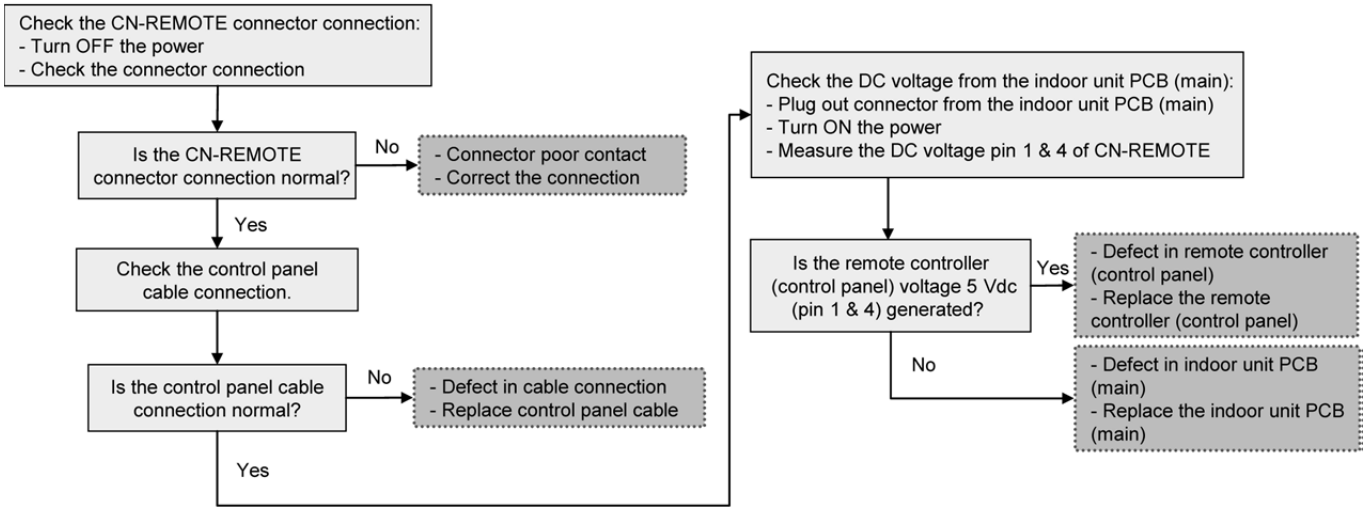
During standby and operation of cooling and heating, indoor-control panel error occur.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty control panel.
3 Faulty indoor unit PCB (main).

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.22 Indoor/Outdoor Abnormal Communication (H90)

Malfunction Decision Conditions:

During operation of cooling and heating, the data received from outdoor unit in indoor unit signal transmission is checked whether it is normal.

Malfunction Caused:

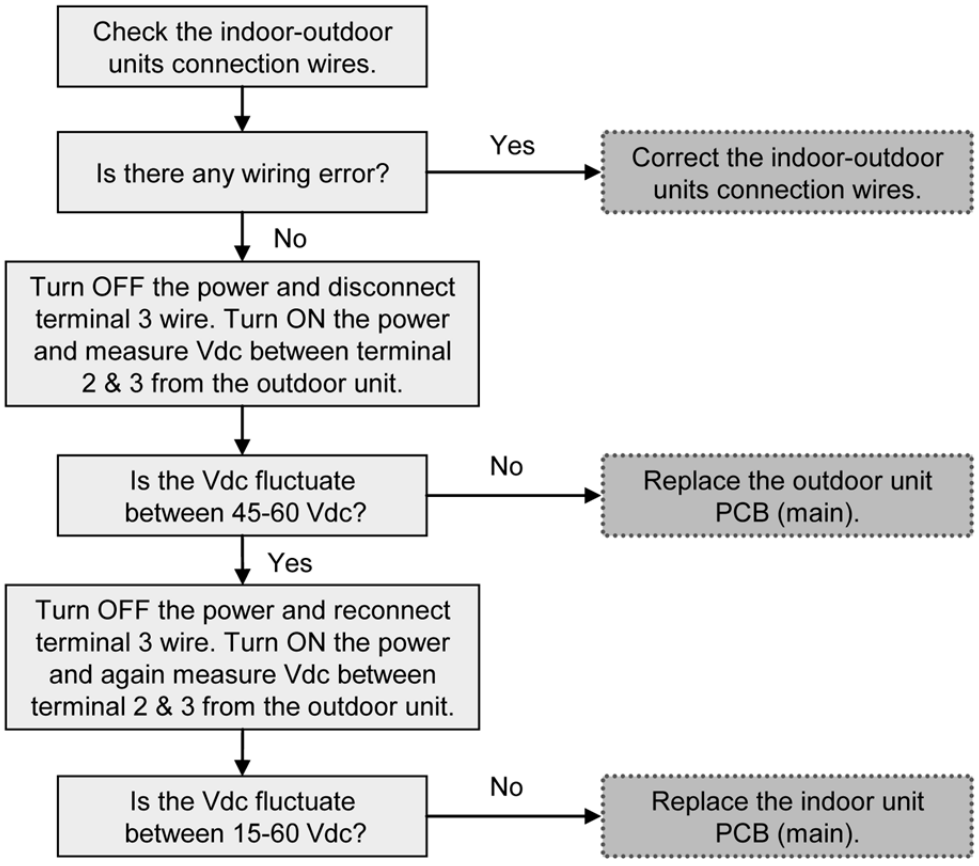
1 Faulty outdoor unit PCB (main).
2 Faulty indoor unit PCB (main).
3 Indoor-outdoor signal transmission error due to wrong wiring.
4 Indoor-outdoor signal transmission error due to breaking of wire in the connection wires between the indoor and outdoor units.
5 Indoor-outdoor signal transmission error due to disturbed power supply waveform.

Abnormality Judgment:

Continue for 1 minute after operation.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.23 Tank Booster Heater OLP Abnormality (H91)

Malfunction Decision Conditions:

During operation of tank booster heater, and tank booster heater OLP open circuit.

Malfunction Caused:

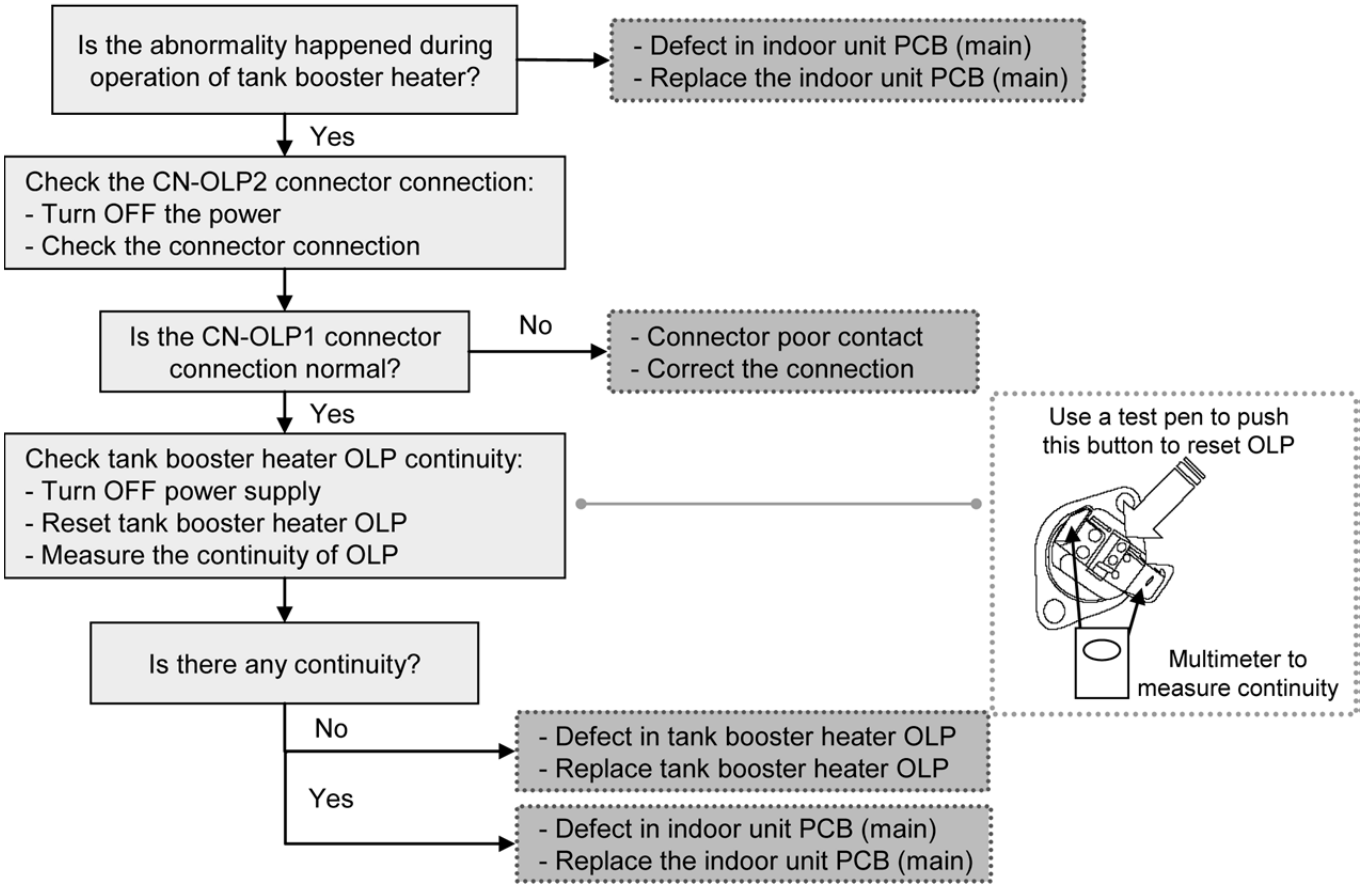
1 Faulty connector connection.
2 Faulty tank booster heater overload protector (OLP).
3 Faulty indoor unit PCB (main).

Abnormality Judgment:

Continue for 60 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.24 Unspecified Voltage between Indoor and Outdoor (H95)

Malfunction Decision Conditions:

The supply power is detected for its requirement by the indoor/outdoor transmission.

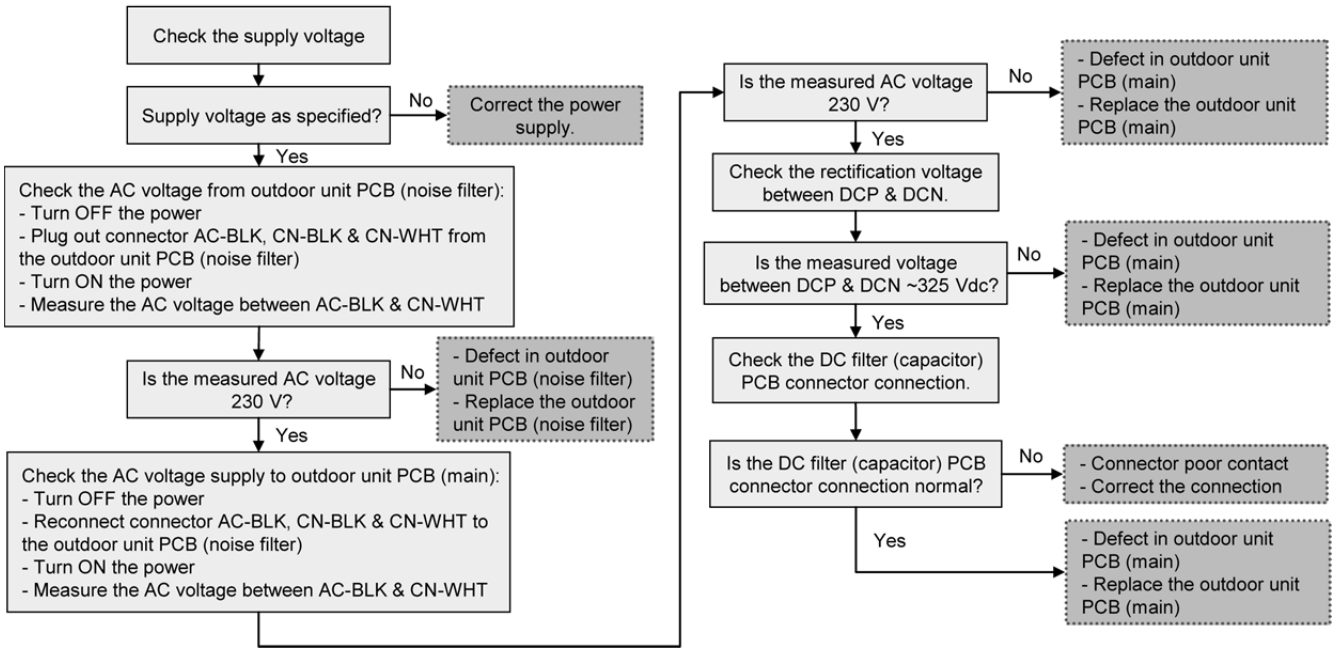
Malfunction Caused:

1 Insufficient power supply.
2 Faulty outdoor unit PCB (noise filter/main).

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



19.5.25 Outdoor High Pressure Protection (H98)

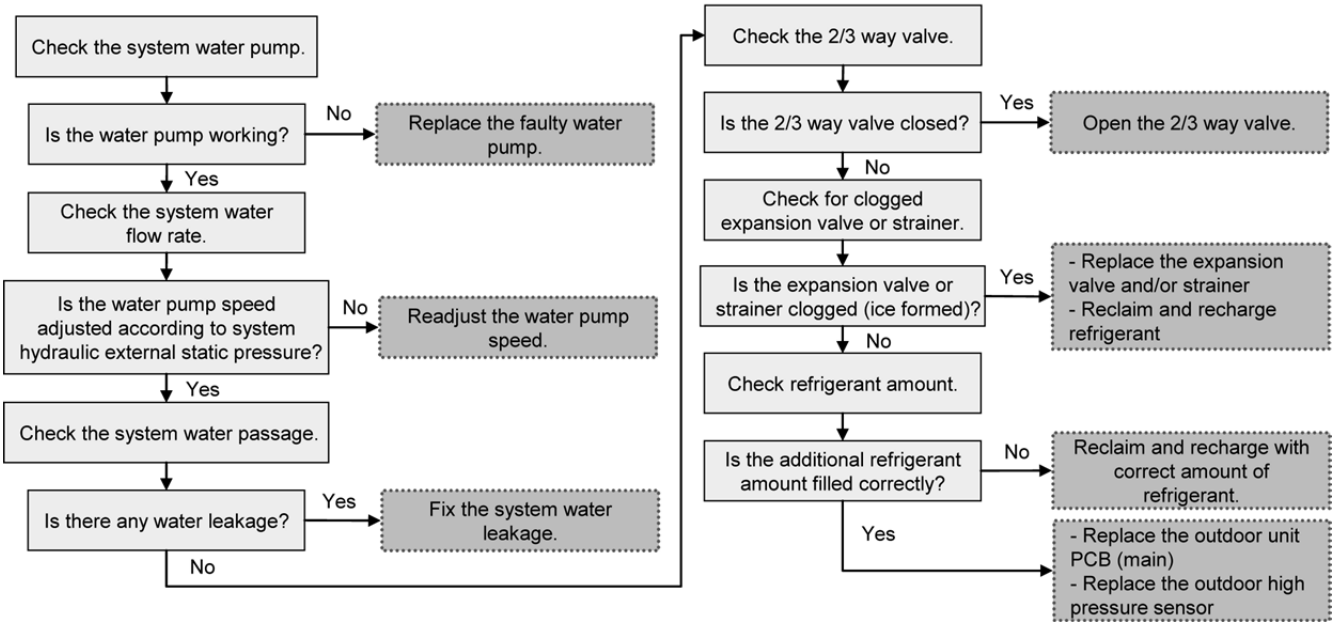
Malfunction Decision Conditions:

During operation of heating, when pressure 4.0 MPa and above is detected by outdoor high pressure sensor.

Malfunction Caused:

1 Faulty water pump.
2 Insufficient water flow rate in system.
3 Water leak in system.
4 2/3 way closed.
5 Clogged expansion valve or strainer.
6 Excessive refrigerant.
7 Faulty outdoor high pressure sensor.
8 Faulty outdoor unit PCB (main).

Troubleshooting: **Caution** For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.26 Indoor Freeze-up Protection (H99)

Malfunction Decision Conditions:

During anti-freezing control in cooling operation, when the indoor refrigerant liquid temperature < 0°C.

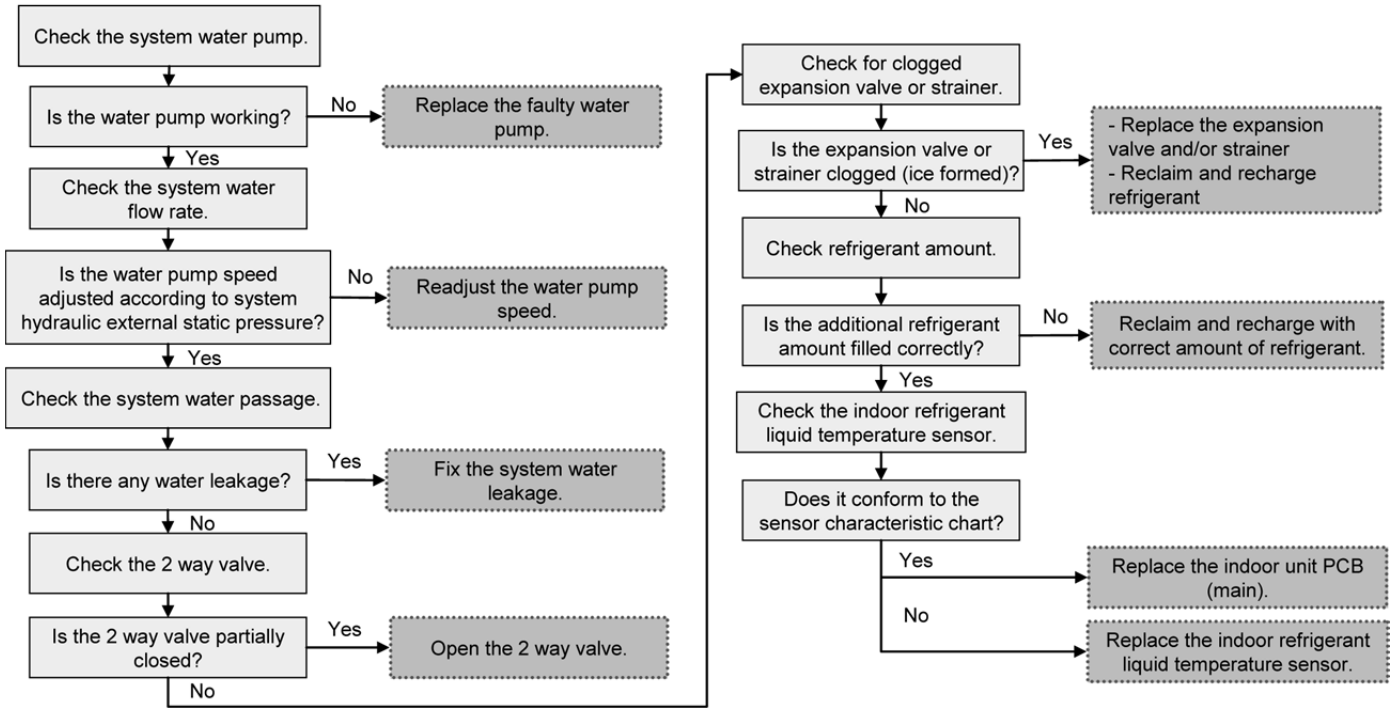
Malfunction Caused:

1 Faulty water pump.
2 Insufficient water flow rate in system.
3 Water leak in system.
4 2 way valve partially closed.
5 Clogged expansion valve or strainer.
6 Refrigerant shortage (refrigerant leakage).
7 Faulty indoor refrigerant liquid temperature sensor.
8 Faulty indoor unit PCB (main).

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



19.5.27 Outdoor High Pressure Switch Activate (F12)

Malfunction Decision Conditions:

During operation of cooling and heating, when pressure 4.5 MPa and above is detected by outdoor high pressure switch.

Malfunction Caused:

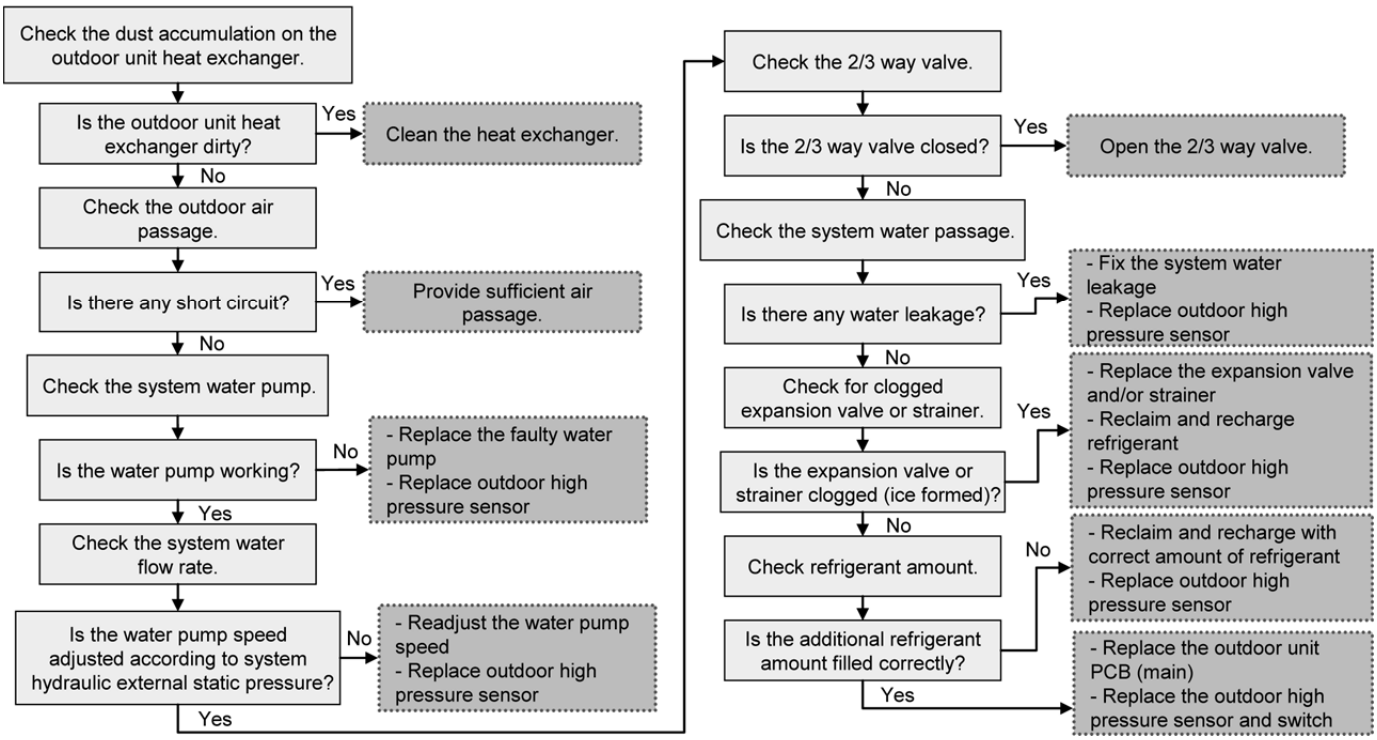
1 Dust accumulation on the outdoor unit heat exchanger.
2 Air short circuit at outdoor unit.
3 Faulty water pump.
4 Insufficient water flow rate in system.
5 Water leak in system.
6 2/3 way valve closed.
7 Clogged expansion valve or strainer.
8 Excessive refrigerant.
9 Faulty outdoor high pressure sensor and switch.
10 Faulty outdoor unit PCB.

Abnormality Judgment:

Continue 4 times in 20 minutes.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.28 Compressor Rotation Failure (F14)

Malfunction Decision Conditions:

A compressor rotation failure is detected by checking the compressor running condition through the position detection circuit.

Malfunction Caused:

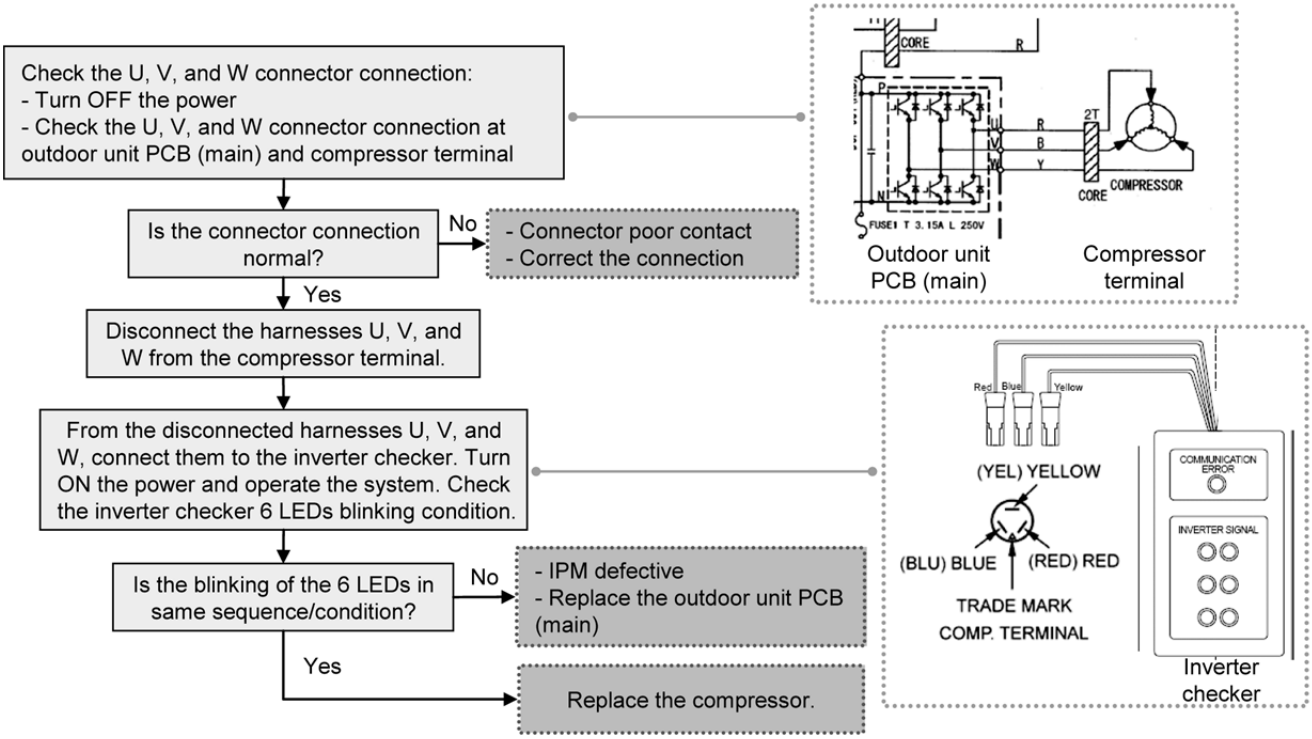
1 Compressor terminal disconnect.
2 Faulty outdoor unit PCB (main).
3 Faulty compressor.

Abnormality Judgment:

Continue 4 times in 20 minutes.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.29 Outdoor Fan Motor (DC Motor) Mechanism Locked (F15)

Malfunction Decision Conditions:

The rotation speed detected by the Hall IC of the fan motor during fan motor operation is used to determine abnormal fan motor (feedback of rotation > 2550 rpm or < 50 rpm).

Malfunction Caused:

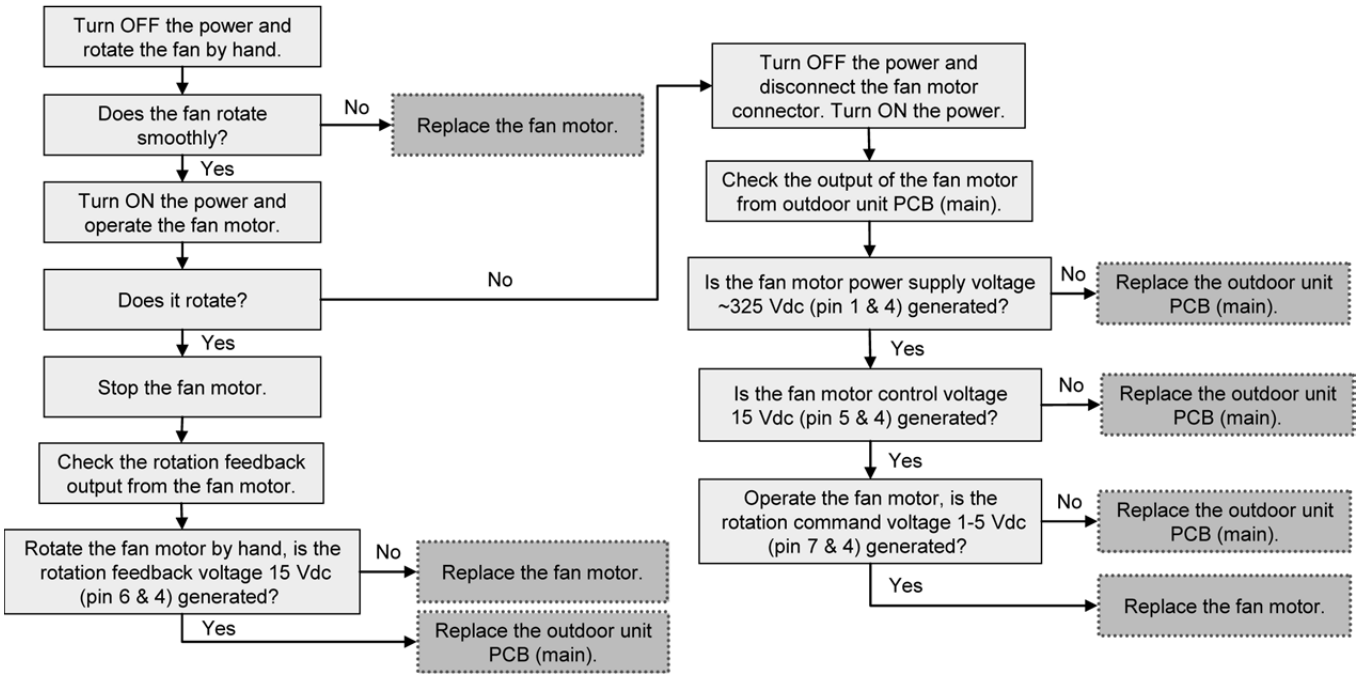
1 Operation stop due to short circuit inside the fan motor winding.
2 Operation stop due to breaking of wire inside the fan motor.
3 Operation stop due to breaking of fan motor lead wires.
4 Operation stop due to fan motor Hall IC malfunction.
5 Operation error due to faulty outdoor unit PCB.

Abnormality Judgment:

Continue 2 times in 30 minutes.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.30 Input Over Current Detection (F16)

Malfunction Decision Conditions:

During operation of cooling and heating, when outdoor current above 24.1 A (Heating) and 16.4 A (Cooling) [UD07FE5)] and 28.1 A (Heating) and 18.1 A (Cooling) [UD09FE5)] is detected by the current transformer (CT) in the outdoor unit PCB.

Malfunction Caused:

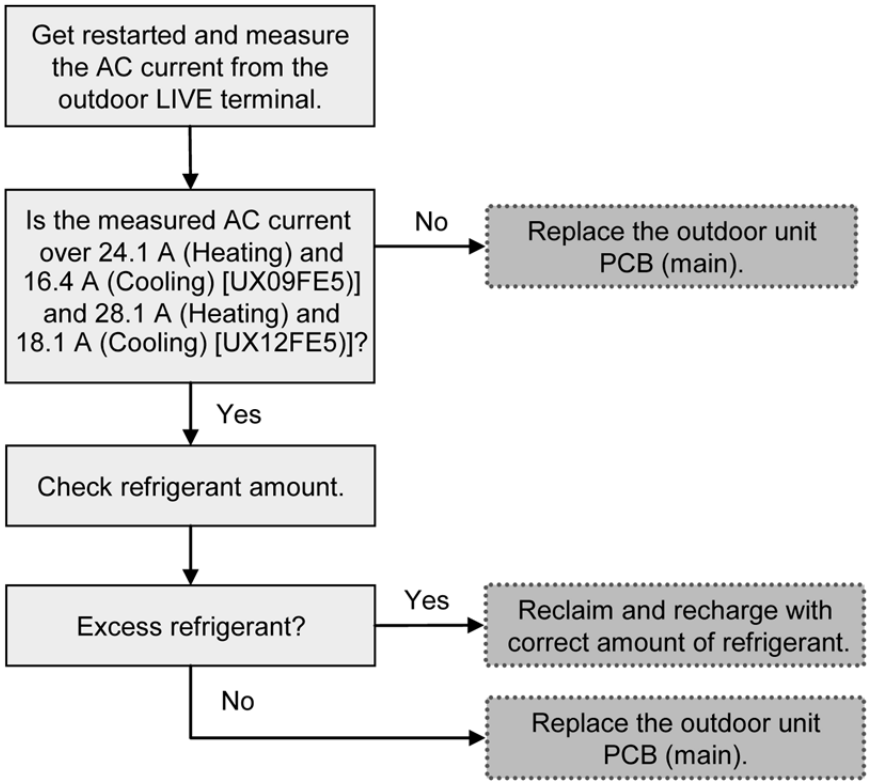
1 Excessive refrigerant.
2 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue 3 times in 20 minutes.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.31 Compressor Overheating (F20)

Malfunction Decision Conditions:

During operation of cooling and heating, when temperature above 112°C is detected by the compressor tank temperature sensor.

Malfunction Caused:

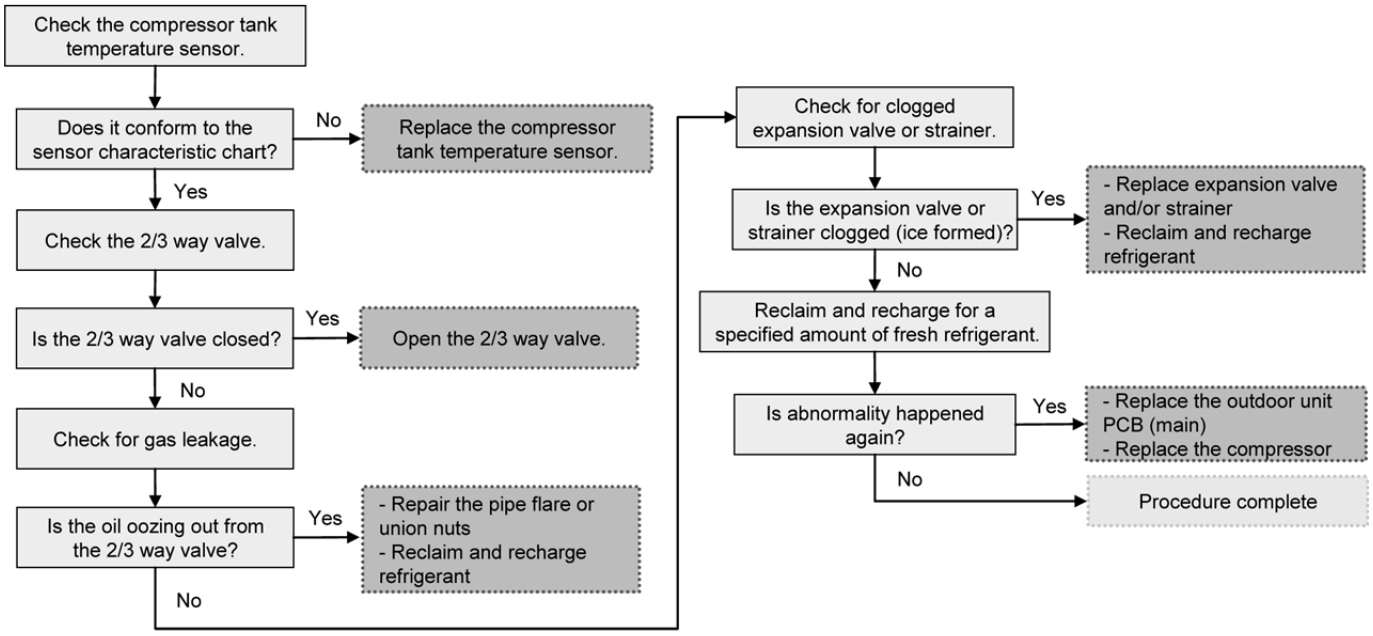
1 Faulty compressor tank temperature sensor.
2 2/3 way valve closed.
3 Refrigerant shortage (refrigerant leakage).
4 Clogged expansion valve or strainer.
5 Faulty outdoor unit PCB (main).
6 Faulty compressor.

Abnormality Judgment:

Continue 4 times in 30 minutes.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.32 IPM Overheating (F22)

Malfunction Decision Conditions:

During operation of cooling and heating, when temperature 95°C is detected by the outdoor IPM temperature sensor.

Malfunction Caused:

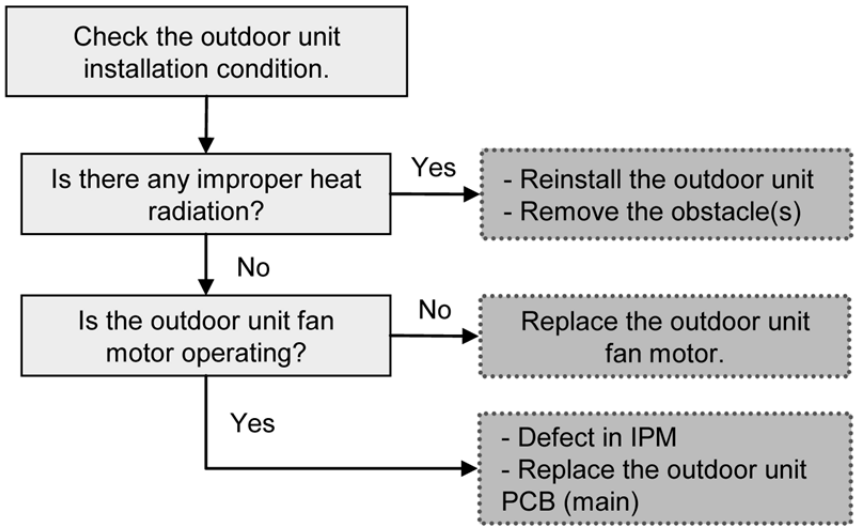
1 Faulty outdoor unit fan motor.
2 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue 3 times in 30 minutes.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.33 Output Over Current Detection (F23)

Malfunction Decision Conditions:

During operation of cooling and heating, when outdoor DC current is above 34 A is detected by the IPM DC Peak sensing circuitry in the outdoor unit PCB (main).

Malfunction Caused:

1 Faulty outdoor unit PCB (main).
2 Faulty compressor.

Abnormality Judgment:

Continue for 7 times.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.

Check the compressor winding resistance:
- Turn OFF the power and disconnect the harnesses U, V, and W
- Measure the winding resistance between U-V, V-W, and W-U

→ Are the compressor's winding resistance (U-V, V-W, or U-W) shorted?

Yes → - Compressor defective due to winding shorted
- Replace the compressor

No → - Outdoor electronic circuit defect due to power transistor shorted
- Replace the outdoor unit PCB (main)

■ Checking the power transistor

- Never touch any live parts for at least 10 minutes after turning off the circuit breaker.
- If unavoidably necessary to touch a live part, make sure the power transistor's supply voltage is below 50 V using the tester.
- For the UVW, make measurements at the Faston terminal on the board or the relay connector.

Tester's negative terminal	Power transistor (+)	UVW	Power transistor (−)	UVW
Tester's positive terminal	UVW	Power transistor (+)	UVW	Power transistor (−)
Normal resistance	Several kohms to several Mohms			
Abnormal resistance	0 or ∞			

19.5.34 Refrigeration Cycle Abnormality (F24) (WH-UD03HE5-1 and WH-UD05HE-1)

Malfunction Decision Conditions:

1 During operation of cooling and heating, compressor frequency > Frated.
2 During operation of cooling and heating, running current: 0.65 A < I < 1.65 A.
3 During operation of cooling, water inlet temperature - indoor refrigerant liquid temperature < 5°C.
4 During operation of heating, indoor refrigerant liquid temperature - water inlet temperature < 5°C.

Malfunction Caused:

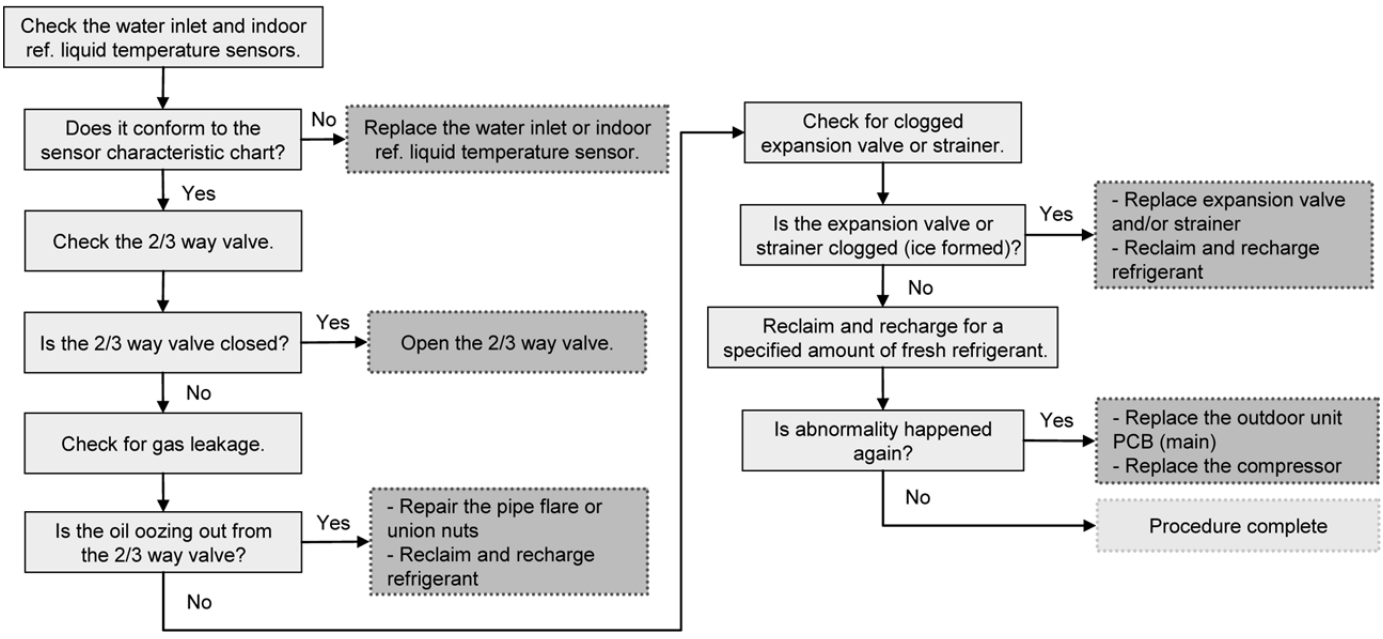
1 Faulty water inlet or indoor refrigerant liquid temperature sensors.
2 2/3 way valve closed.
3 Refrigerant shortage (refrigerant leakage).
4 Clogged expansion valve or strainer.
5 Faulty outdoor unit PCB (main).
6 Poor compression of compressor.

Abnormality Judgment:

Continue 2 times in 20 minutes.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.35 Refrigeration Cycle Abnormality (F24) (WH-UD07HE5-1 and WH-UD09HE-1)

Malfunction Decision Conditions:

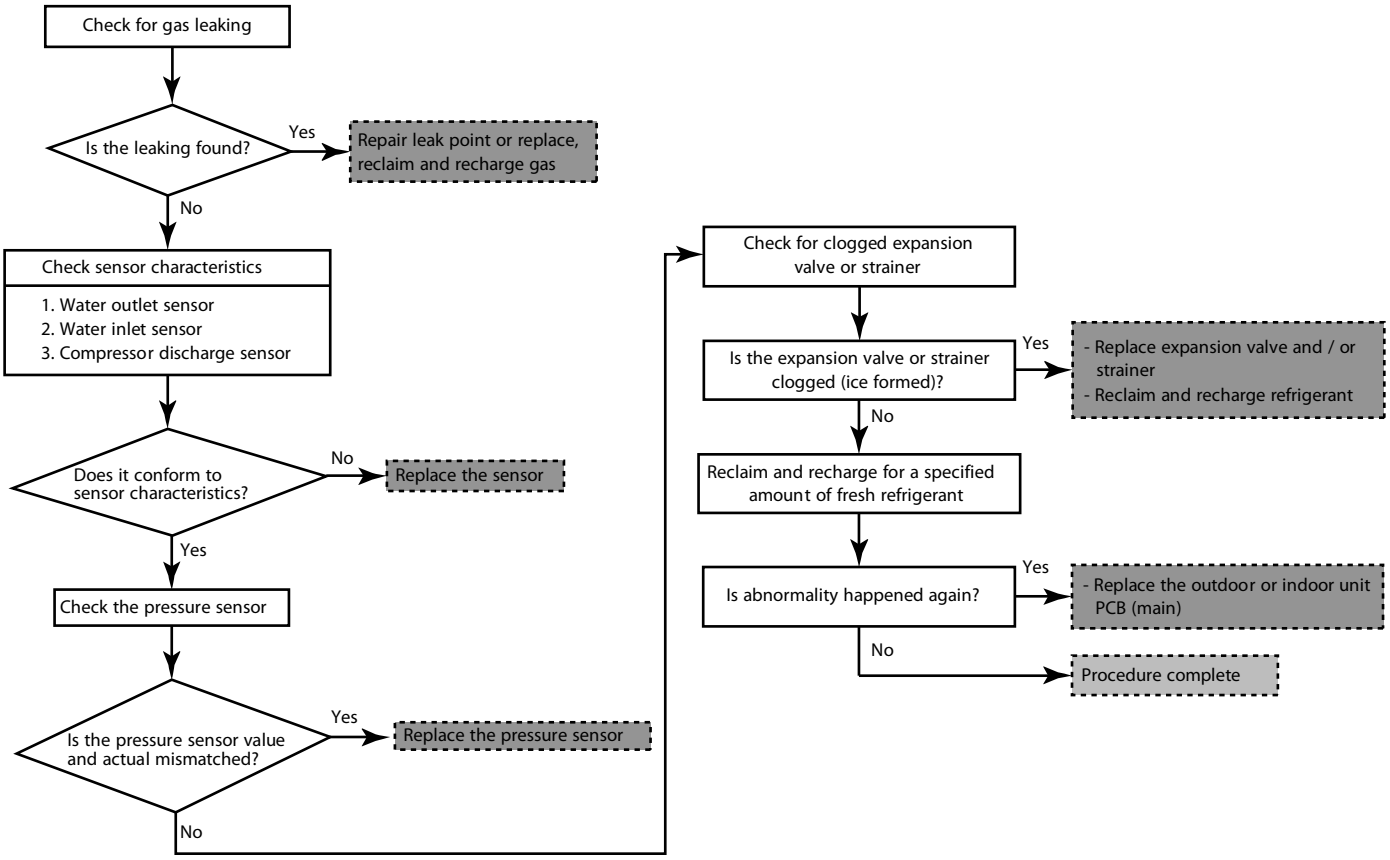
1 During operation running (heating / cooling) for more than 10 minutes expect deice, pumpdown and test mode.
2 During heating / cooling, water outlet and water inlet difference is less than 1°C.
3 During heating / cooling, high pressure < 1MPa (143 Psi) for more than 10 minutes or < 0.2 MPa (28 Psi) for more than 5 minutes.
4 During heating / cooling, discharge temperature - saturation temperature of high pressure ≥ 60°C.

Malfunction Caused:

1 Refrigerant shortage (refrigerant leakage).
2 Faulty indoor water inlet, indoor water outlet, compressor discharge temp sensor or high pressure sensor.
3 2/3 way valve closed.
4 Clogged expansion valve or strainer.
5 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue 2 times in 20 minutes.



19.5.36 Four Way Valve Abnormality (F25)

Malfunction Decision Conditions:

1 During heating operation, when the indoor pipe temperature of thermostat ON indoor unit < 0°C.
2 During cooling operation, when the indoor pipe temperature of thermostat ON indoor unit > 45°C.

Malfunction Caused:

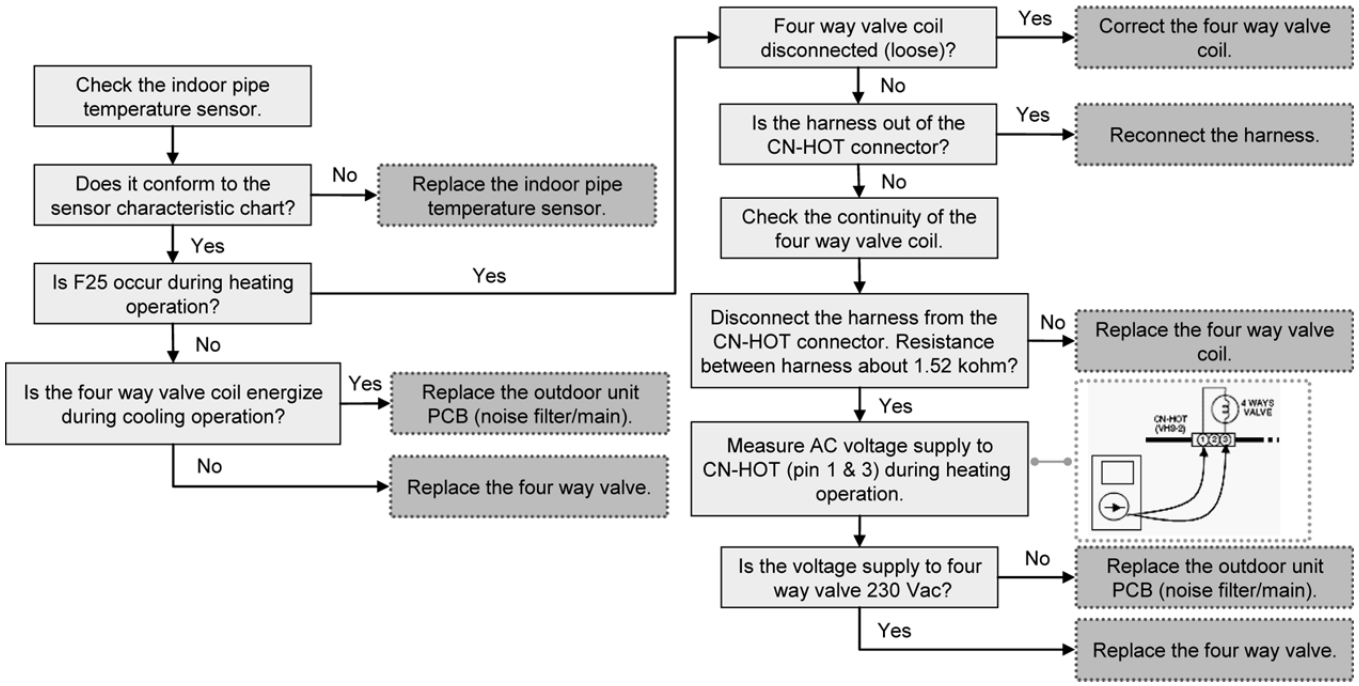
1 Faulty sensor.
2 Faulty connector connection.
3 Faulty outdoor unit PCB (noise filter/main).
4 Faulty four way valve.

Abnormality Judgment:

Continue 4 times in 30 minutes.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.37 Outdoor High Pressure Switch Abnormal (F27)

Malfunction Decision Conditions:

During compressor stop, and outdoor high pressure switch is remain opened.

Malfunction Caused:

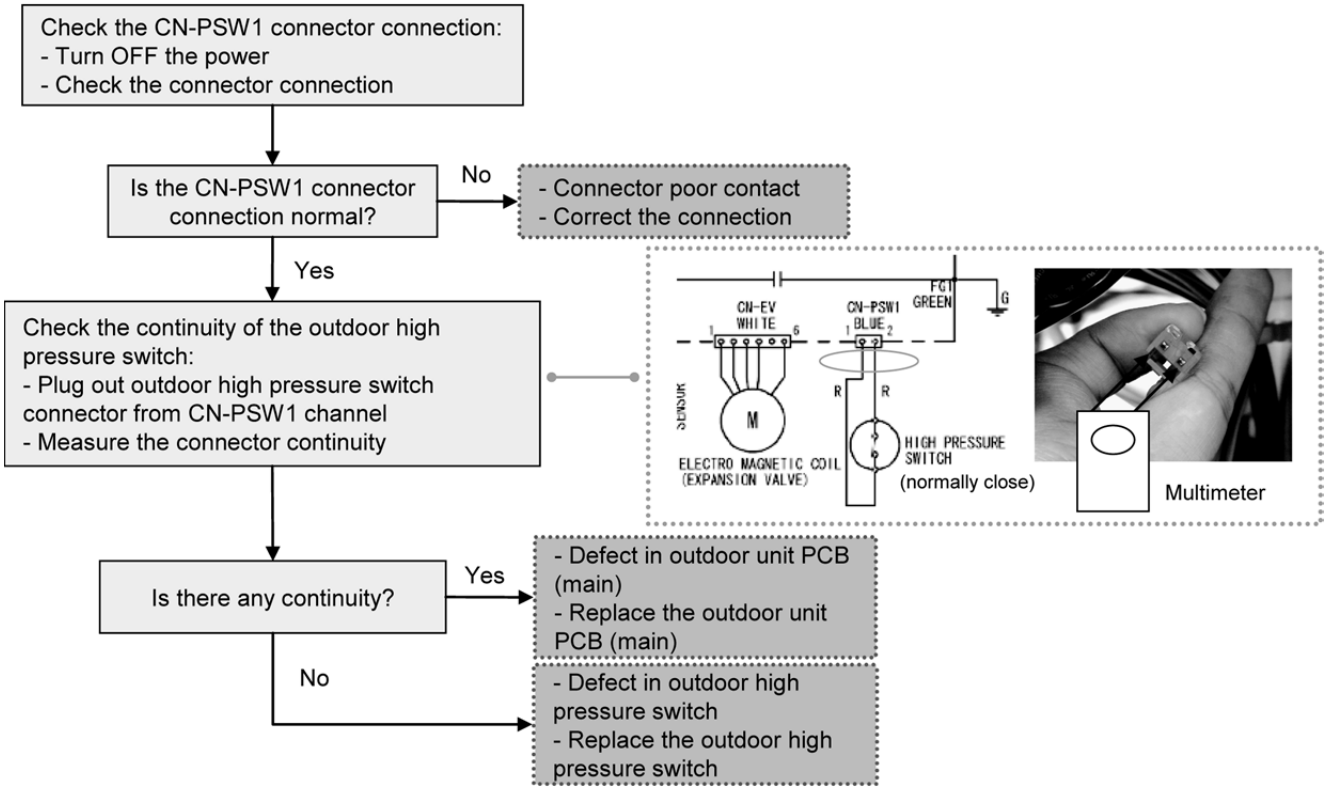
1 Faulty connector connection.
2 Faulty switch.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue for 1 minute.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.38 Low Discharge Superheat (F29)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the outdoor pipe temperature sensor are used to determine sensor error.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).
4 Faulty High Pressure Switch
5 Refrigerant shortage (refrigerant leakage).

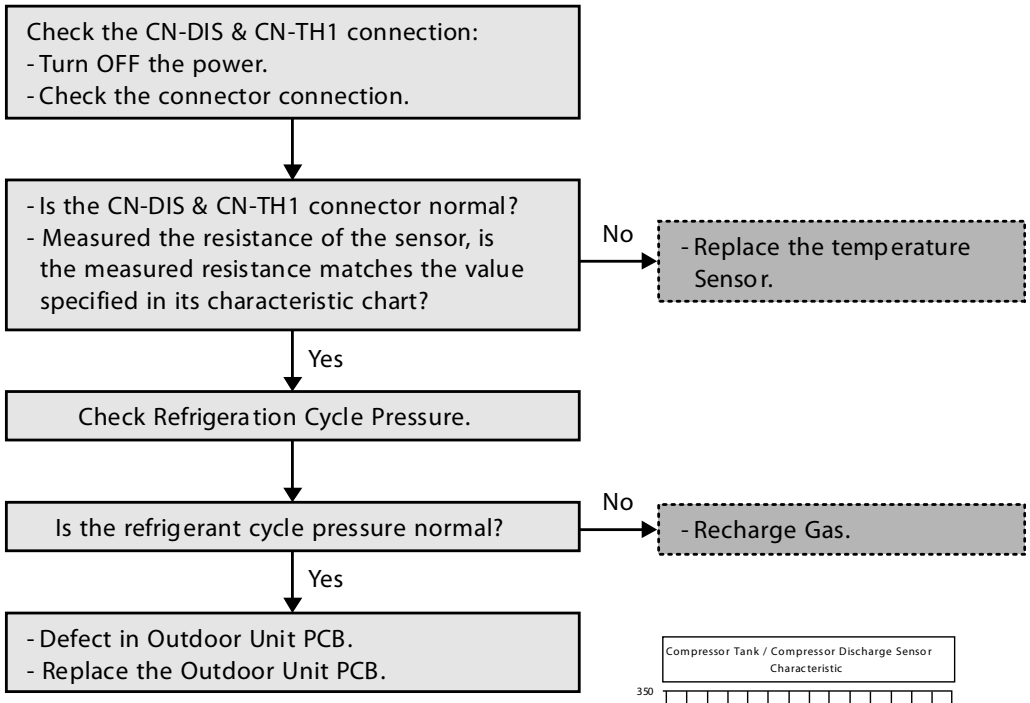
Abnormality Judgment:

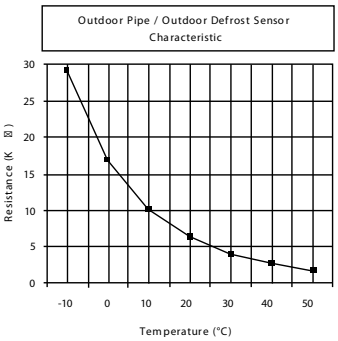
1 times occurrence within 2550 minutes.

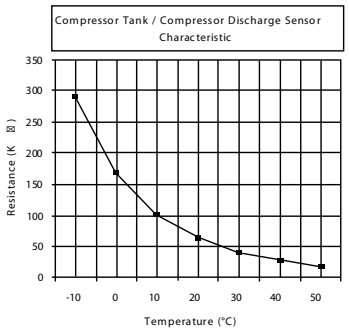
Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch offthe power before remove and connect the component.







19.5.39 Indoor Water Outlet Temperature Sensor 2 Abnormality (F30)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the indoor water outlet temperature sensor 2 are used to determine sensor error.

Malfunction Caused:

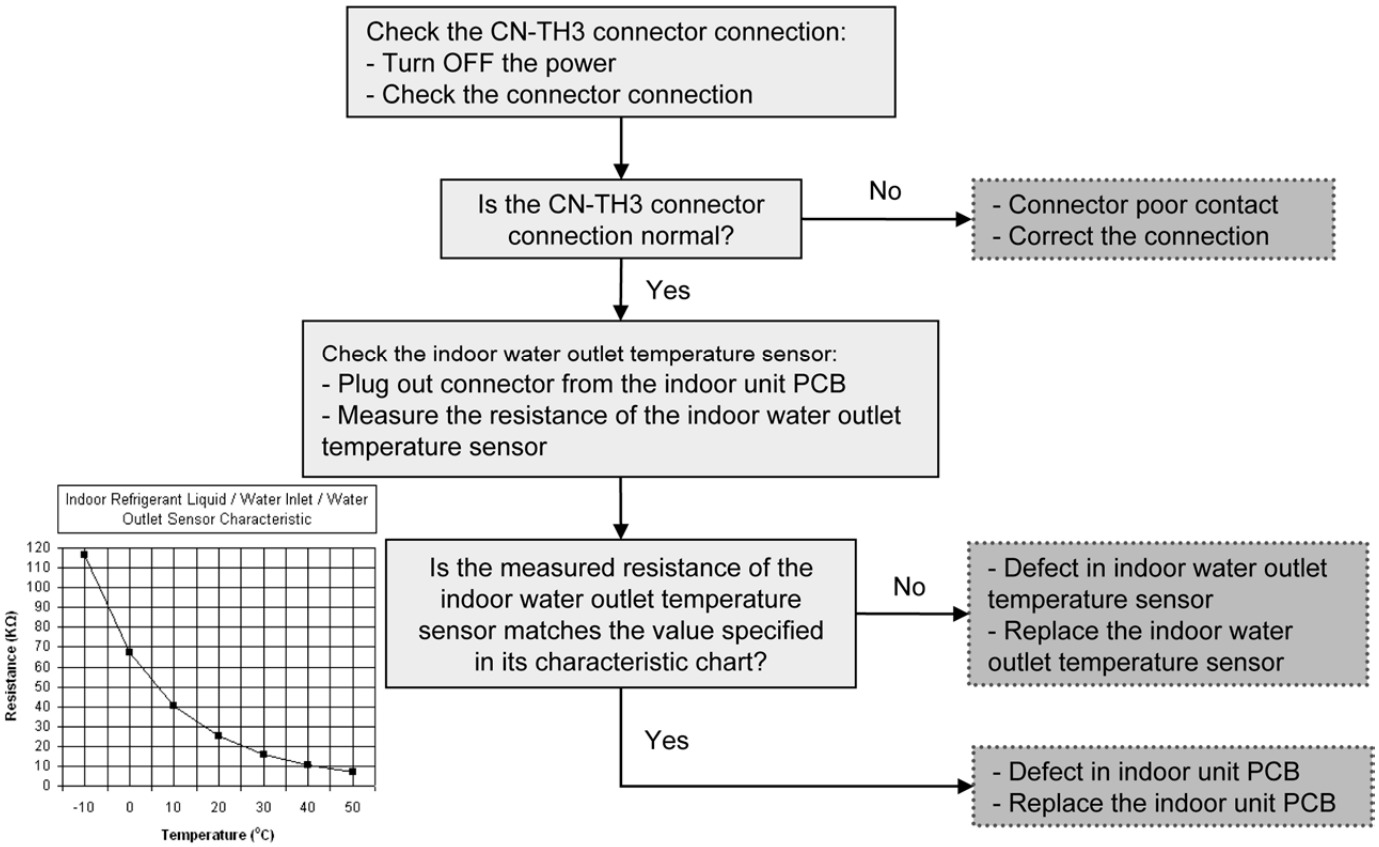
6 Faulty connector connection.
7 Faulty sensor.
8 Faulty indoor unit PCB.

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.40 Outdoor Air Temperature Sensor Abnormality (F36)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the outdoor air temperature sensor are used to determine sensor error.

Malfunction Caused:

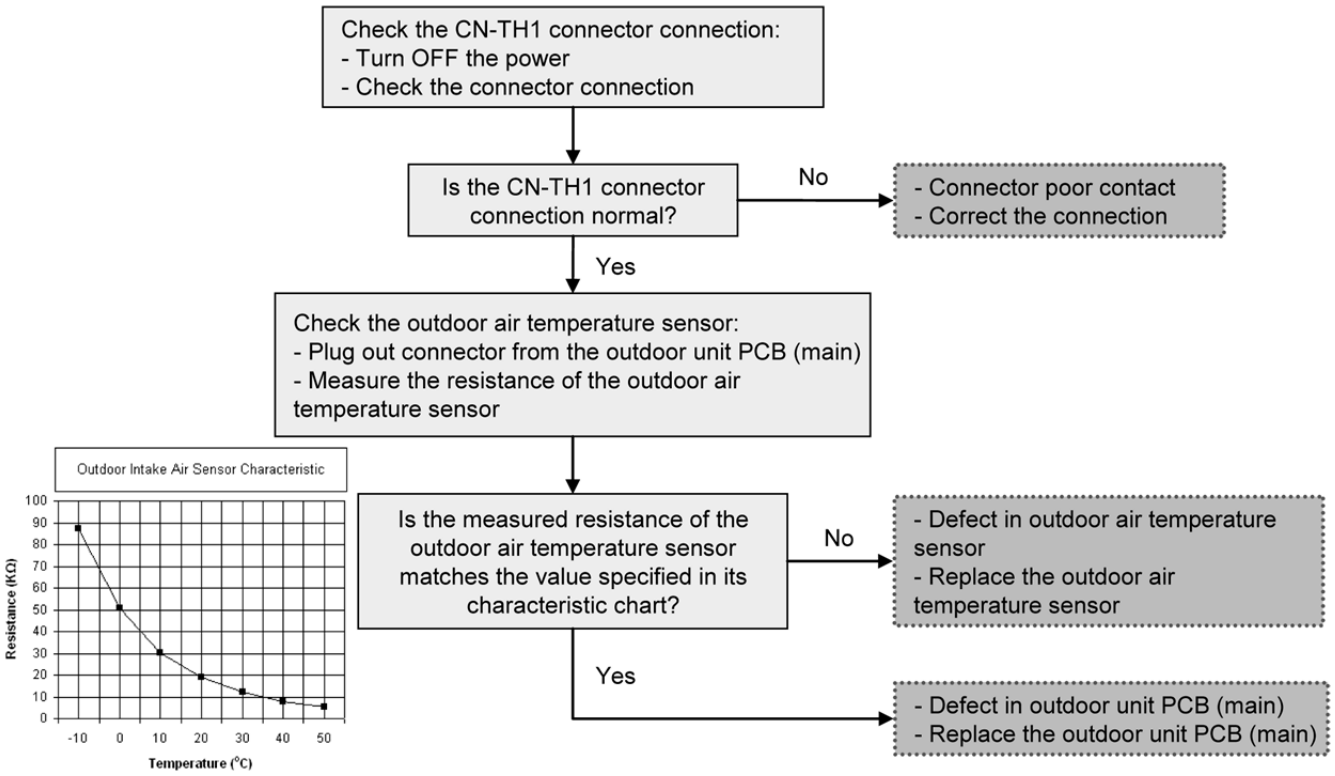
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.41 Indoor Water Inlet Temperature Sensor Abnormality (F37)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the indoor water inlet temperature sensor are used to determine sensor error.

Malfunction Caused:

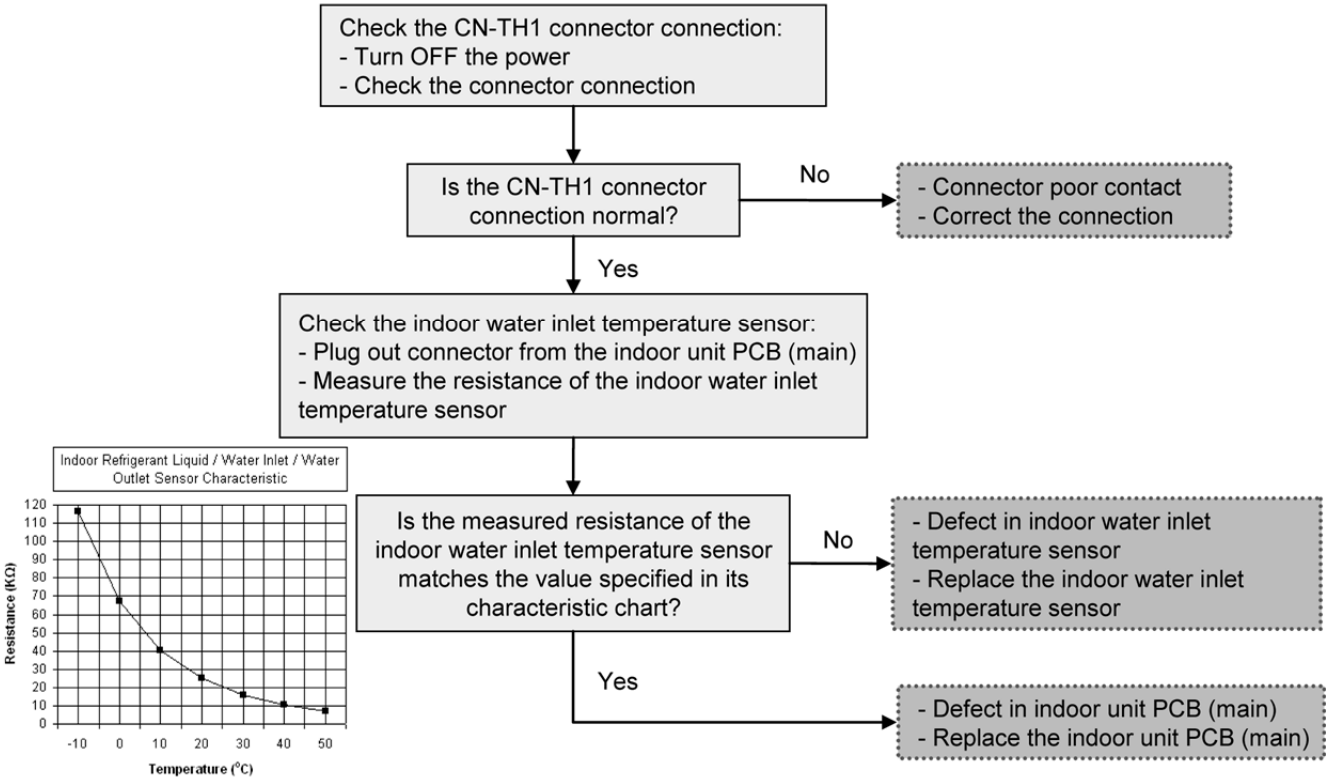
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty indoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.42 Outdoor Discharge Pipe Temperature Sensor Abnormality (F40)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the outdoor discharge pipe temperature sensor are used to determine sensor error.

Malfunction Caused:

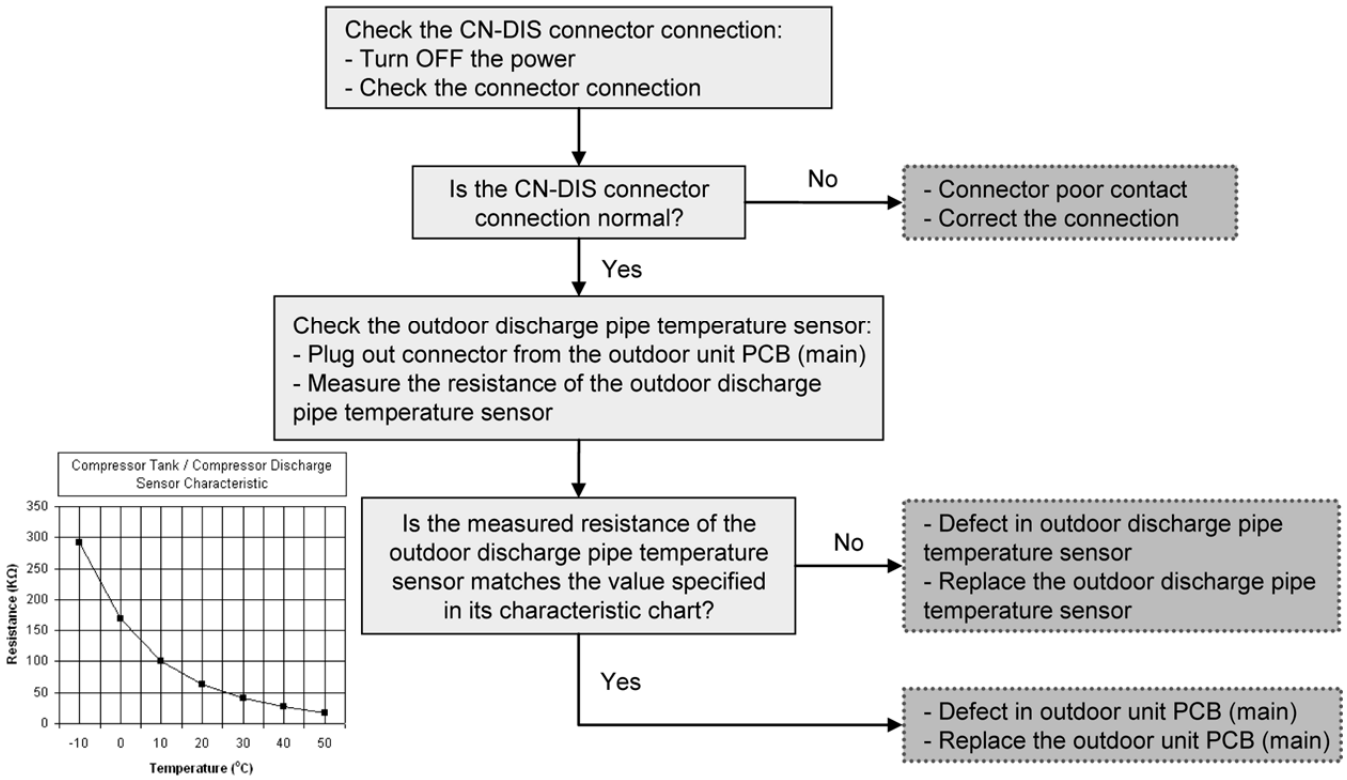
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.43 Power Factor Correction (PFC) Abnormality (F41)

Malfunction Decision Conditions:

During operation of cooling and heating, when the PFC protection circuitry in the outdoor unit PCB (main) senses abnormal high DC voltage level.

Malfunction Caused:

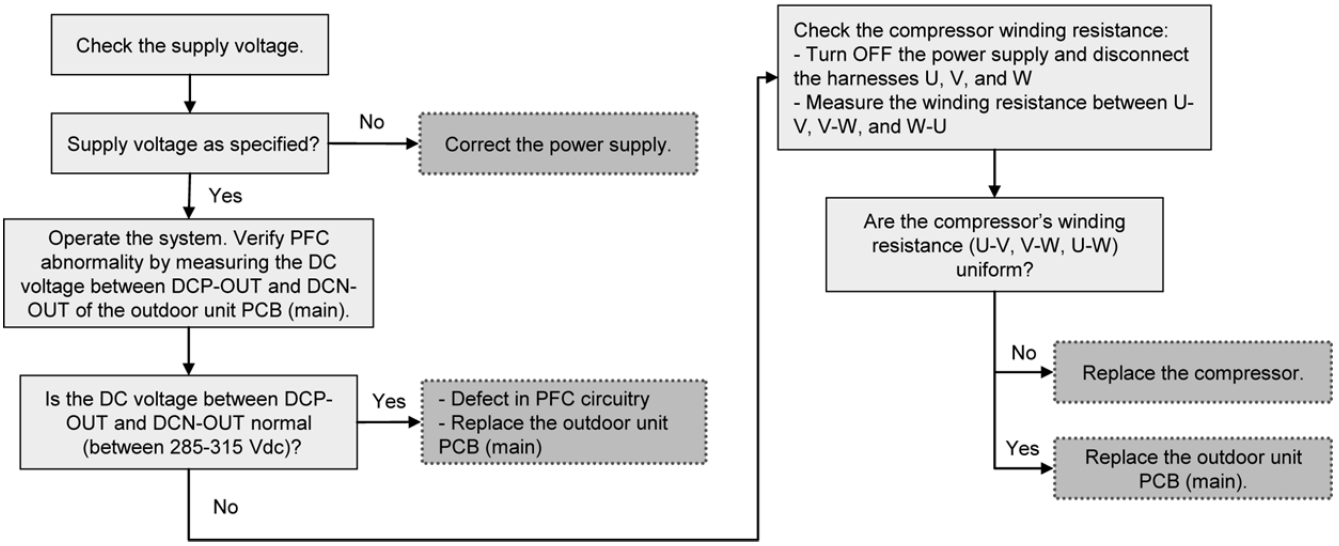
1 Power supply surge.
2 Compressor windings not uniform.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue 4 times in 10 minutes.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.44 Outdoor Pipe Temperature Sensor Abnormality (F42)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the outdoor pipe temperature sensor are used to determine sensor error.

Malfunction Caused:

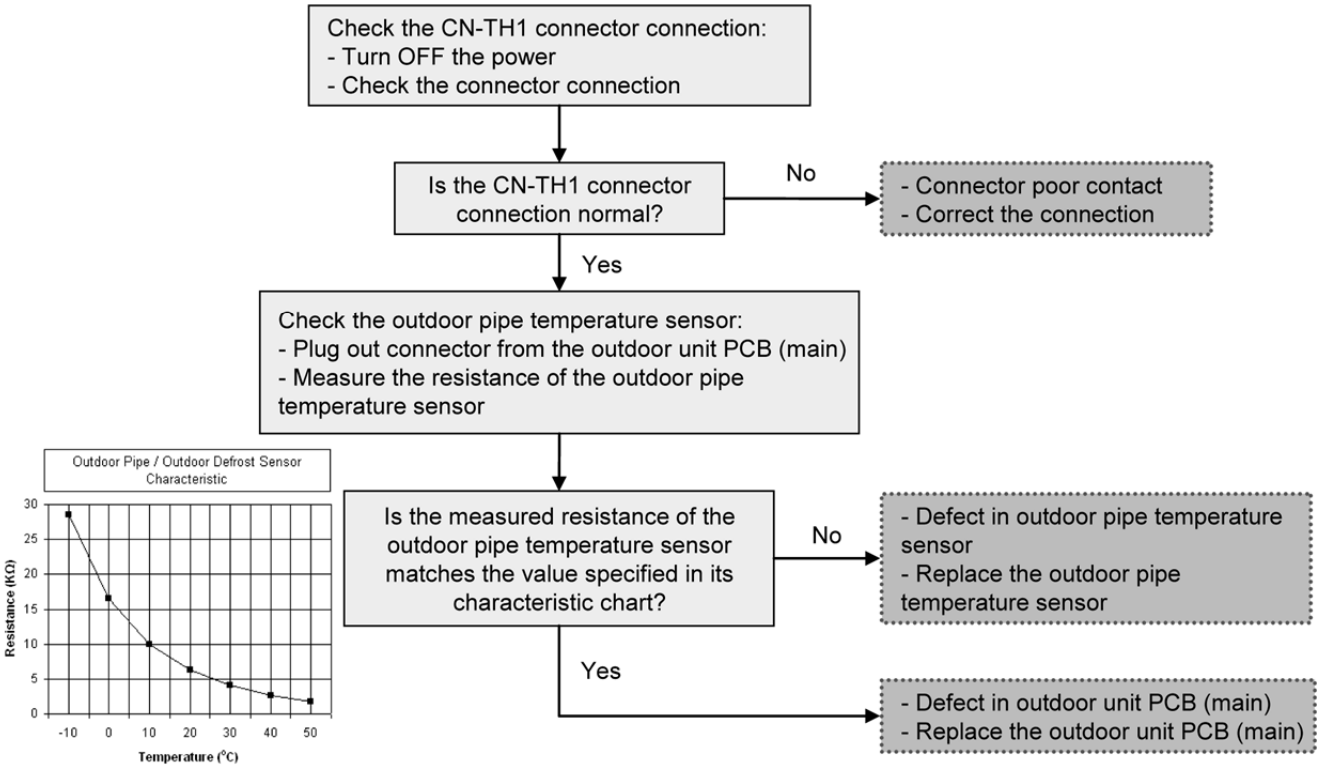
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.45 Outdoor Defrost Temperature Sensor Abnormality (F43)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the outdoor defrost temperature sensor are used to determine sensor error.

Malfunction Caused:

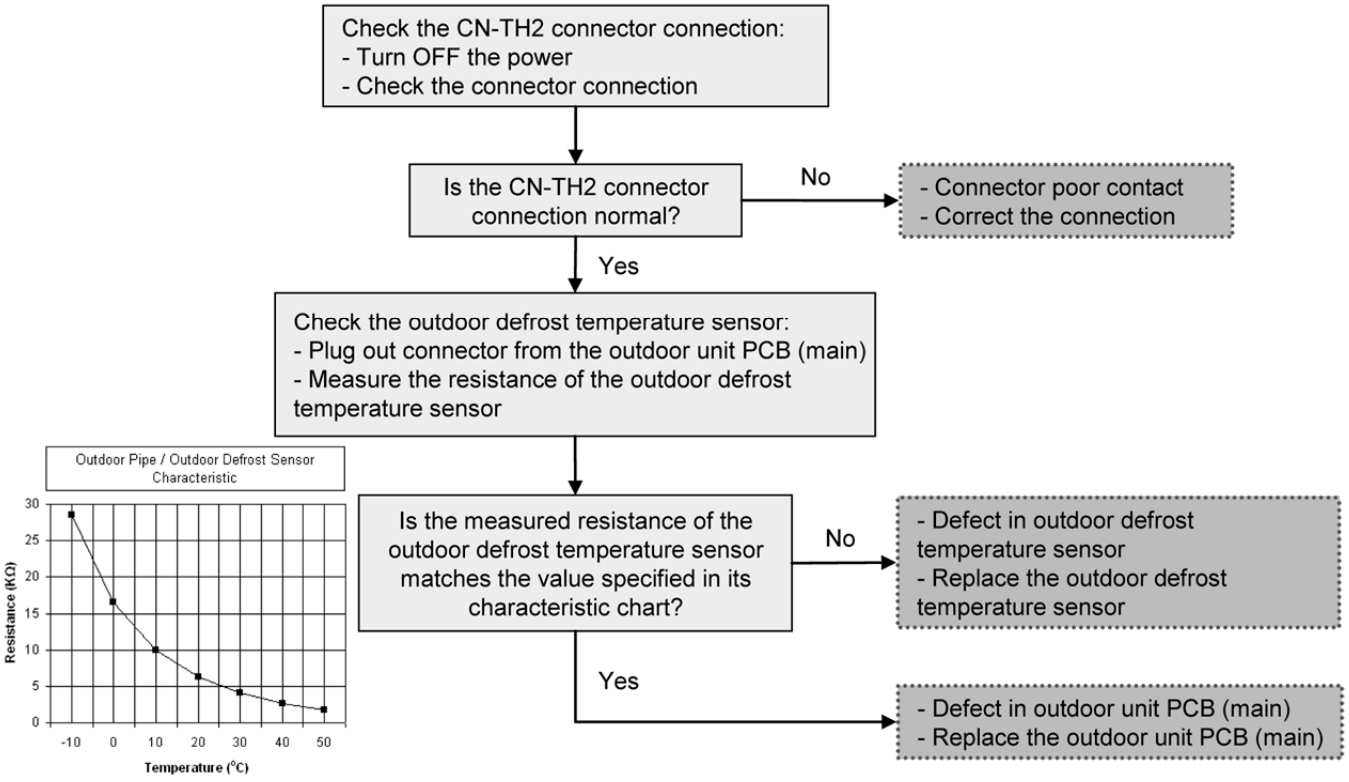
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.46 Indoor Water Outlet Temperature Sensor Abnormality (F45)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the indoor water outlet temperature sensor are used to determine sensor errors.

Malfunction Caused:

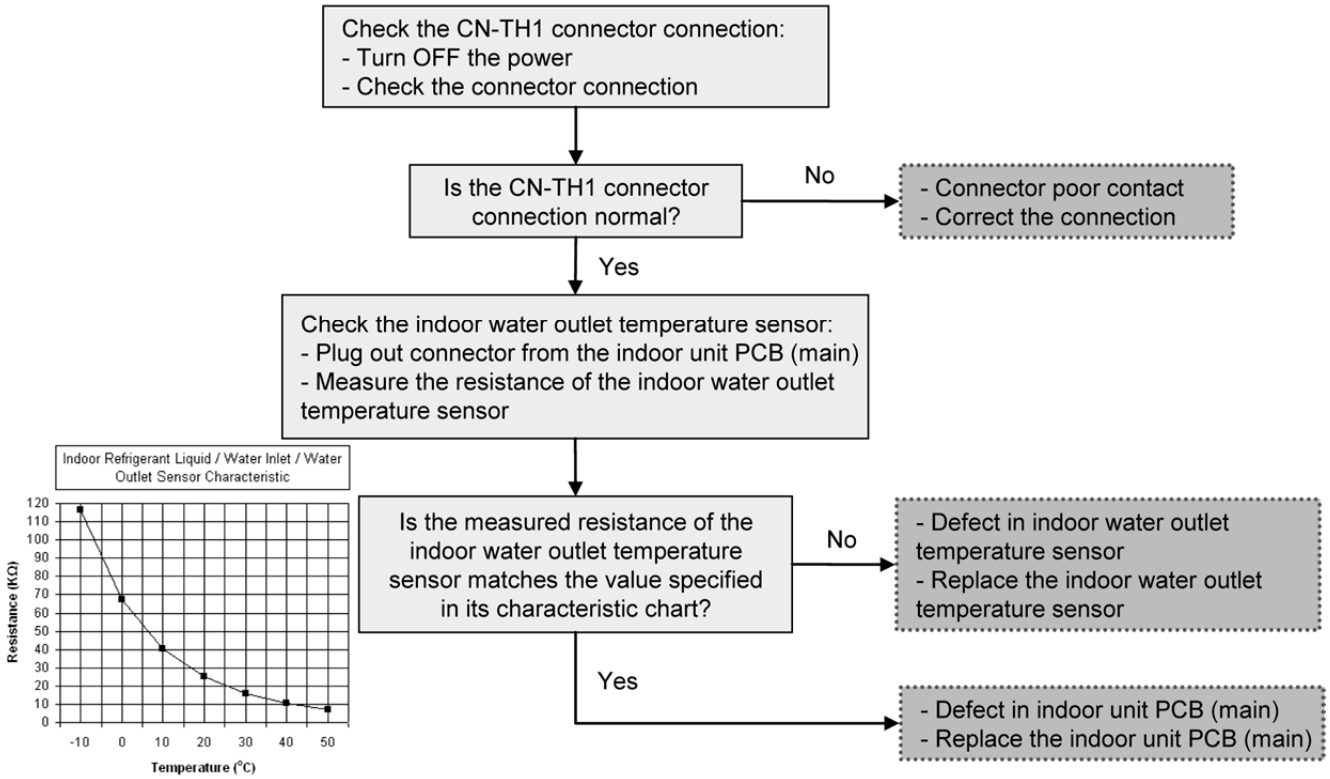
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty indoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.47 Outdoor Current Transformer Open Circuit (F46)

Malfunction Decision Conditions:

A current transformer (CT) open circuit is detected by checking the compressor running frequency (⩾ rated frequency) and CT detected input current (< 0.65 A) for continuously 20 seconds.

Malfunction Caused:

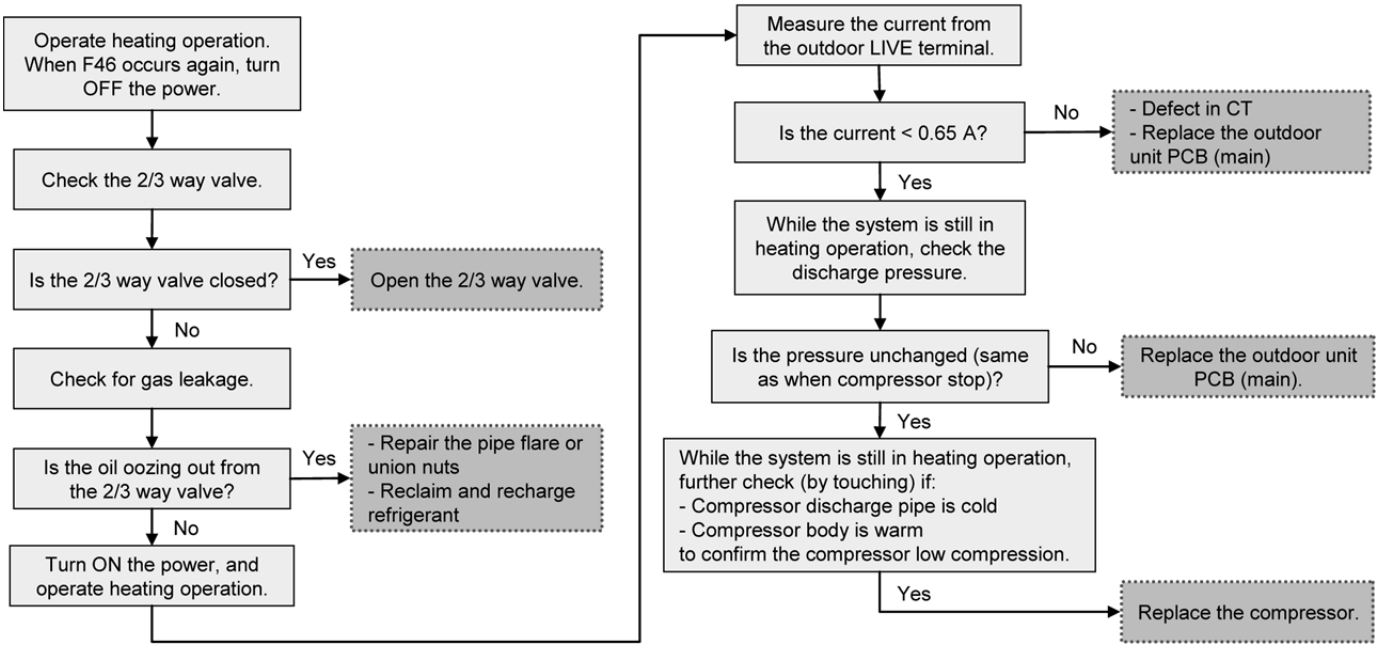
1 CT defective.
2 Faulty outdoor unit PCB (main).
3 Compressor defective (low compression).

Abnormality Judgment:

Continue 3 times in 20 minutes.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.48 Cooling High Pressure Overload Protection (F95)

Malfunction Decision Conditions:

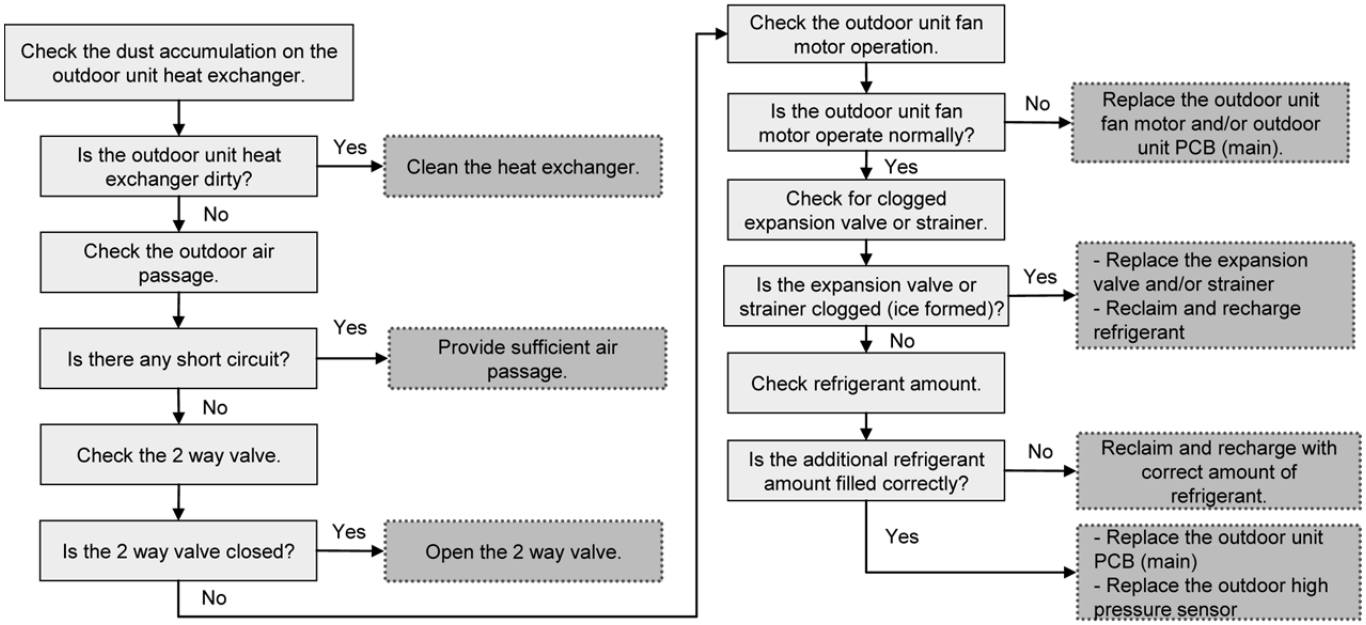
During operation of cooling, when pressure 4.0 MPa and above is detected by outdoor high pressure sensor.

Malfunction Caused:

1 Dust accumulation in the outdoor unit heat exchanger.
2 Air short circuit at outdoor unit.
3 2 way valve closed.
4 Faulty outdoor unit fan motor.
5 Clogged expansion valve or strainer.
6 Excessive refrigerant.
7 Faulty outdoor high pressure sensor.
8 Faulty outdoor unit PCB (main).

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



20. Disassembly and Assembly Instructions

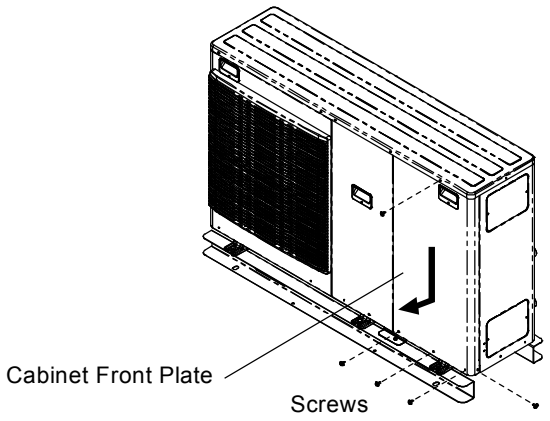
⚠ WARNING

High Voltage are generated in the electrical parts area by the capacitor. Ensure that the capacitor has discharged sufficiently before proceeding with repair work. Failure to heed this caution may result in electric shocks.

⚠ WARNING

Be sure to switch off all the power supply before performing each of the below action.

20.1 To Remove The Cabinet Front Plate

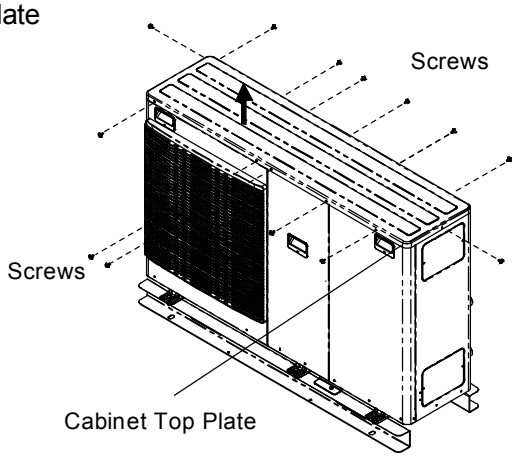


1. Remove 5 mounting screws.

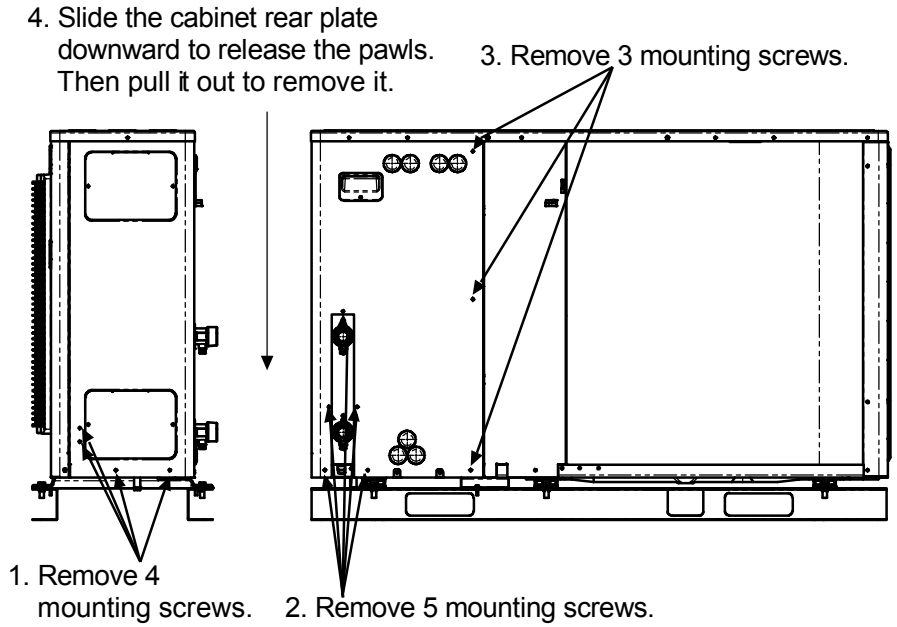
2. Slide the cabinet front plate downward to release the pawls Then, pull it toward front to remove it.

20.2 To Remove The Cabinet Top Plate

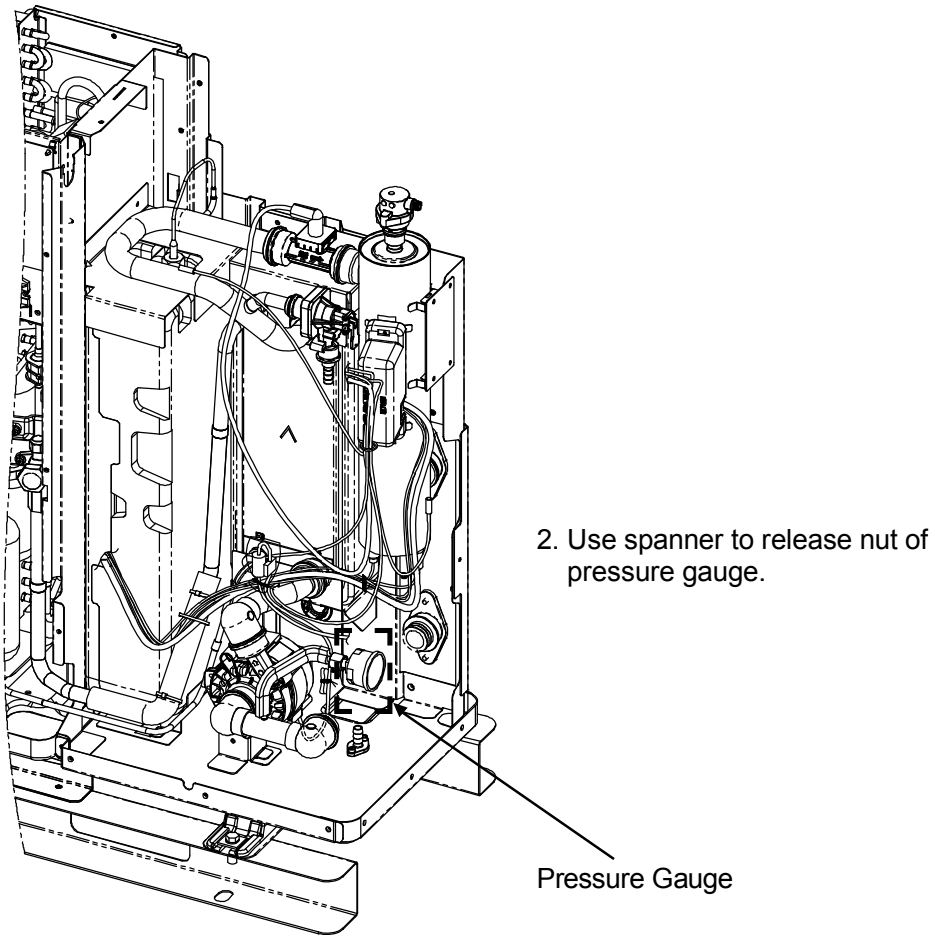
1. Remove the 12 mounting screws
2. Lift the cabinet top plate upward to remove it.



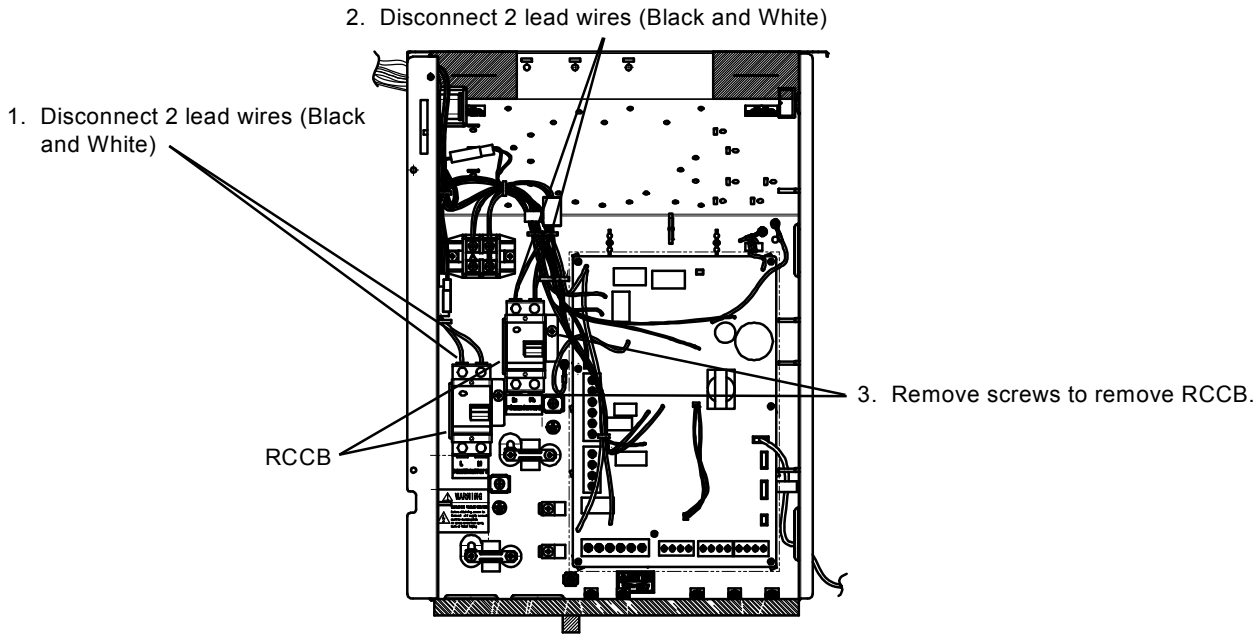
20.3 To Remove The Cabinet Rear Plate



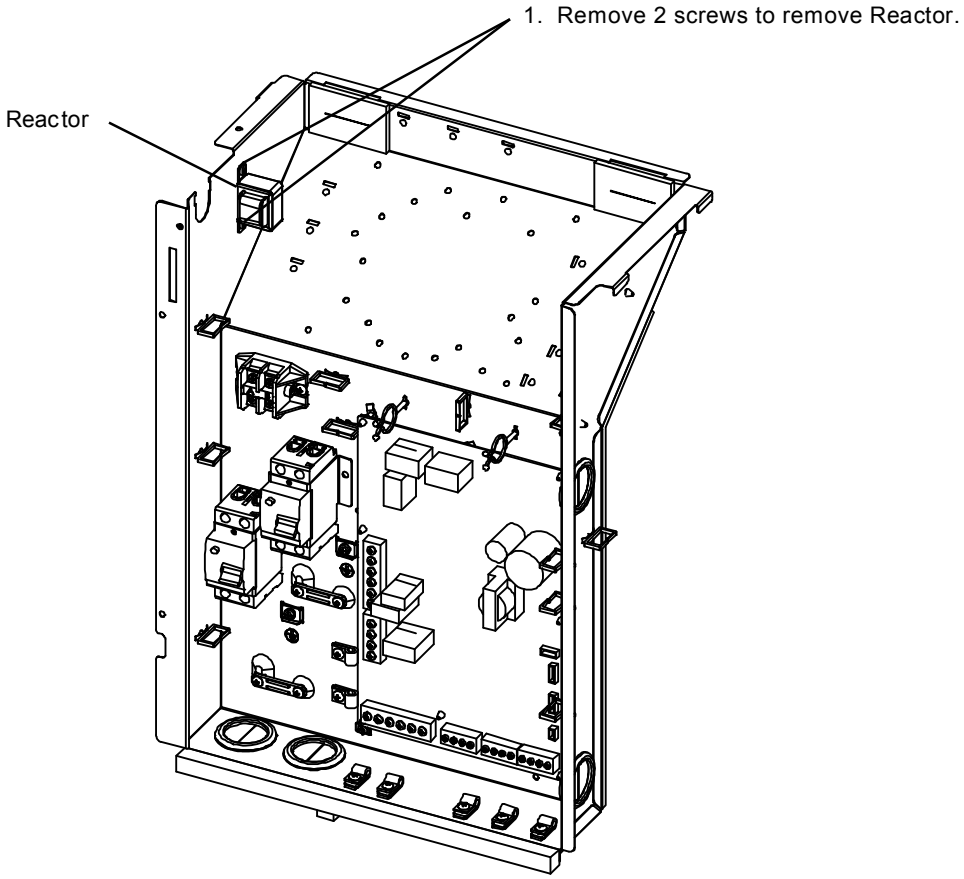
20.4 To Remove Pressure Gauge



20.5 To Remove RCCB

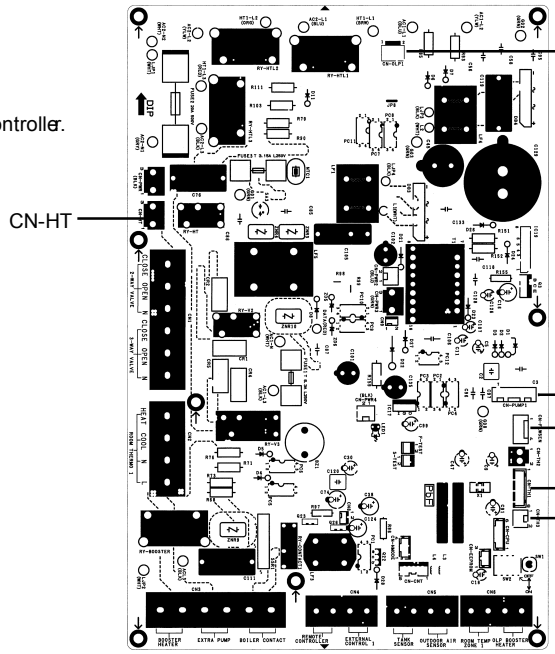


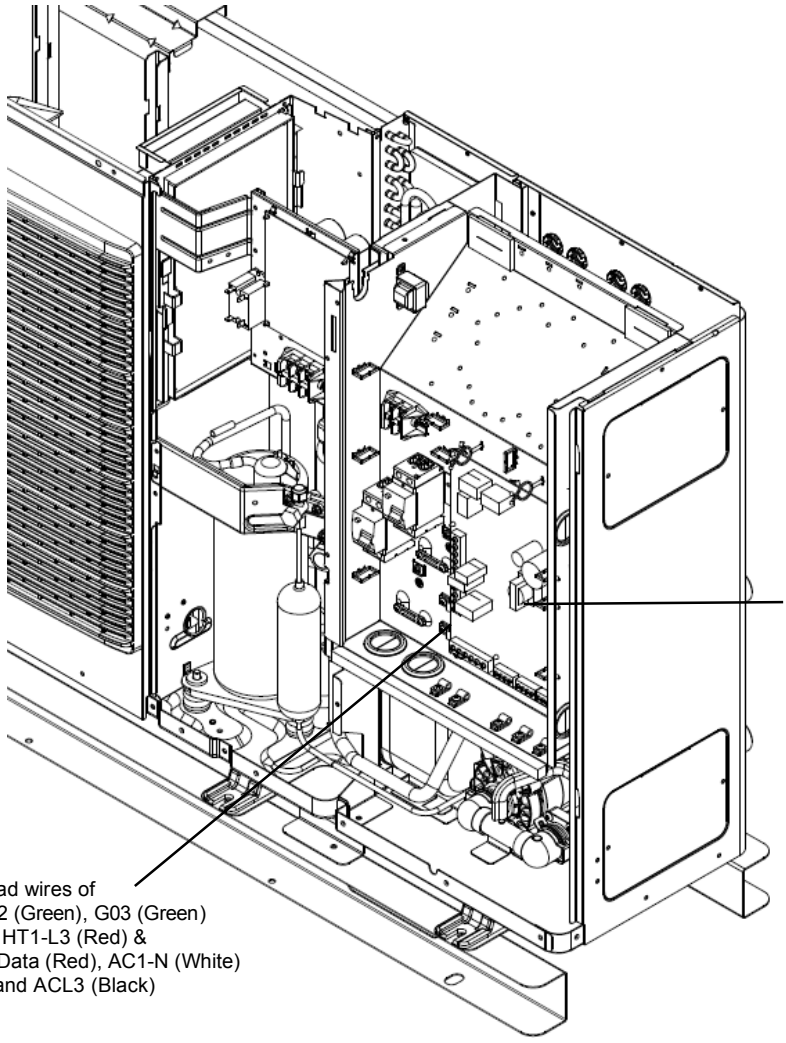
20.6 To Remove Reactor



20.7 To Remove Water System Electronic Controller Board

1. Remove all the connectors from the Electronic Controller.



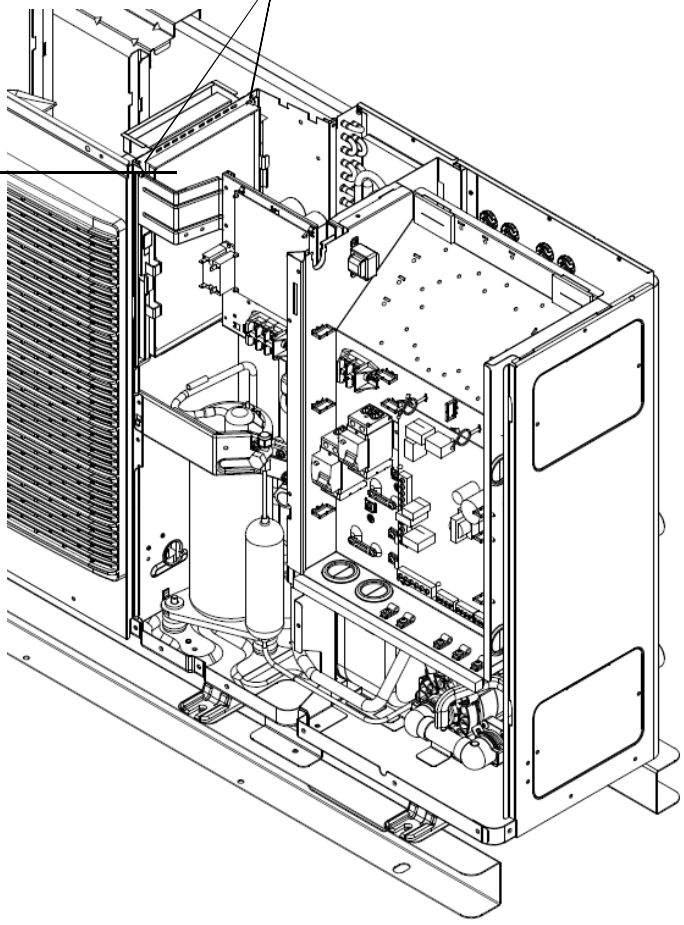


2. Disconnect all lead wires of
   G01 (Green), G02 (Green), G03 (Green)
   AC2-N2 (White), HT1-L3 (Red) &
   AC2-L3 (Black), Data (Red), AC1-N (White)
   AC1-L3 (Black) and ACL3 (Black)

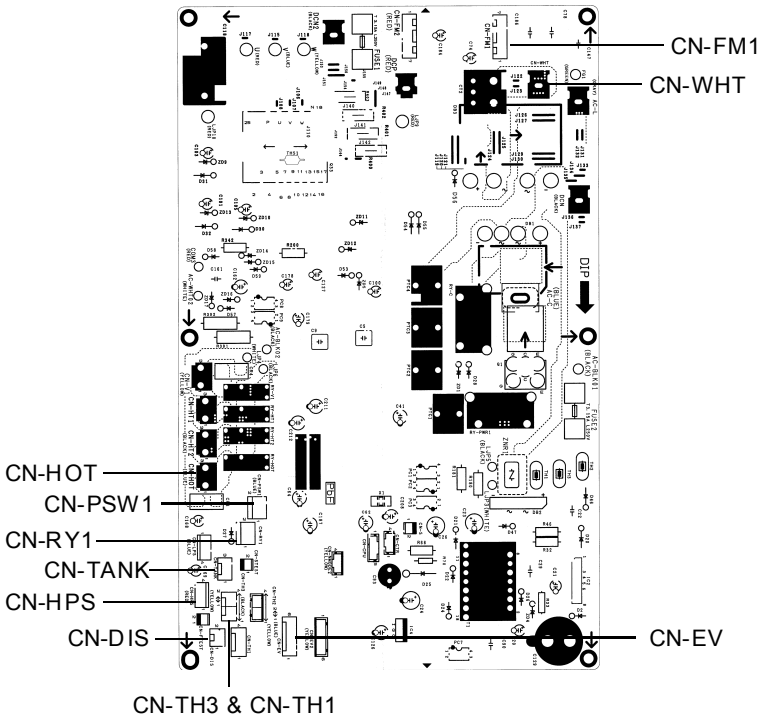
20.8 To Remove Refrigerant Side Control Board Complete

3. Remove 2 screws to remove the Control Board Complete

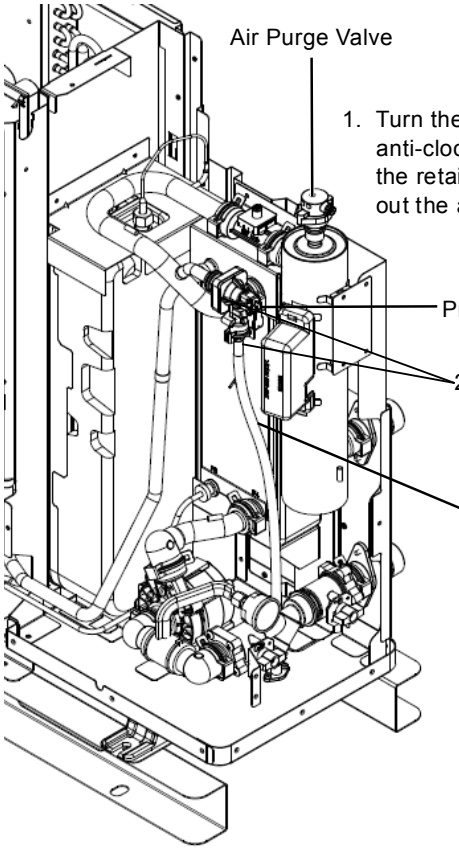
1. Remove all lead wire FG1 (Green)
AC-L (Grey), DCP (Red), DCN2 (Black),
DCN (Black), AC-C (Blue), AC-BLK01 (Black),
AC-BLK02 (Black), COM3 (Red), AC-WHT02 (White)
and U (Red) V (Blue) W (Yellow) connector.



2. Disconnect all the connectors
from the electronic control board.



20.9 To Remove Pressure Relief Valve and Air Purge Valve



⚠ During reinstall Air Purge Valve.

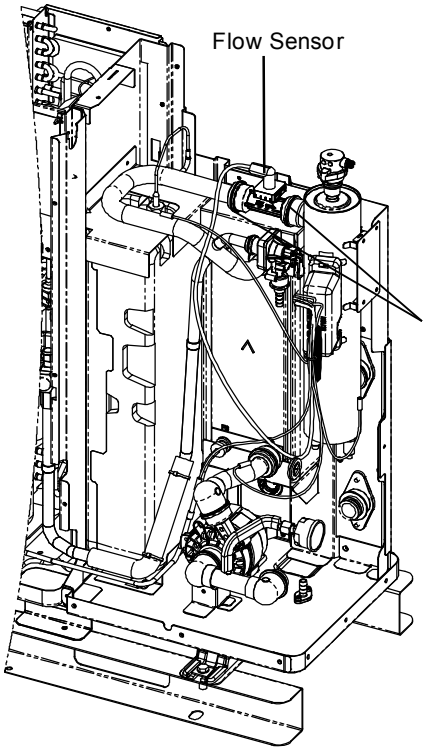
Make sure plug is fully close in clockwise direction.



Packing

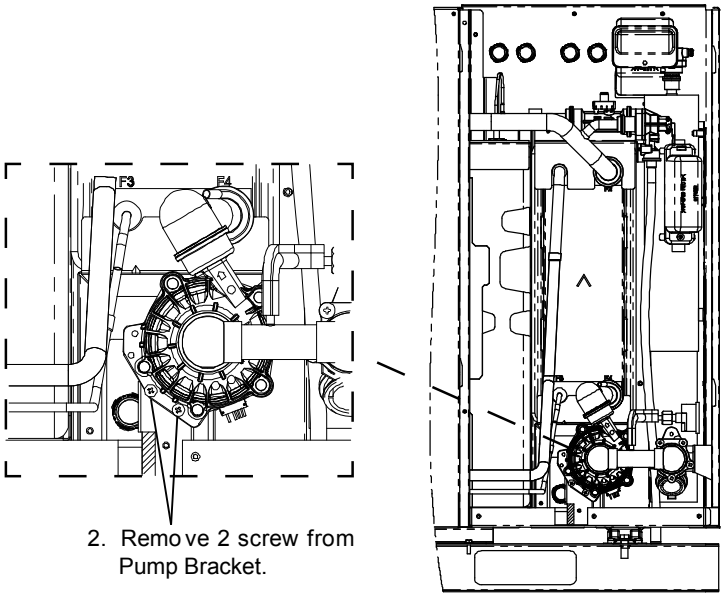
Apply some water based grease on packing surface before fix with shut off valve.

20.10 To Remove Flow Sensor



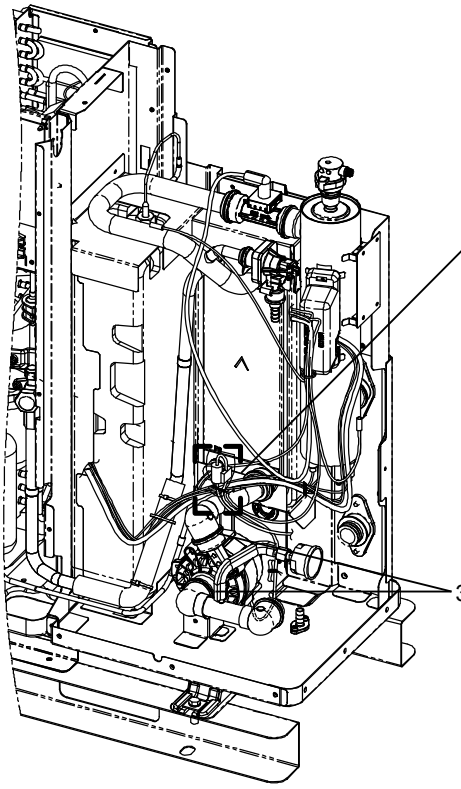
⚠ When reinstall the water pipe, use grease or water at the joining

20.11 To Remove Water Pump

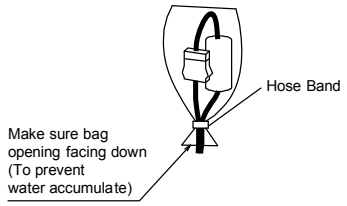


1. Disconnect CN-PUMP1 connector from Electronic Controller Board.

2. Remove 2 screw from Pump Bracket.



⚠ If pump is replace, please follow below instruction.

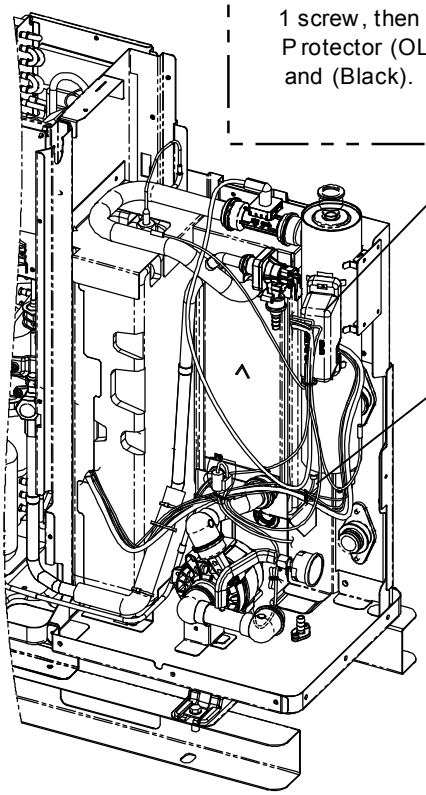


1. Please fix bag around the connector as shown, and seal it with a band.

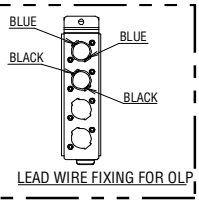
3. Remove 2 retaining ring, then slowly pull out the water pump.

⚠ When reinstall the water pipe, use grease or water at the joining.

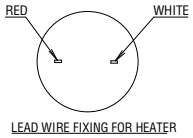
20.12 To Remove Heater

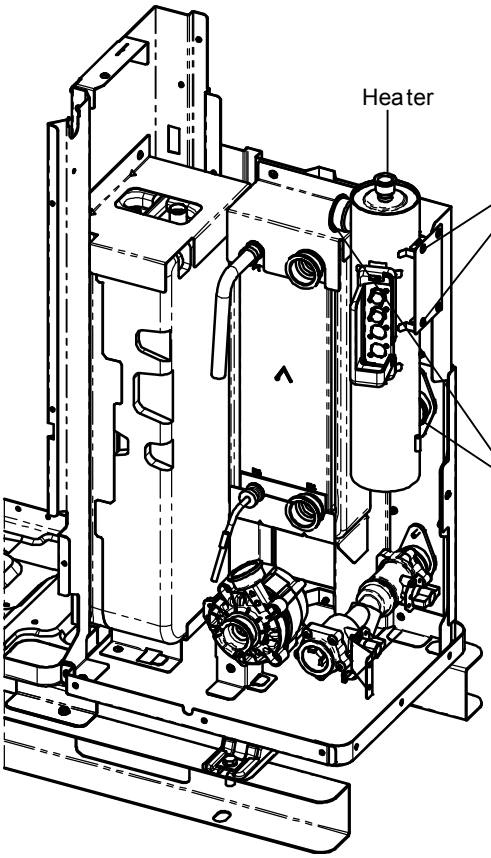


2. Remove terminal cover by releasing 1 screw, then remove the Overload Protector (OLP) lead wire (Blue) and (Black).



1. Remove 2 lead wire, HT-L3 (Red) and AC2-N2 (White) from below the Heater.

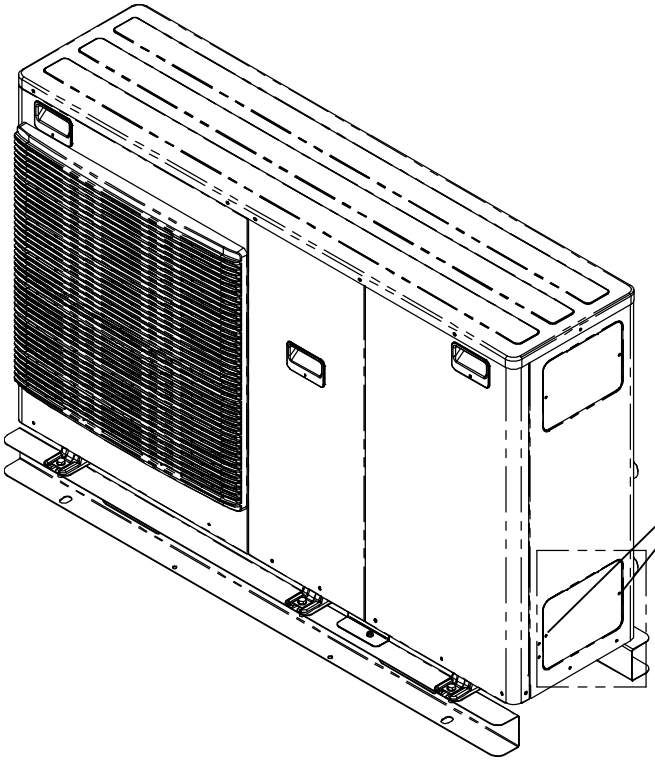


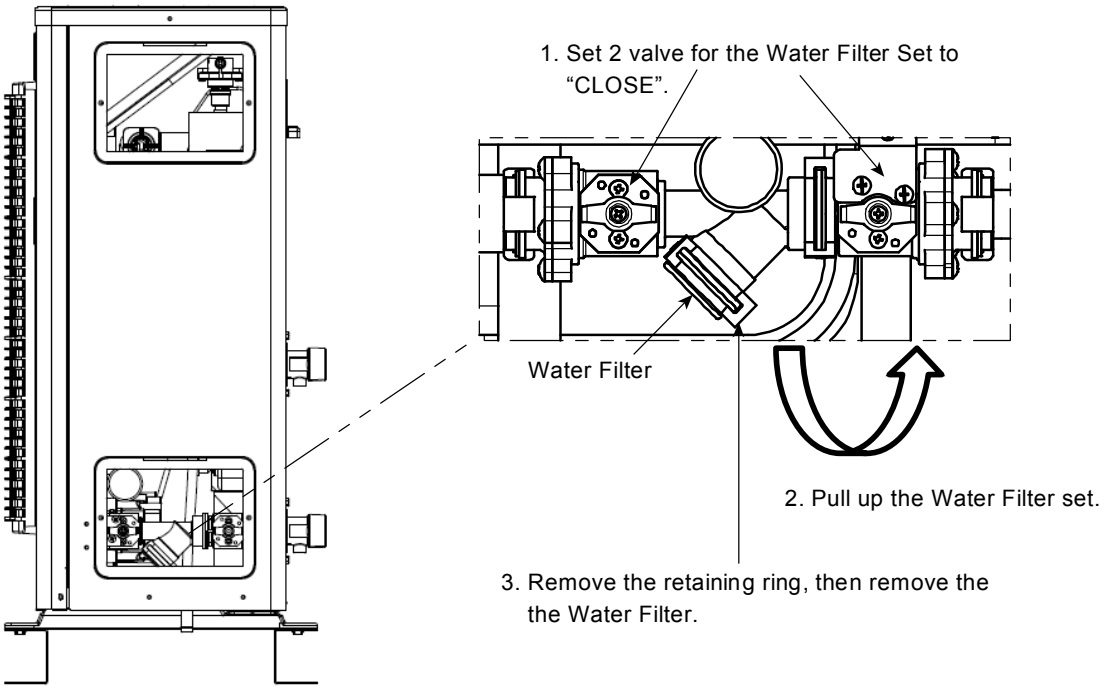


3. Remove 2 screw from the Heater Bracket

4. Release 2 retaining ring, and slowly remove the Heater Complete.

20.13 To Remove Filter





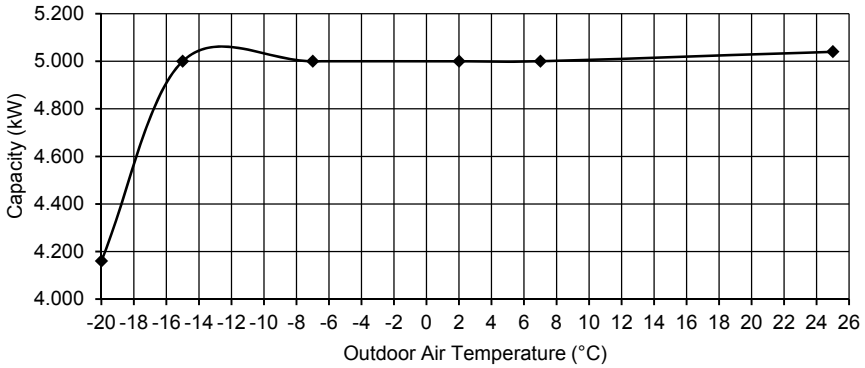
21. Technical Data

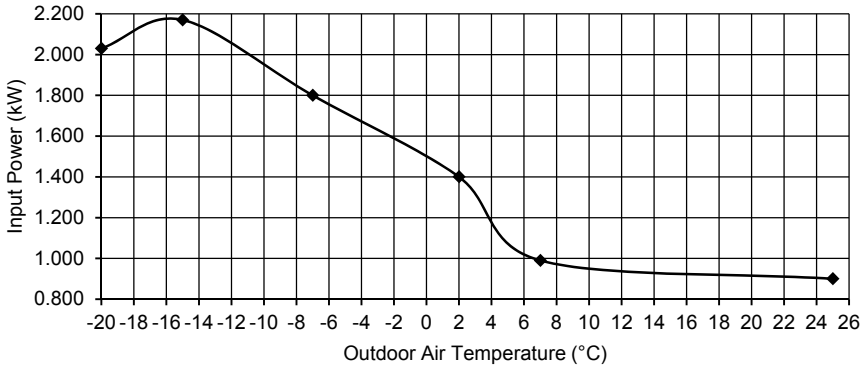
21.1 Operation Characteristics

21.1.1 WH-MDC05J3E5

Heating Characteristics at Different Outdoor Air Temperature

Condition
Outdoor air temperature : 7°C (DBT), 6°C (WBT)
Indoor water inlet temperature : 30°C
Indoor water outlet temperature : 35°C





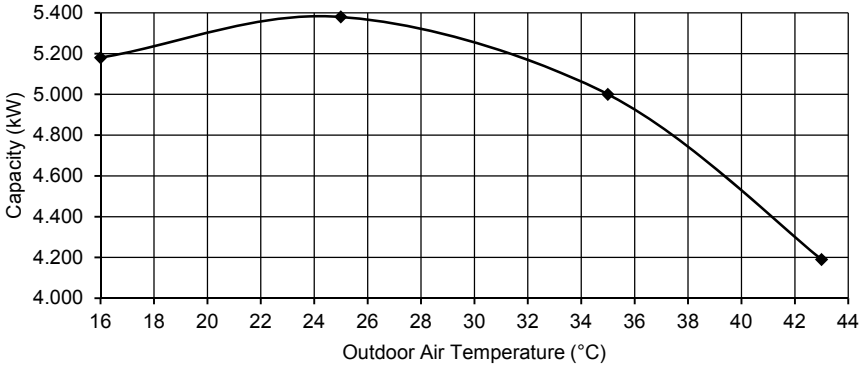
Cooling Characteristics at Different Outdoor Air Temperature

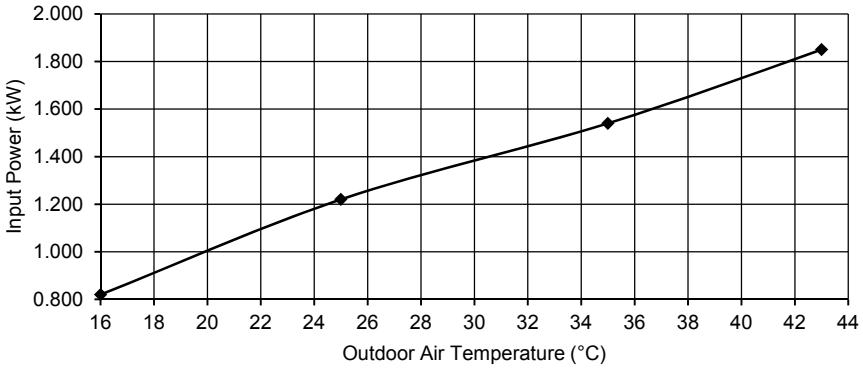
Condition

Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 12°C

Indoor water outlet temperature : 7°C





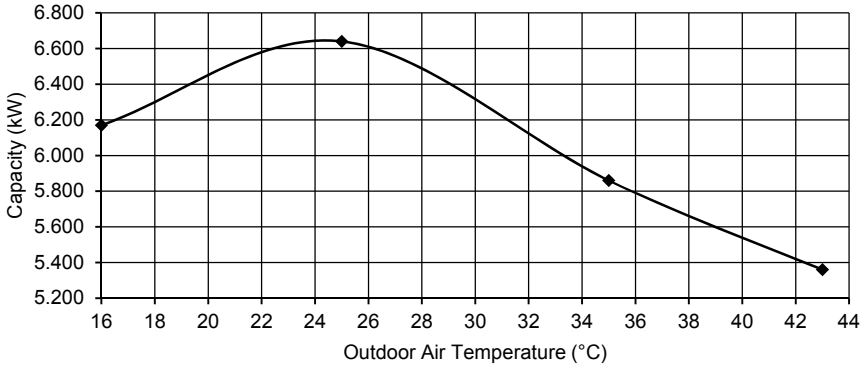
Cooling Characteristics at Different Outdoor Air Temperature

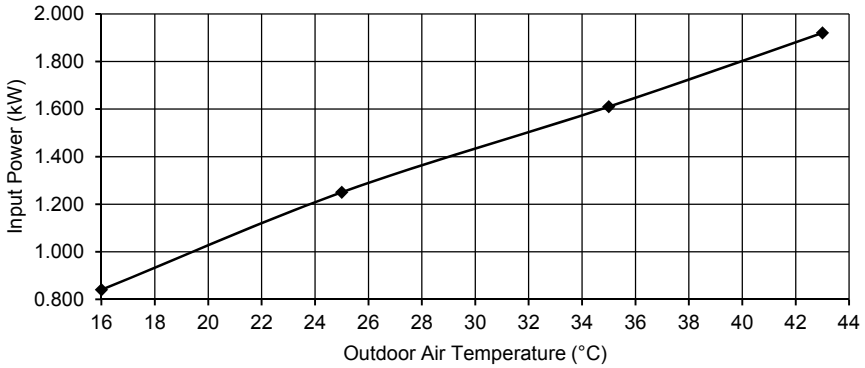
Condition

Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 19°C

Indoor water outlet temperature : 14°C





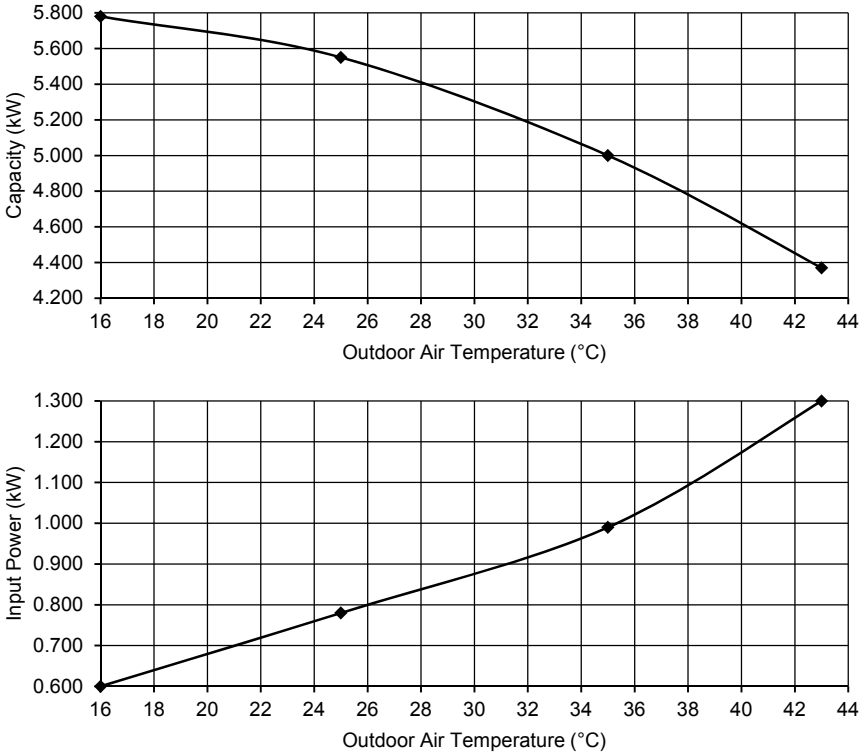
Cooling Characteristics at Different Outdoor Air Temperature

Condition

Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 23°C

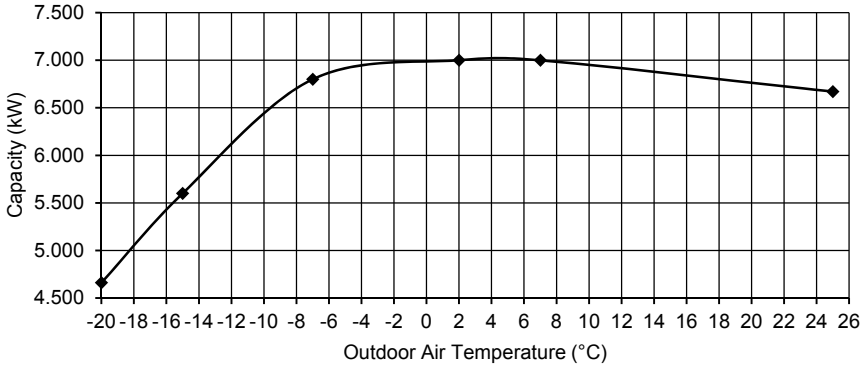
Indoor water outlet temperature : 18°C

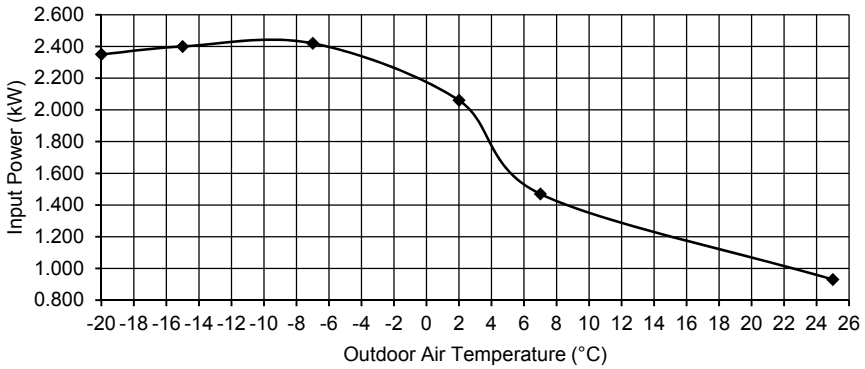


21.1.2 WH-MDC07J3E5

Heating Characteristics at Different Outdoor Air Temperature

Condition
Outdoor air temperature : 7°C (DBT), 6°C (WBT)
Indoor water inlet temperature : 30°C
Indoor water outlet temperature : 35°C





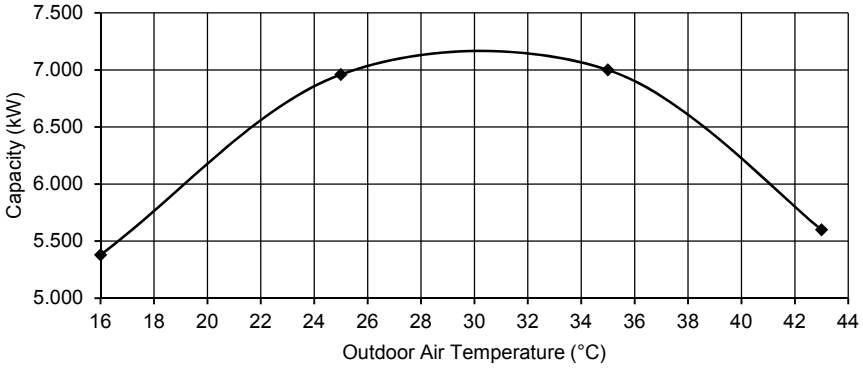
Cooling Characteristics at Different Outdoor Air Temperature

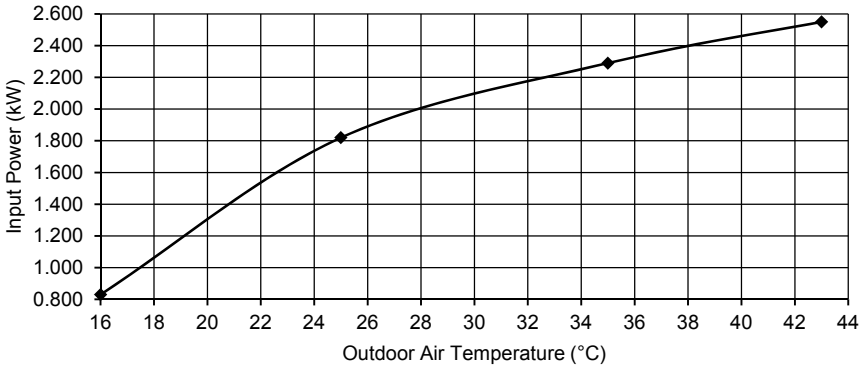
Condition

Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 12°C

Indoor water outlet temperature : 7°C





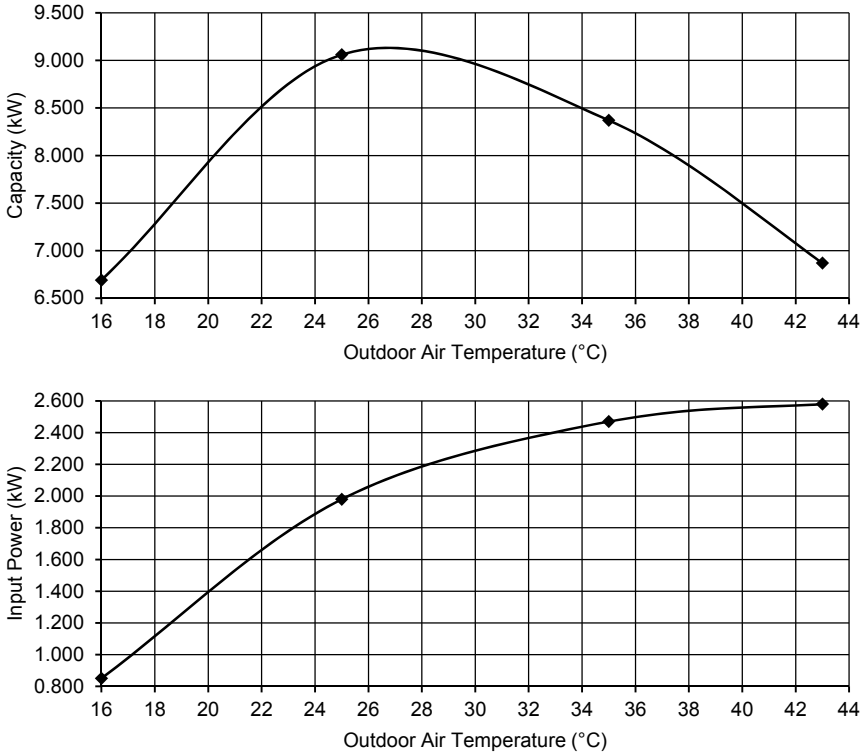
Cooling Characteristics at Different Outdoor Air Temperature

Condition

Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 19°C

Indoor water outlet temperature : 14°C



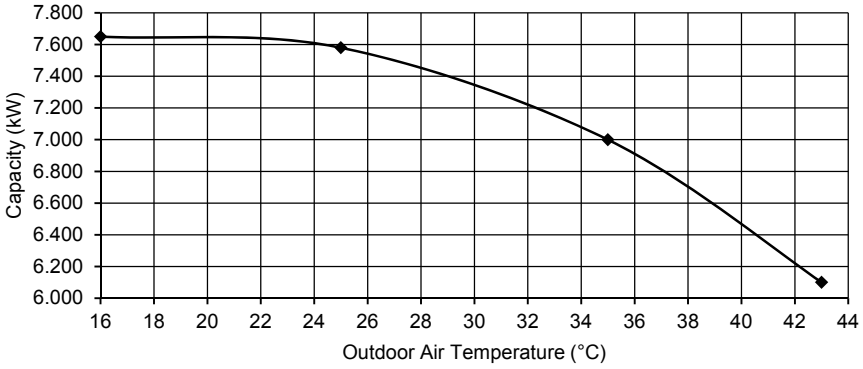
Cooling Characteristics at Different Outdoor Air Temperature

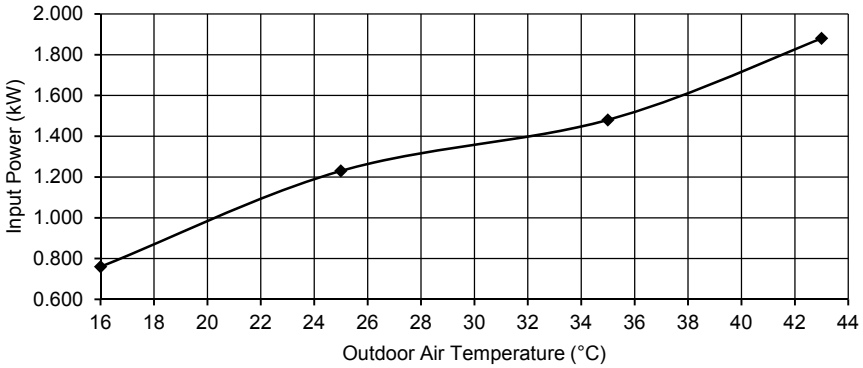
Condition

Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 23°C

Indoor water outlet temperature : 18°C

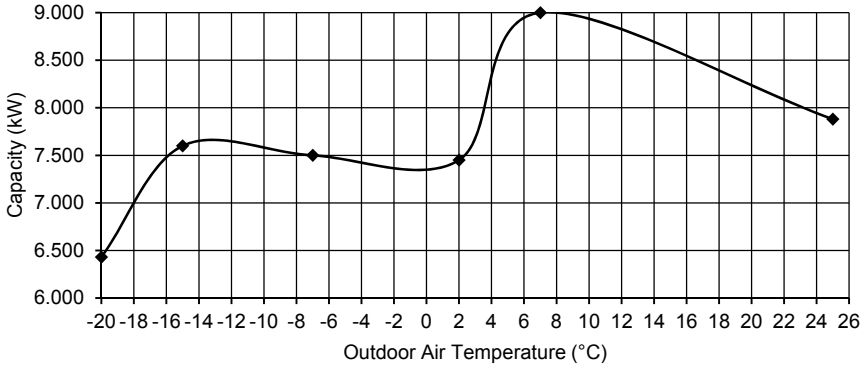


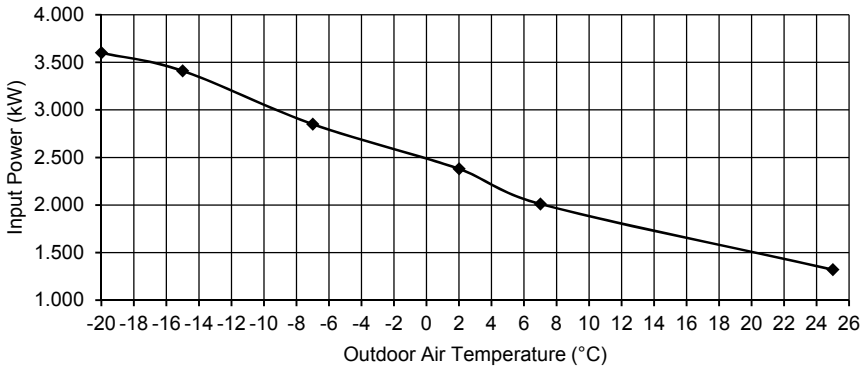


21.1.3 WH-MDC09J3E5

Heating Characteristics at Different Outdoor Air Temperature

Condition
Outdoor air temperature : 7°C (DBT), 6°C (WBT)
Indoor water inlet temperature : 30°C
Indoor water outlet temperature : 35°C





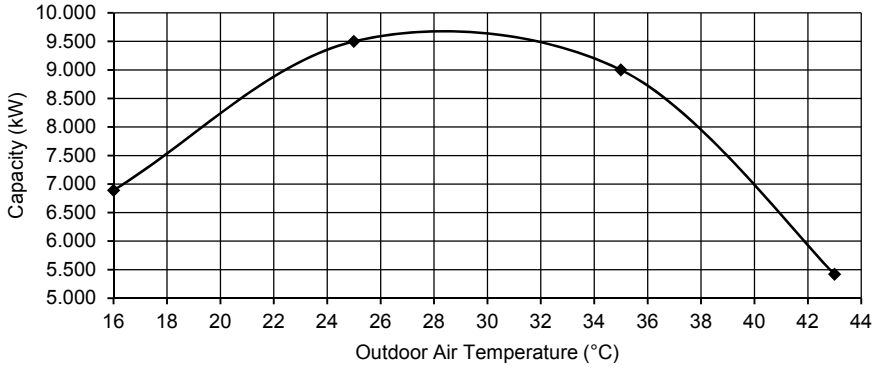
Cooling Characteristics at Different Outdoor Air Temperature

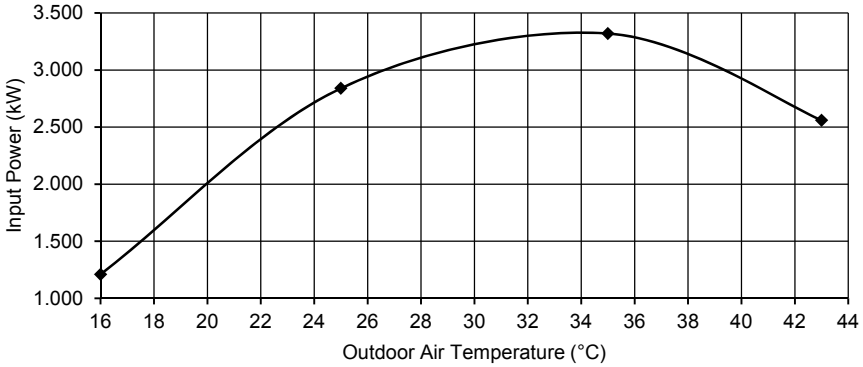
Condition

Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 12°C

Indoor water outlet temperature : 7°C





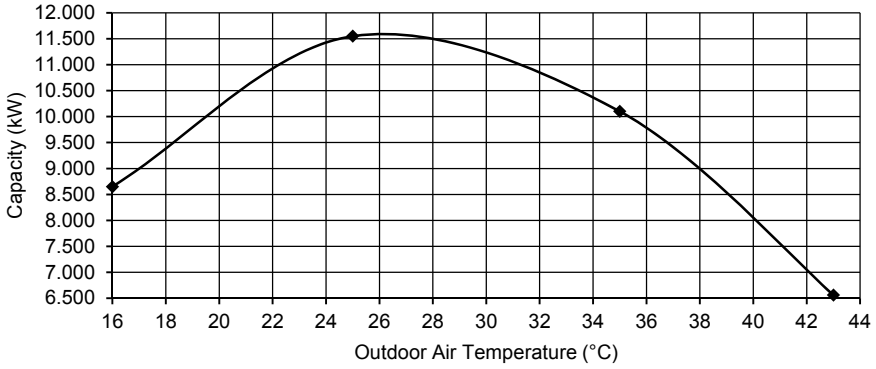
Cooling Characteristics at Different Outdoor Air Temperature

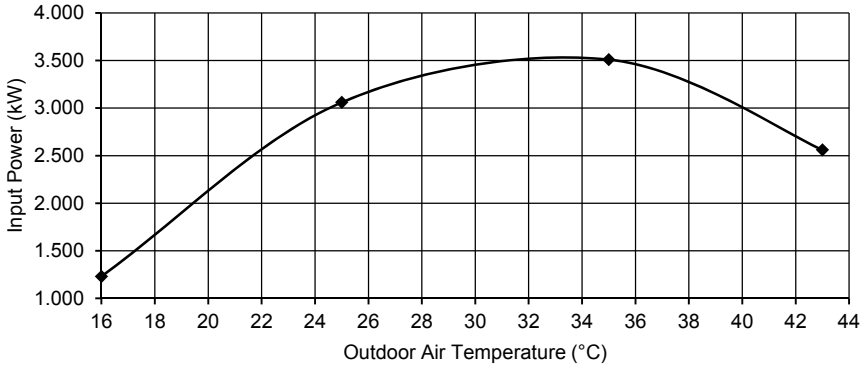
Condition

Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 19°C

Indoor water outlet temperature : 14°C





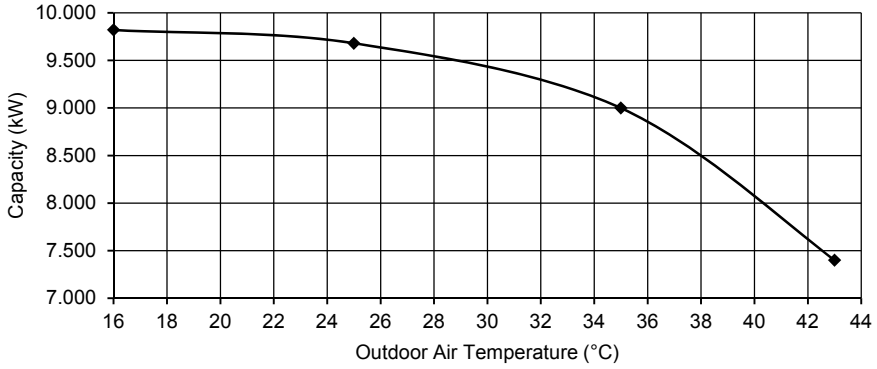
Cooling Characteristics at Different Outdoor Air Temperature

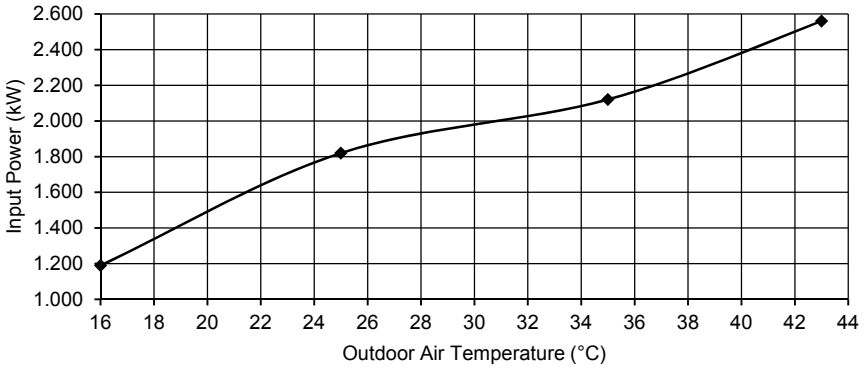
Condition

Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 23°C

Indoor water outlet temperature : 18°C





21.2 Heating Capacity Table

21.2.1 WH-MDC05J3E5

Water Out	25ºC		35ºC		45ºC		55ºC		60ºC	
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)
-20	4370	1730	4160	2030	3840	2370	3430	2640		
-15	5130	1780	5000	2170	4750	2510	3700	2450		
-7	5170	1490	5000	1800	4800	2160	5000	2700	4680	2710
2	5000	1110	5000	1400	5000	1810	5000	2200	4800	2400
7	5090	780	5000	990	5000	1310	5000	1660	4580	1900
25	4960	770	5040	900	5310	1160	5610	1340	5150	1330

21.2.2 WH-MDC07J3E5

Water Out	25ºC		35ºC		45ºC		55ºC		60ºC	
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)
-20	4860	2030	4660	2350	4440	2750	4230	3130		
-15	5800	2110	5600	2400	5300	2840	5000	3320		
-7	6760	2070	6800	2420	6300	2820	6300	3390	4740	2760
2	6830	1660	7000	2060	6850	2500	6300	2920	4800	2400
7	7320	1190	7000	1470	7000	1960	7000	2480	6180	2440
25	6800	640	6670	930	6790	1380	6700	1800	6220	1780

21.2.3 WH-MDC09J3E5

Water Out	25ºC		35ºC		45ºC		55ºC		60ºC	
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)
-20	5330	2360	6430	3600	5780	3830	4830	3640		
-15	7760	3200	7600	3410	7000	3710	5600	3800		
-7	7390	2450	7500	2850	7300	3370	7000	3890	6440	3670
2	7380	1890	7450	2380	7000	2850	7000	3300	5460	2720
7	9150	1590	9000	2010	9000	2610	8950	3220	7250	2870
25	8020	980	7880	1320	8460	1860	7600	2030	6300	1870

21.3 Cooling Capacity Table

21.3.1 WH-MDC05J3E5

OD	Water Out 7ºC		Water Out 14ºC		Water Out 18ºC	
	IQ	IP	IQ	IP	IQ	IP
16	5180	820	6170	840	5780	600
25	5380	1220	6640	1250	5550	780
35	5000	1540	5860	1610	5000	990
43	4190	1850	5360	1920	4370	1300

21.3.2 WH-MDC07J3E5

OD	Water Out 7ºC		Water Out 14ºC		Water Out 18ºC	
	IQ	IP	IQ	IP	IQ	IP
16	5380	830	6690	850	7650	760
25	6960	1820	9060	1980	7580	1230
35	7000	2290	8370	2470	7000	1480
43	5600	2550	6870	2580	6100	1880

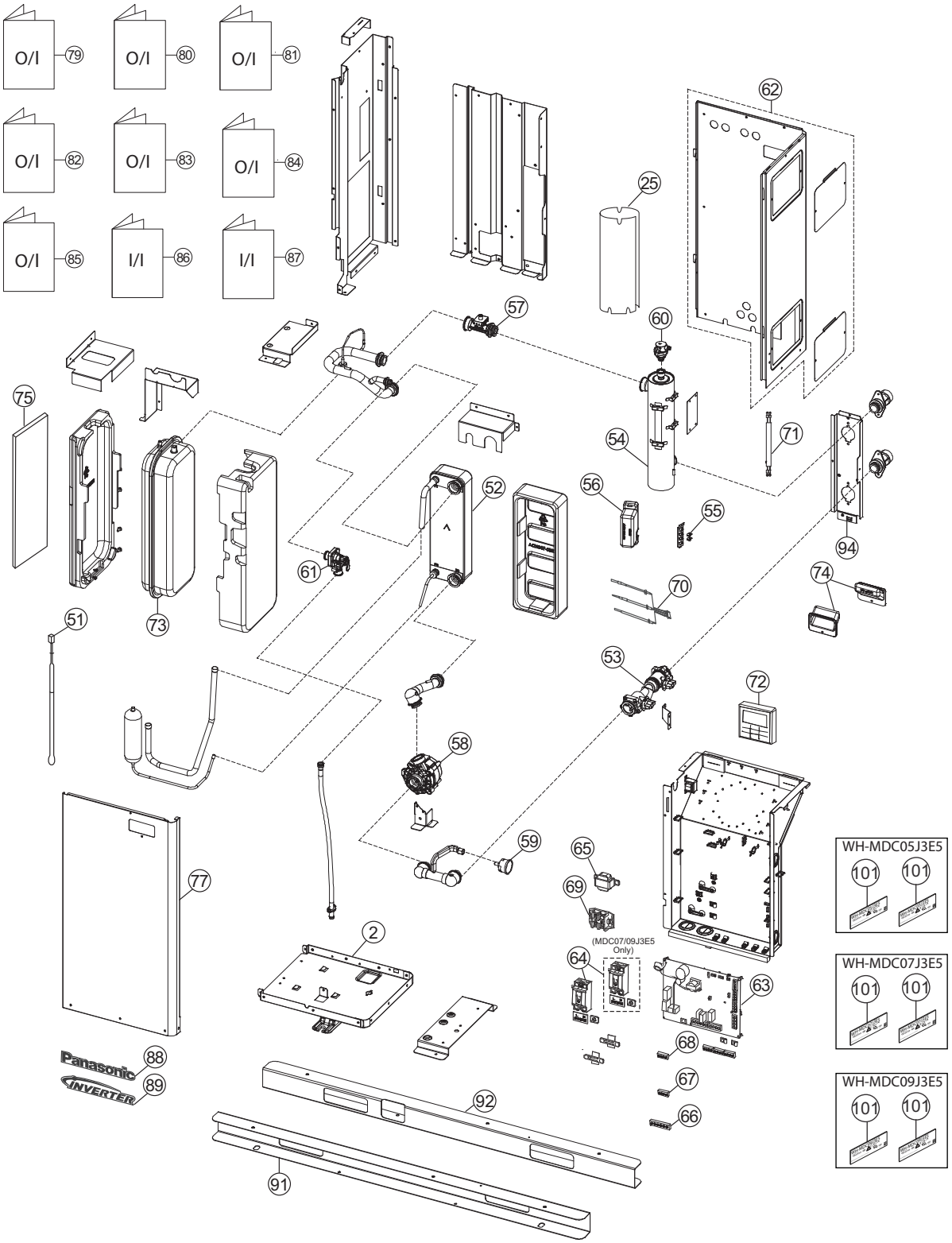
21.3.3 WH-MDC09J3E5

OD	Water Out 7ºC		Water Out 14ºC		Water Out 18ºC	
	IQ	IP	IQ	IP	IQ	IP
16	6890	1210	8650	1230	9820	1190
25	9500	2840	11550	3060	9680	1820
35	9000	3320	10100	3510	9000	2120
43	5420	2560	6560	2560	7400	2560

IQ = Capacity (W)
IP = Input Power (W)

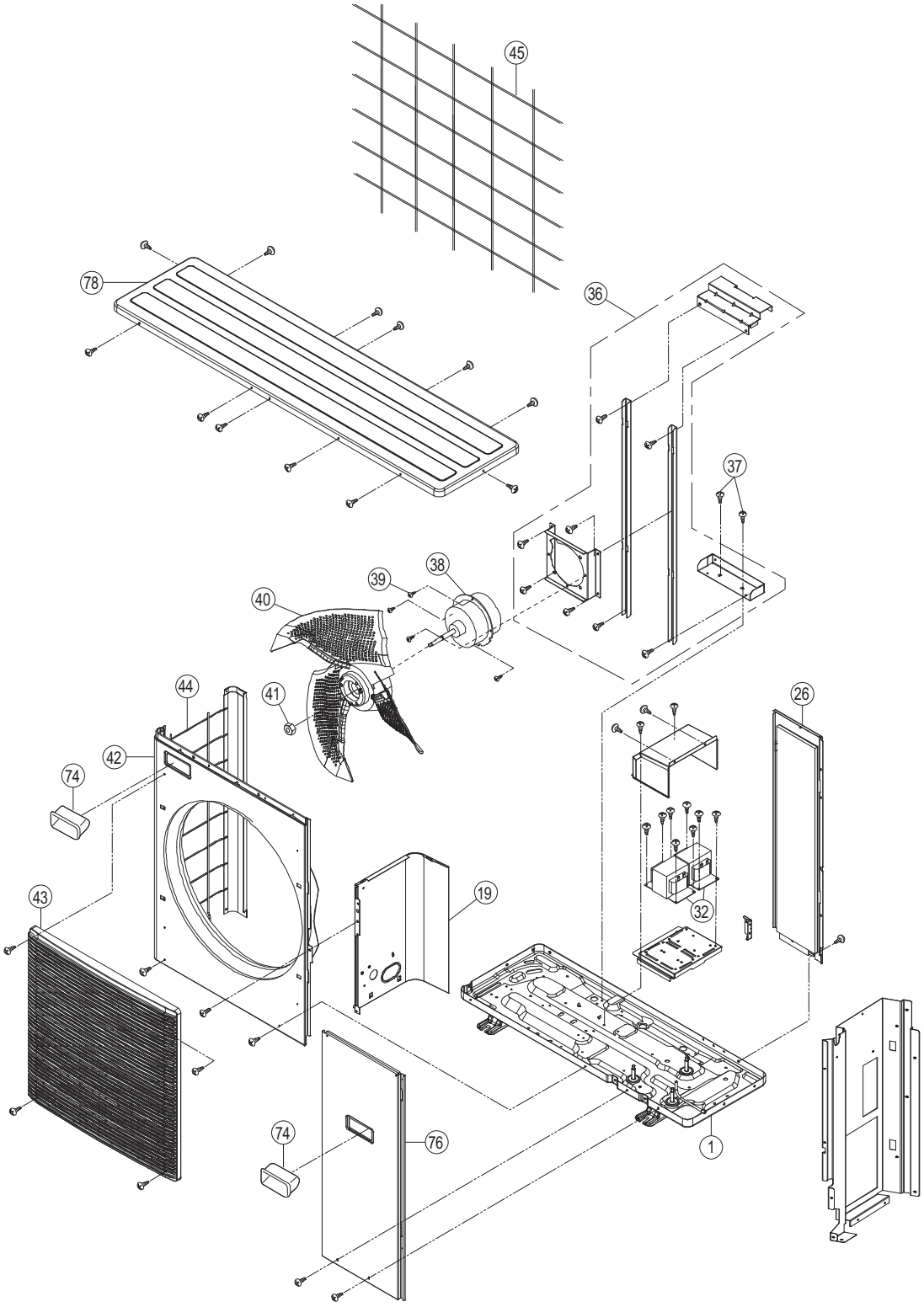
22. Exploded View and Replacement Parts List

22.1 Water System

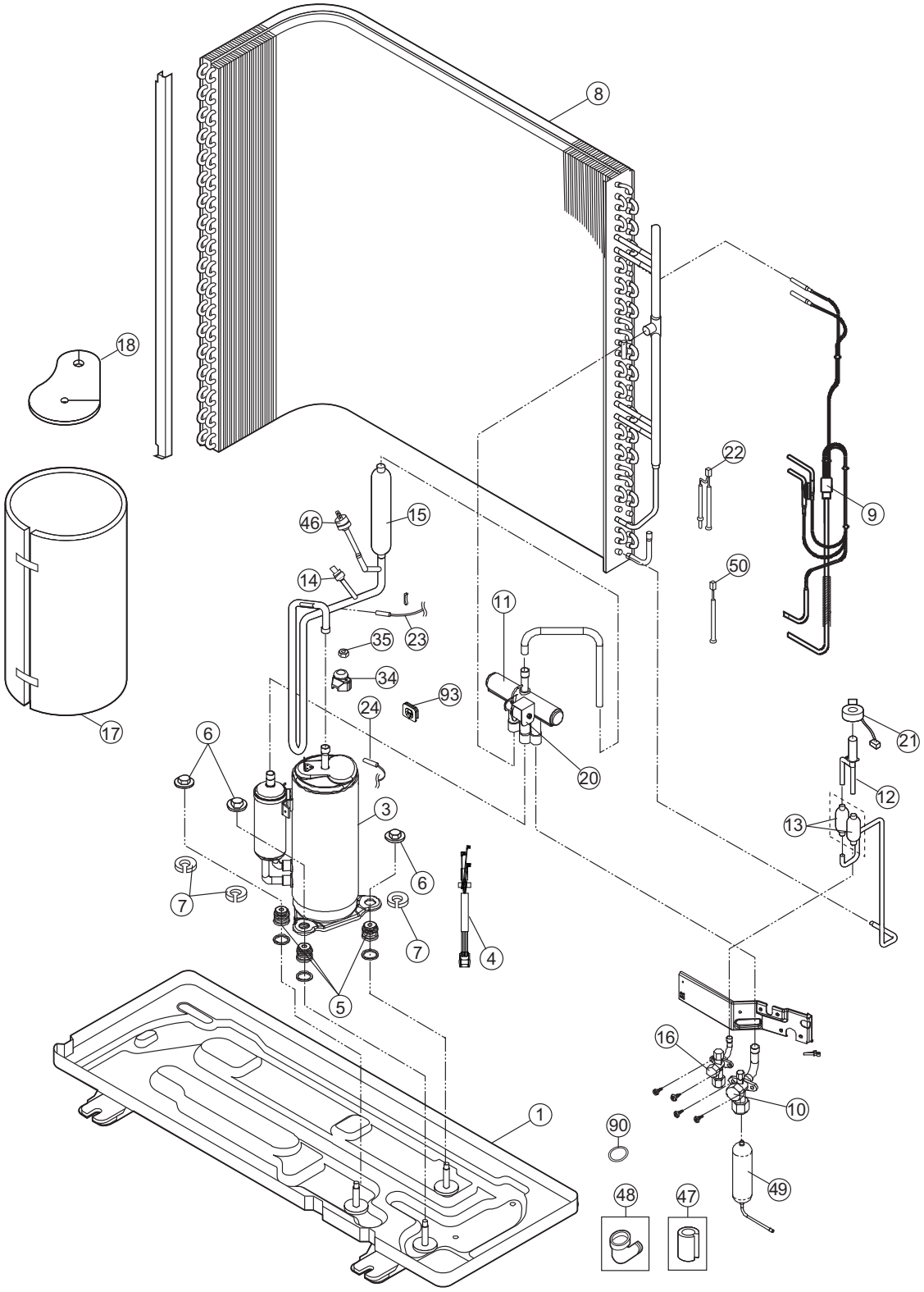


Note:
The above exploded view is for the purpose of parts disassembly and replacement.
The non-numbered parts are not kept as standard service parts.

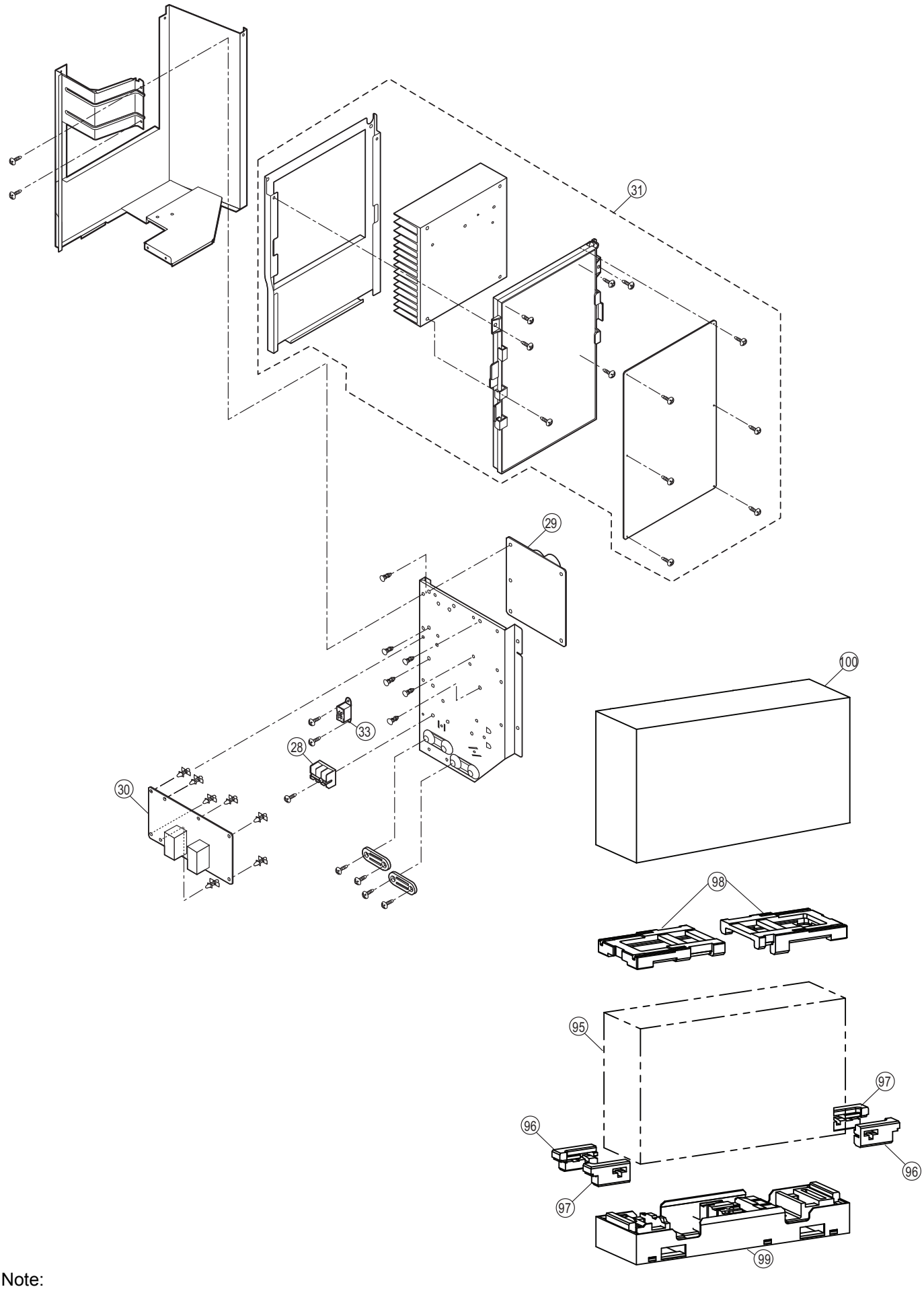
22.2 Refrigerant System



Note:
The above exploded view is for the purpose of parts disassembly and replacement.
The non-numbered parts are not kept as standard service parts.



Note:
The above exploded view is for the purpose of parts disassembly and replacement.
The non-numbered parts are not kept as standard service parts.



Note:
The above exploded view is for the purpose of parts disassembly and replacement.
The non-numbered parts are not kept as standard service parts.

<Model: WH-MDC05J3E5>

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-MDC05J3E5	REMARK
	1	BASE PAN ASS'Y (1)	1	CWD52K1282	
	2	BASE PAN ASS'Y (2)	1	ACXD52K03580	
⚠	3	COMPRESSOR	1	9KD240XBD21	O
⚠	4	LEADWIRE COMPRESSOR	1	CWA68C1810	O
	5	BUSHING - COMPRESSOR MOUNT	3	CWH50055	
	6	NUT-COMPRESSOR MOUNT	3	CWH561049	
	7	GASKET FOR TERMINAL COVER	3	CWB811017	
	8	CONDENSER CO.	1	ACXB32C09150	
	9	TUBE ASSY (CAP.TUBE)	1	CWT07K1547	
	10	3-WAYS VALVE (GAS)	1	CWB011700	
	11	4-WAYS VALVE	1	CWB001026J	
	12	EXPANSION VALVE	1	CWB051029	
	13	STRAINER	2	CWB111024	
	14	HIGH PRESSURE SENSOR CN-HPS	1	CWA501463	
	15	DISCHARGE MUFFLER	1	CWB121013	
	16	3-WAYS VALVE(LIQUID)	1	CWB011698	
	17	SOUND PROOF MATERIAL - COMP. BODY	1	ACXG30-10121	
	18	SOUND PROOF MATERIAL - COMP. TOP	1	CWG302246	
	19	SOUND PROOF BOARD	1	CWH151429	
⚠	20	V-COIL COMPLETE(4 WAY VALVE)	1	CWA43C2169J	O
⚠	21	V-COIL COMPLETE(MAIN EXPANSION VALVE)	1	CWA43C2342	O
⚠	22	SENSOR-COMP.(OUTDOOR AIR,PIPE TEMP)	1	CWA50C2652	O
⚠	23	SENSOR-COMP.(DISCHARGE TEMP)	1	CWA50C2528	O
⚠	24	SENSOR-COMP.(DEFROST TEMP)	1	CWA50C2665	O
	25	SOUND PROOF MATERIAL	1	ACXG30-10190	
	26	CABINET REAR PLATE-COMPLETE	1	ACXE02C01210	
⚠	28	TERMINAL BOARD ASS'Y(1,2,3,4,5)	1	CWA28K1251	O
⚠	29	ELECTRONIC CO.(CAPACITOR)	1	CWA746184	O
⚠	30	ELECTRONIC CO.(NOISE FILTER)	1	ACXA73-34930	O
⚠	31	ELECTRONIC CONTROLLER (MAIN) - REFRIGERANT	1	ACXA73C58260R	O
⚠	32	REACTOR	1	G0C193J00021	O
⚠	33	ELECTRO MAGNETIC SWITCH	-	-	O
	34	TERMINAL COVER	1	CWH171039A	
	35	NUT - TERMINAL COVER	1	CWH7080300J	
	36	FAN MOTOR BRACKET	1	ACXD54K02150	
	37	SCREW-FAN MOTOR BRACKET	4	CWH551040J	
⚠	38	FAN MOTOR	1	EHDS80C60AC	O
	39	SCREW-FAN MOTOR MOUNT	4	CWH551323	
	40	PROPELLER FAN ASSY	1	CWH00K1006	
	41	NUT	1	CWH561092	
	42	CABINET FRONT PLATE	1	ACXE06-03400A	
	43	DISCHARGE GRILLE	1	CWE201154	
	44	CABINET SIDE PLATE ASSY	1	CWE04K1019A	
	45	WIRE NET	1	CWD041059	
	46	HIGH PRESSURE SWITCH	1	CWA101013	

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-MDC05J3E5	REMARK
	47	ACCESSORY CO.(RUBBER CAP)	1	CWH82C1839	
	48	ACCESSORY CO.(DRAIN ELBOW)	1	CWG87C900	
	49	RECEIVER	1	ACXB14-00670	
⚠	50	SENSOR-COMPLETE	1	CWA50C3379	O
⚠	51	BASE PAN HEATER	1	CWA341071	O
	52	HOT WATER COIL-COMPLETE	1	ACXB90C00960	
	53	FILTER COMPLETE	1	ACXB51C00100	
⚠	54	HEATER ASS'Y	1	ACXA34K00240	O
⚠	55	THERMOSTAT	2	CWA151074	O
	56	TERMINAL COVER	1	CWH171051	
	57	FLOW SENSOR	1	ACXB62-00911	
⚠	58	PUMP	1	ACXB53-00390	O
	59	PRESSURE GAUGE	1	CWB070003	
	60	AIR PURGE VALVE	1	ACXB62-00130	
	61	PRESSURE RELIEF VALVE	1	ACXB62-00740	
	62	CABINET SIDE PLATE	1	ACXE04-11720A	
⚠	63	ELECTRONIC CONTROLLER (MAIN) - WATER	1	ACXA73C58300R	O
	64	CIRCUIT BREAKER	1	CWA181005	O
⚠	65	REACTOR	1	G0C103Z00003	O
⚠	66	TERMINAL BOARD ASS'Y(1,2,3,4,5,6)	1	CWA28K1341	O
⚠	67	TERMINAL BOARD ASS'Y(7,8,9,10,11,12)	2	CWA28K1340	O
⚠	68	TERMINAL BOARD ASS'Y(13,14,15,16,17,18)	3	CWA28K1343	O
⚠	69	TERMINAL BOARD ASS'Y(A,B)	1	CWA28K1238	O
⚠	70	SENSOR-COMPLETE(WATER IN,OUT, REF TEMP)	1	ACXA50C06750	O
⚠	71	SENSOR-COMPLETE(HEX WATER OUTLET)	1	ACXA50C06660	O
⚠	72	REMOTE CONTROL COMPLETE	1	ACXA75C15750	O
	73	RECEIVER	1	CWB141066	
	74	HANDLE	4	CWE161021	
	75	SOUND PROOF MATERIAL	1	ACXG30-10210	
	76	CABINET FRONT PLATE CO.	1	ACXE06C03790	
	77	CABINET FRONT PLATE	1	ACXE06-04500A	
	78	CABINET TOP PLATE CO.	1	ACXE03C01980	
	79	OPERATING INSTRUCTION	1	ACXF55-26660	
	80	OPERATING INSTRUCTION	1	ACXF55-26670	
	81	OPERATING INSTRUCTION	1	ACXF55-26680	
	82	OPERATING INSTRUCTION	1	ACXF55-26690	
	83	OPERATING INSTRUCTION	1	ACXF55-26700	
	84	OPERATING INSTRUCTION	1	ACXF55-26710	
	85	OPERATING INSTRUCTION	1	ACXF55-27620	
	86	INSTALLATION INSTRUCTION	1	ACXF60-38370	
	87	INSTALLATION INSTRUCTION	1	ACXF60-38380	
	88	PANASONIC BADGE	1	CWE373439	
	89	INVERTER BADGE	1	CWE373441	
	90	PACKING-L.TUBE	1	CWB81012	
	91	CONNECTING BAR	1	CWE261213A	
	92	CONNECTING BAR	1	CWE261214A	
	93	GASKET FOR TERMINAL COVER	1	CWB811177	

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-MDC05J3E5	REMARK
	94	HOLDER COUPLING	1	ACXH35-01880A	
	95	BAG	1	CWG861483	
	96	SHOCK ABSORBER	2	CWG713660	
	97	SHOCK ABSORBER	2	CWG713661	
	98	SHOCK ABSORBER	2	CWG713662	
	99	BASE BOARD-COMPLETE	1	ACXG62C02080	
	100	C.C.CASE	1	ACXG50-55100	
	101	MODEL LABEL	2	ACXF85-31560	

Note:

- All parts are supplied from PAPAMY, Malaysia (Vendor Code: 00029488).
- “O” marked parts are recommended to be kept in stock.

<Model: WH-MDC07J3E5 WH-MDC09J3E5>

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-MDC07J3E5	WH-MDC09J3E5	REMARK
	1	BASE PAN ASS'Y (1)	1	CWD52K1282	←	
	2	BASE PAN ASS'Y (2)	1	ACXD52K03580	←	
⚠	3	COMPRESSOR	1	9KD240XBD21	←	O
⚠	4	LEADWIRE COMPRESSOR	1	CWA68C1810	←	O
	5	BUSHING - COMPRESSOR MOUNT	3	CWH50055	←	
	6	NUT-COMPRESSOR MOUNT	3	CWH561049	←	
	7	GASKET FOR TERMINAL COVER	3	CWB811017	←	
	8	CONDENSER CO.	1	ACXB32C09150	←	
	9	TUBE ASSY (CAP.TUBE)	1	CWT07K1547	←	
	10	3-WAYS VALVE (GAS)	1	CWB011700	←	
	11	4-WAYS VALVE	1	CWB001026J	←	
	12	EXPANSION VALVE	1	CWB051029	←	
	13	STRAINER	2	CWB111024	←	
	14	HIGH PRESSURE SENSOR CN-HPS	1	CWA501463	←	
	15	DISCHARGE MUFFLER	1	CWB121013	←	
	16	3-WAYS VALVE(LIQUID)	1	CWB011698	←	
	17	SOUND PROOF MATERIAL - COMP. BODY	1	ACXG30-10121	←	
	18	SOUND PROOF MATERIAL - COMP. TOP	1	CWG302246	←	
	19	SOUND PROOF BOARD	1	CWH151429	←	
⚠	20	V-COIL COMPLETE(4 WAY VALVE)	1	CWA43C2169J	←	O
⚠	21	V-COIL COMPLETE(MAIN EXPANSION VALVE)	1	CWA43C2342	←	O
⚠	22	SENSOR-COMP.(OUTDOOR AIR,PIPE TEMP)	1	CWA50C2652	←	O
⚠	23	SENSOR-COMP.(DISCHARGE TEMP)	1	CWA50C2528	←	O
⚠	24	SENSOR-COMP.(DEFROST TEMP)	1	CWA50C2665	←	O
	25	SOUND PROOF MATERIAL	1	ACXG30-10190	←	
	26	CABINET REAR PLATE-COMPLETE	1	ACXE02C01210	←	
⚠	28	TERMINAL BOARD ASS'Y(1,2,3,4,5)	1	CWA28K1251	←	O
⚠	29	ELECTRONIC CO.(CAPACITOR)	1	CWA746184	←	O
⚠	30	ELECTRONIC CO.(NOISE FILTER)	1	ACXA73-34920	←	O
⚠	31	ELECTRONIC CONTROLLER (MAIN) - REFRIGERANT	1	ACXA73C58270R	ACXA73C58280R	O
⚠	32	REACTOR	1	G0C592J00005	←	O
⚠	33	ELECTRO MAGNETIC SWITCH	1	CWA001023	←	O
	34	TERMINAL COVER	1	CWH171039A	←	
	35	NUT - TERMINAL COVER	1	CWH7080300J	←	
	36	FAN MOTOR BRACKET	1	ACXD54K02150	←	
	37	SCREW-FAN MOTOR BRACKET	4	CWH551040J	←	
⚠	38	FAN MOTOR	1	EHDS80C60AC	←	O
	39	SCREW-FAN MOTOR MOUNT	4	CWH551323	←	
	40	PROPELLER FAN ASSY	1	CWH00K1006	←	
	41	NUT	1	CWH561092	←	
	42	CABINET FRONT PLATE	1	ACXE06-03400A	←	
	43	DISCHARGE GRILLE	1	CWE201154	←	
	44	CABINET SIDE PLATE ASSY	1	CWE04K1019A	←	
	45	WIRE NET	1	CWD041059	←	
	46	HIGH PRESSURE SWITCH	1	CWA101013	←	
	47	ACCESSORY CO.(RUBBER CAP)	1	CWH82C1839	←	

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-MDC07J3E5	WH-MDC09J3E5	REMARK
	48	ACCESSORY CO.(DRAIN ELBOW)	1	CWG87C900	←	
	49	RECEIVER	1	ACXB14-00670	←	
⚠	50	SENSOR-COMPLETE	1	CWA50C3379	←	O
⚠	51	BASE PAN HEATER	1	CWA341071	←	O
	52	HOT WATER COIL-COMPLETE	1	ACXB90C00960	←	
	53	FILTER COMPLETE	1	ACXB51C00100	←	
⚠	54	HEATER ASS'Y	1	ACXA34K00240	←	O
⚠	55	THERMOSTAT	2	CWA151074	←	O
	56	TERMINAL COVER	1	CWH171051	←	
	57	FLOW SENSOR	1	ACXB62-00911	←	
⚠	58	PUMP	1	ACXB53-00390	←	O
	59	PRESSURE GAUGE	1	CWB070003	←	
	60	AIR PURGE VALVE	1	ACXB62-00130	←	
	61	PRESSURE RELIEF VALVE	1	ACXB62-00740	←	
	62	CABINET SIDE PLATE	1	ACXE04-11720A	←	
⚠	63	ELECTRONIC CONTROLLER (MAIN) - WATER	1	ACXA73C58310R	ACXA73C58320R	O
	64	CIRCUIT BREAKER	2	CWA181005	←	O
⚠	65	REACTOR	1	G0C103Z00003	←	O
⚠	66	TERMINAL BOARD ASS'Y(1,2,3,4,5,6)	1	CWA28K1341	←	O
⚠	67	TERMINAL BOARD ASS'Y(7,8,9,10,11,12)	2	CWA28K1340	←	O
⚠	68	TERMINAL BOARD ASS'Y(13,14,15,16,17,18)	3	CWA28K1343	←	O
⚠	69	TERMINAL BOARD ASS'Y(A,B)	1	CWA28K1238	←	O
⚠	70	SENSOR-COMPLETE(WATER IN,OUT, REF TEMP)	1	ACXA50C06750	←	O
⚠	71	SENSOR-COMPLETE(HEX WATER OUTLET)	1	ACXA50C06660	←	O
⚠	72	REMOTE CONTROL COMPLETE	1	ACXA75C15750	←	O
	73	RECEIVER	1	CWB141066	←	
	74	HANDLE	4	CWE161021	←	
	75	SOUND PROOF MATERIAL	1	ACXG30-10210	←	
	76	CABINET FRONT PLATE CO.	1	ACXE06C03790	←	
	77	CABINET FRONT PLATE	1	ACXE06-04500A	←	
	78	CABINET TOP PLATE CO.	1	ACXE03C01980	←	
	79	OPERATING INSTRUCTION	1	ACXF55-26660	←	
	80	OPERATING INSTRUCTION	1	ACXF55-26670	←	
	81	OPERATING INSTRUCTION	1	ACXF55-26680	←	
	82	OPERATING INSTRUCTION	1	ACXF55-26690	←	
	83	OPERATING INSTRUCTION	1	ACXF55-26700	←	
	84	OPERATING INSTRUCTION	1	ACXF55-26710	←	
	85	OPERATING INSTRUCTION	1	ACXF55-27620	←	
	86	INSTALLATION INSTRUCTION	1	ACXF60-38370	←	
	87	INSTALLATION INSTRUCTION	1	ACXF60-38380	←	
	88	PANASONIC BADGE	1	CWE373439	←	
	89	INVERTER BADGE	1	CWE373441	←	
	90	PACKING-L.TUBE	1	CWB81012	←	
	91	CONNECTING BAR	1	CWE261213A	←	
	92	CONNECTING BAR	1	CWE261214A	←	
	93	GASKET FOR TERMINAL COVER	1	CWB811177	←	
	94	HOLDER COUPLING	1	ACXH35-01880A	←	

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-MDC07J3E5	WH-MDC09J3E5	REMARK
	95	BAG	1	CWG861483	←	
	96	SHOCK ABSORBER	2	CWG713660	←	
	97	SHOCK ABSORBER	2	CWG713661	←	
	98	SHOCK ABSORBER	2	CWG713662	←	
	99	BASE BOARD-COMPLETE	1	ACXG62C02080	←	
	100	C.C.CASE	1	ACXG50-55100	←	
	101	MODEL LABEL	2	ACXF85-31570	ACXF85-31580	

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